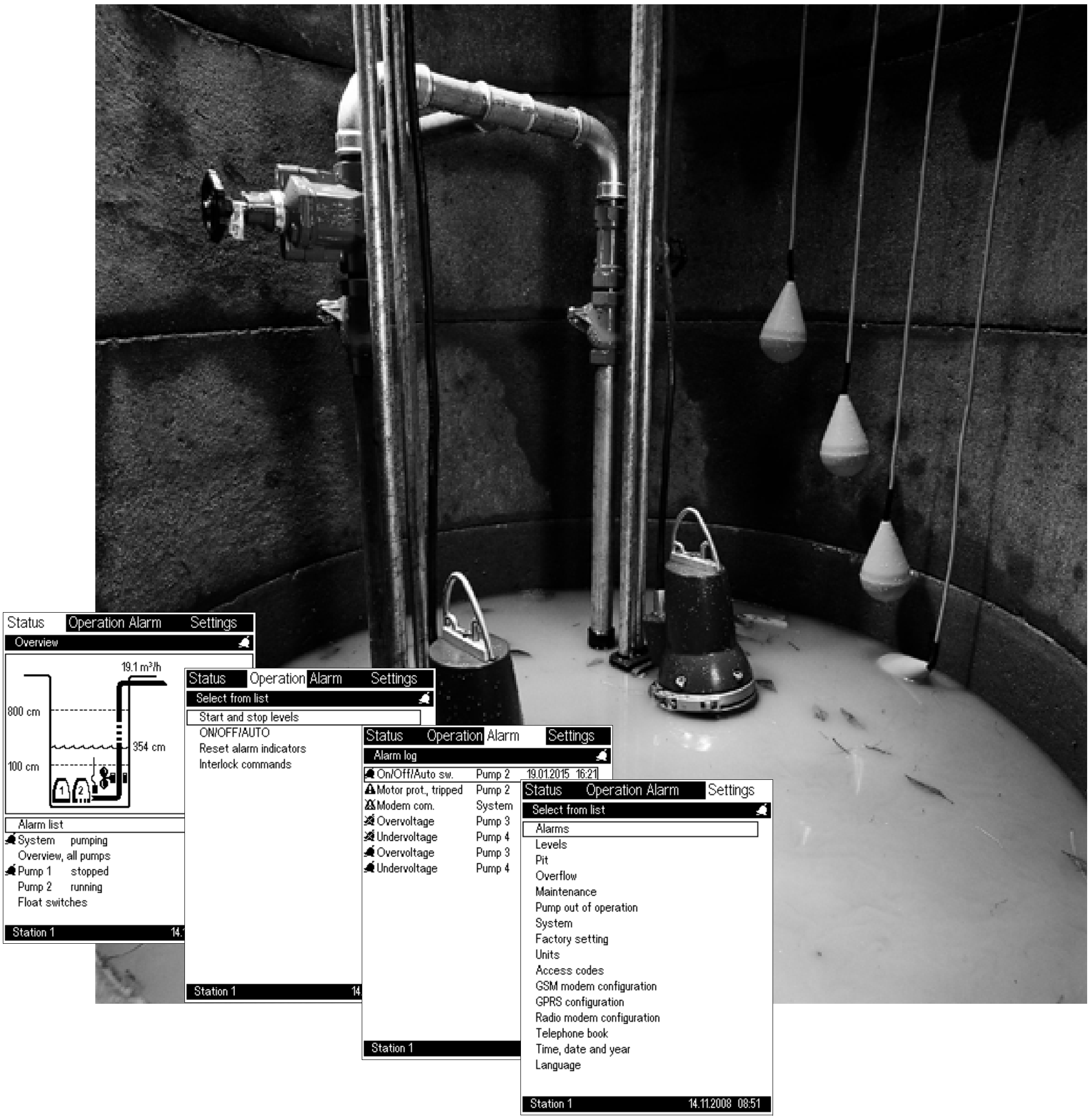
GRUNDFOS **INSTRUCTIONS**

Grundfos Modular Controls

for wastewater pumping stations
WW 2 application, version V03.00

GB **Installation and operating instructions**



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CONTENTS

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

All wires to units outside the control panel must be of the type H05VV-F according to CENELEC HD21 (to avoid injury from touching wires).

1. Reading instructions

It is recommended to read the installation and operating instructions for Grundfos Modular Controls as follows:

1. First read the entire installation and operating instructions.
2. Then use the installation and operating instructions as a reference book.

Sections	Contain
2. to 7.	general information about installation, mutual position of control unit and modules, electrical connection, program card and operator display
8. to 14.	a description of the displays appearing in the operator display
16. to 18.	an overview of inputs and outputs, fault finding chart and dimensions

For electronic reading, section *8.2 OD 401 menu overview* is recommended. The menu overview has links for all headlines, allowing quick navigation between the various sections.

Note: The displays shown in this installation and operating instructions are dynamic, changing according to the system configuration. They are therefore to be considered as examples.

2. General description

Modular Controls is a Grundfos control system designed for the control and monitoring of a number of Grundfos pumps or pumps of another make via digital and analog inputs and outputs.

The Modular Controls control system WW2 consists of a CU 401 control unit connected to an IO 403 module and one or more IO 401 modules. Being the "brain" of the system, the CU 401 unit must be incorporated in all installations. The number of modules in the system depends on the number of pumps to be monitored and controlled.

The control system controls the pumps by means of contactors; electronic pumps are controlled via GENIbus. Contactors, other power components and cables should be positioned at the greatest possible distance from the control system and signal cables.

3. Mounting

The Modular Controls system should be mounted in a closed cabinet.

The ambient temperature for the control system must be kept between –20 °C and +60 °C for maximum life.

3.1 Further documentation

For installation and specific technical data, see the installation and operating instructions of the individual units. See also the individual instructions when replacing flash cards, SIM cards etc.

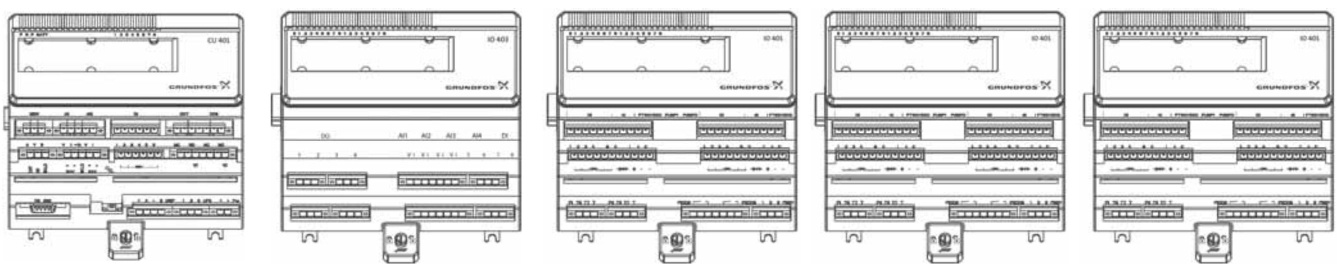
4. Positioning

Fig. 1 shows the mutual position of control unit and modules of a system incorporating up to six wastewater pumps.

- Always position the CU 401 control unit to the left of the modules.
- Position the IO 403 module as the first module to the right of the CU 401.
- Position the IO 401 modules to the right of the IO 403 module.
- There must be sufficient space around the unit and the modules to allow a modem, program card and communication bus to be fitted.

Note: Remember to insert the battery into the CU 401.

CU 401	IO 403	IO 401	IO 401	IO 401



Control unit (See *5.5 Wiring diagram for system connections (CU 401)*)	**Additional equipment to be connected** (See *5.6 Wiring diagram for IO 403 connections (example)*)	**Pumps 1 and 2 connected** (See *5.7 Wiring diagram for two pumps via IO 401 module*)	**Pumps 3 and 4 connected** (See *5.7 Wiring diagram for two pumps via IO 401 module*)	**Pumps 5 and 6 connected** (See *5.7 Wiring diagram for two pumps via IO 401 module*)
1 to 2 pumps				
1 to 4 pumps				
1 to 6 pumps				

Fig. 1 Position of control unit and modules

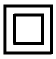
5. Electrical connection and safety requirements

5.1 Float switches

If used, float switches must be connected from left to right, meaning that the lowest float switch must be connected to input DI1, the second float switch from the bottom must be connected to input DI2, etc.. See the wiring diagrams, pages 5 to 8.

Float switches must be with reinforced isolation.

Symbol:



5.2 Alarm relays

If external alarms are connected to the alarm relays and high-level-alarm level sensor, a reset button should be fitted on the DI6 terminals. Otherwise the alarms can only be reset as shown on the display in section *11.2.1 Alarm list - resetting/approving system alarms and warnings*.

Note: Physical connection of sensors and control buttons is not sufficient. The units must also be enabled in the alarm displays. See section *14.2.1 System alarms and warnings - enable/disable*.

5.3 Earthing

If electric noise or vibrations are expected, the Modular Controls control unit and modules should be earthed by means of screws on a metallic back plate, earthed with the most suitable connection to earth. This must be done to avoid injury to persons and prevent undesired electric noise.

Signal cables must be securely earthed. The optimum solution is to fit a jumper on the electrically conducting back plate direct across and in contact with the cable screen.

Each pump must have separate protection in the form of a motor-protective circuit breaker or similar protection.

OD 401 operator display cables running outside a closed cabinet must be of the H05VV-F type.

5.4 Connection of float switches

The table refers to the digital inputs in the wiring diagrams on pages 5 to 8.

Control type	Digital inputs					
	1	2	3	4	5	6
Float switches	Lowest float switch	Second float switch from the bottom	Third float switch from the bottom	Fourth float switch from the bottom	Fifth float switch from the bottom	Alarm relay reset button
Analog sensor with dry-running float switch	Dry-running float switch	–	Overflow float switch*	Fault switch**	Mixer feedback***	Alarm relay reset button
Analog sensor with high-level float switch	High-level float switch	–	Overflow float switch*	Fault switch**	Mixer feedback***	Alarm relay reset button
Analog sensor with dry-running and high-level float switches	Dry-running float switch	High-level float switch	Overflow float switch*	Fault switch**	Mixer feedback***	Alarm relay reset button

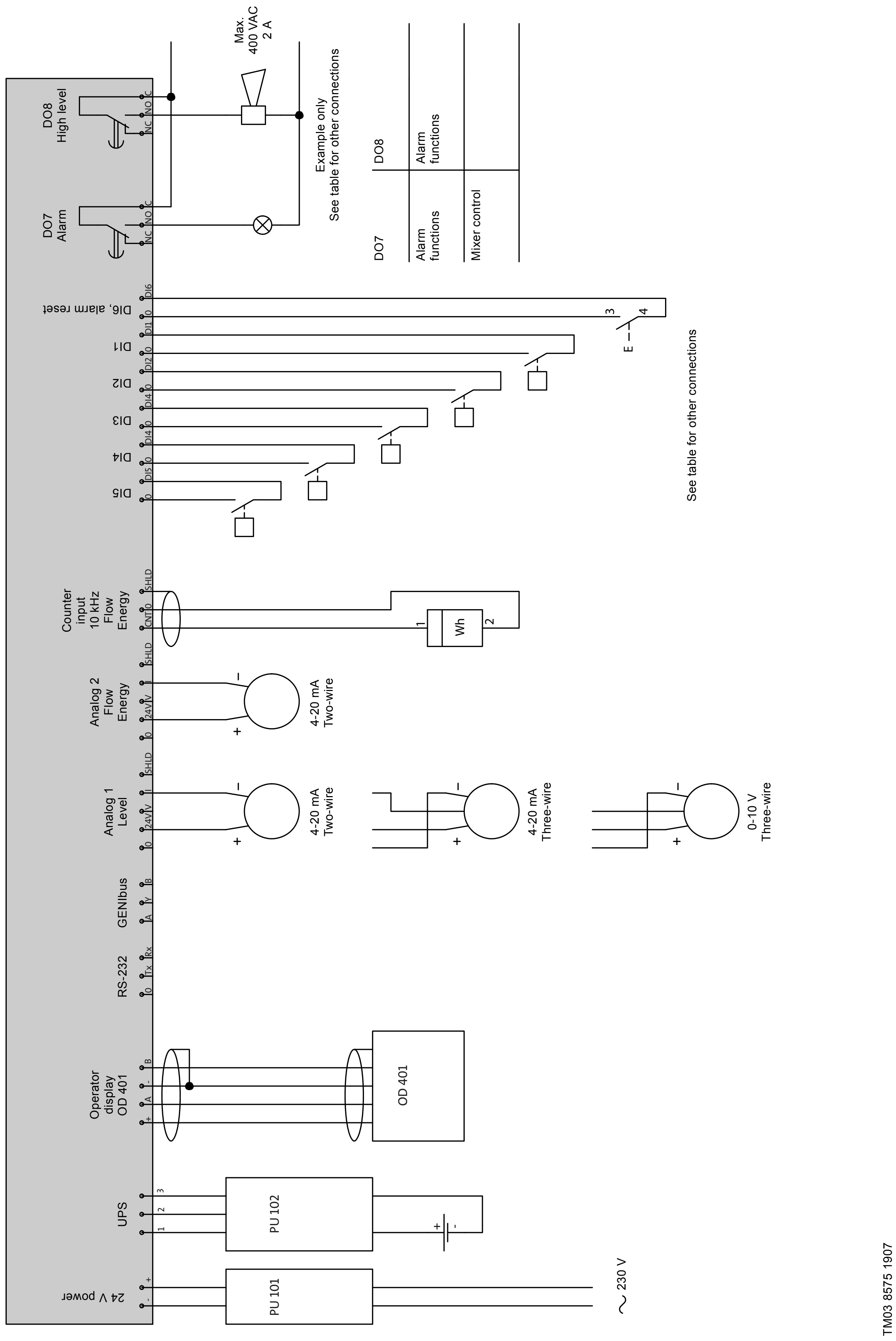
* Function which can be added as additional accuracy for overflow system 1.

** Function which can be added as additional safety for overflow to tank.

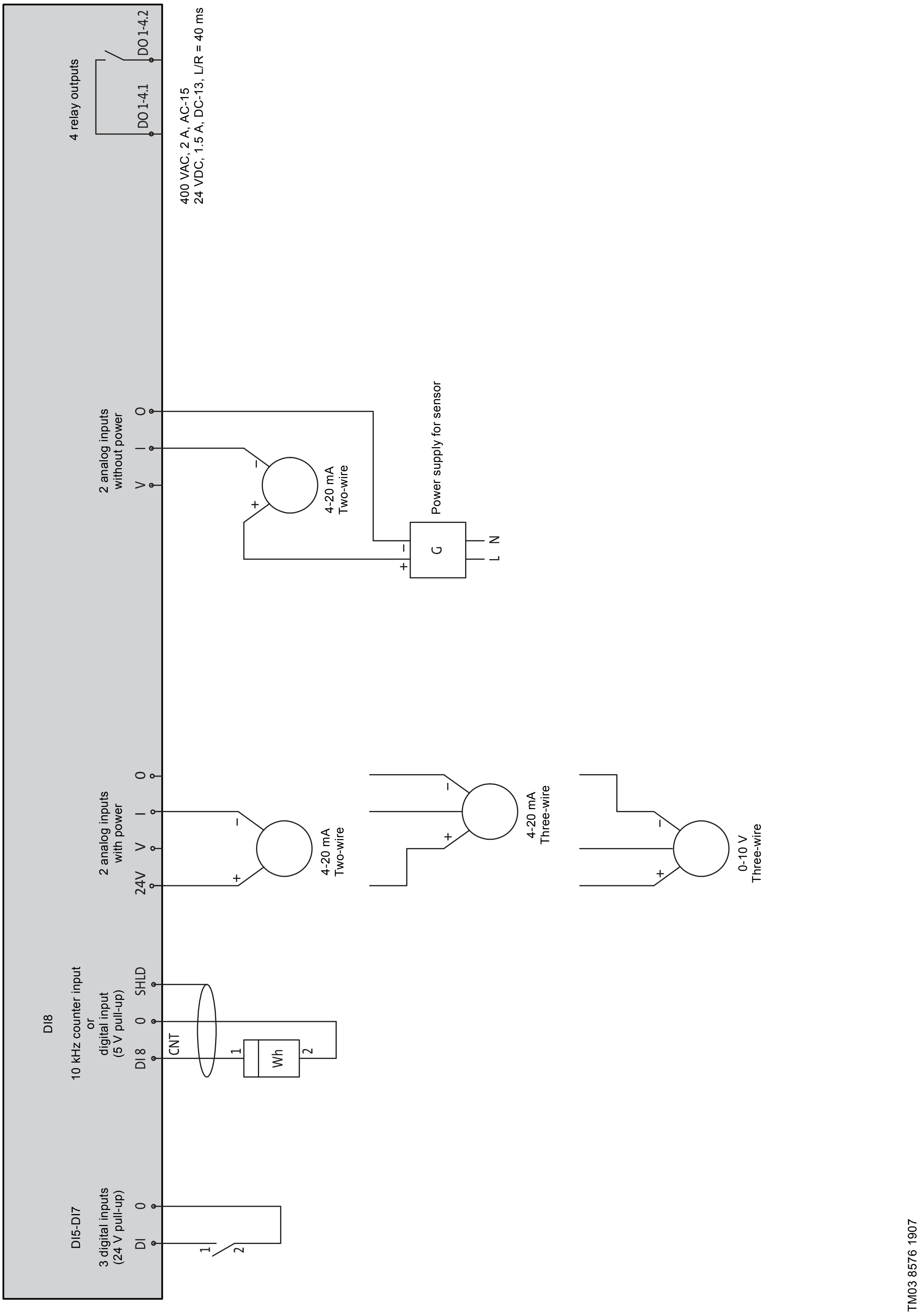
*** Function which can be added as additional safety in mixer installations.

5.5 Wiring diagram for system connections (CU 401)

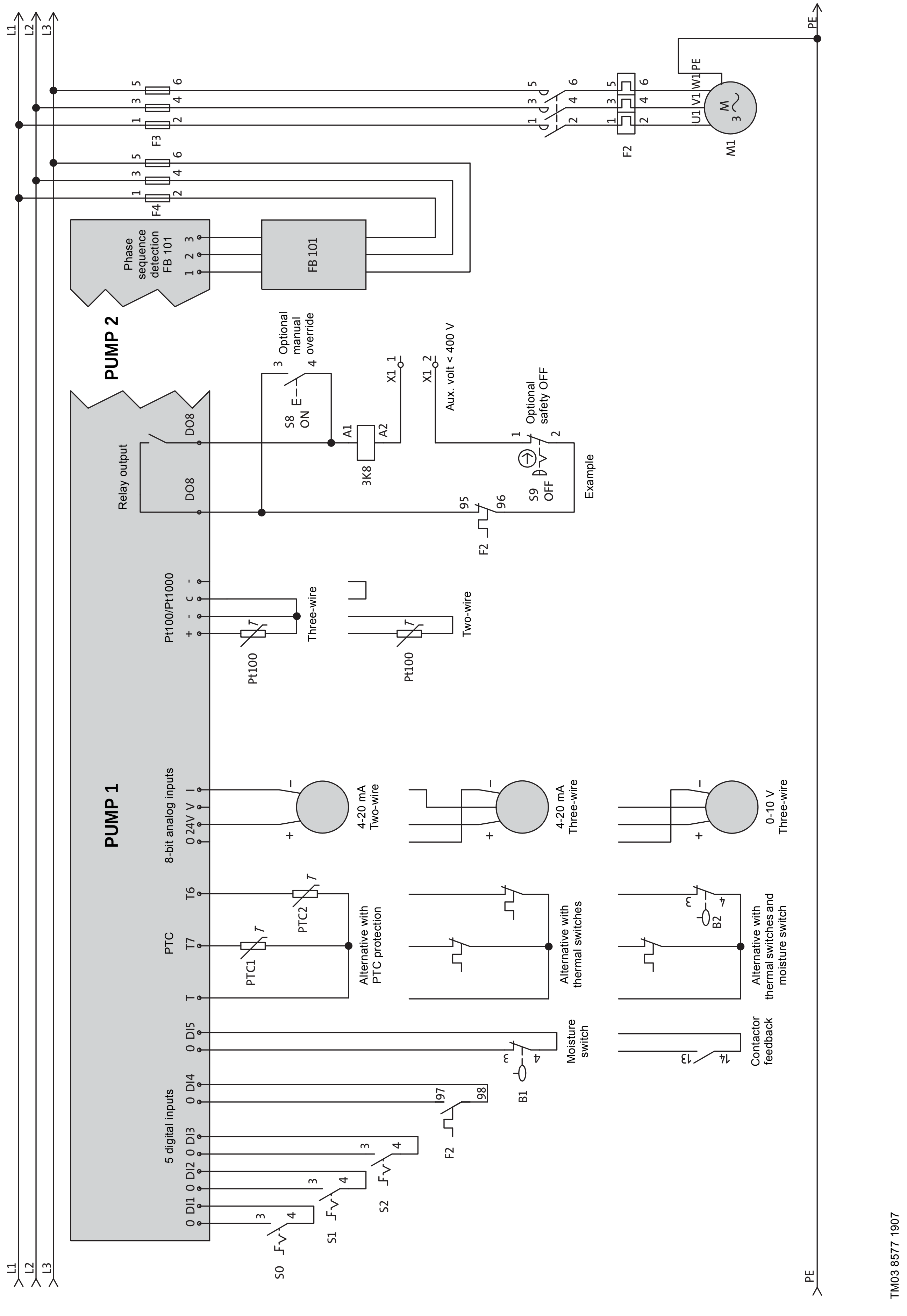
Note: The wiring diagram shown below should be considered as an example.



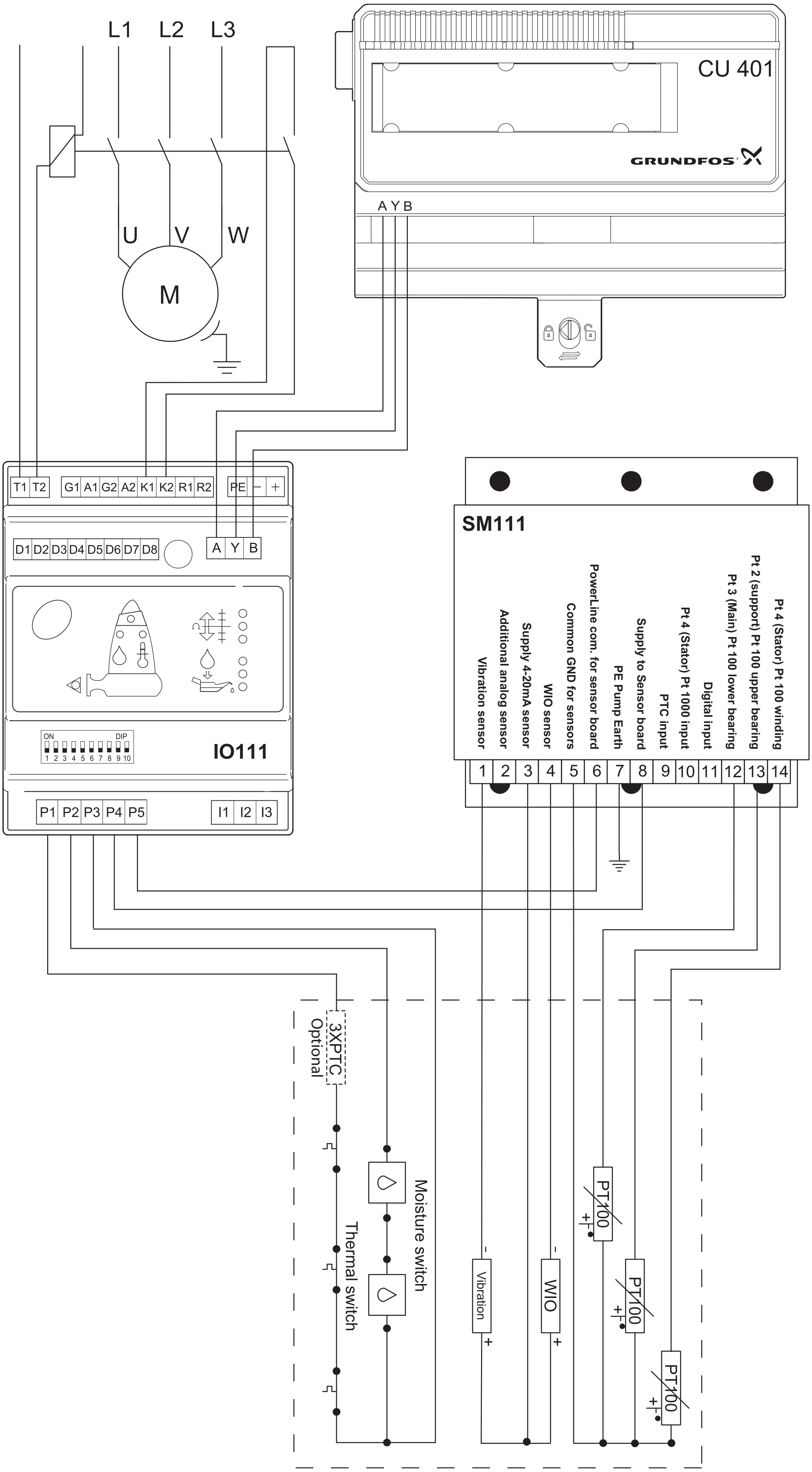
5.6 Wiring diagram for IO 403 connections (example)



5.7 Wiring diagram for two pumps via IO 401 module



5.8 Wiring diagram for SM 111 with IO 111 module



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Note: For further information about electrical connection, see installation and operating instructions for the IO 111 and section *5.10.8 Inputs on the SM 111*.

5.9 Use of inputs when connecting multiple units

Connected unit	IO 403	IO 401		MP 204		IO 111		IO 111 and SM 111	
Water in oil	–	–	AI	–	–	X [1]	GENIbus	X [3]	GENIbus
Latest current	–	–	AI	X [1]	GENIbus	–	–	–	–
Insulation resistance	–	–	–	X [1]	GENIbus	[4]	GENIbus	[4]	GENIbus
Temperature, Pt 1	–	X [1]	Pt100/ Pt1000	–	–	–	–	–	–
Temperature, Pt 2	–	–	–	[4]	GENIbus	X [1]	GENIbus	X [1]	GENIbus
Temperature, Pt 3	–	–	–	–	–	–	–	X [1]	Pt100/ Pt1000
Temperature, Pt 4	–	–	–	–	–	X [1]	GENIbus	X [1]	GENIbus
Moisture in motor	–	X [1]	DI	–	–	X [1]	GENIbus	X [1]	GENIbus
Overtemp., sensor 1	–	X [1]	DI	–	–	X [1]	GENIbus	X [1]	GENIbus
Vibration sensor	AI1 [2]	–	–	–	–	–	–	X [1]	GENIbus
Depth gauge	AI2 [2]	–	–	–	–	–	–	–	–
Pressure gauge	AI3 [2]	–	–	–	–	–	–	–	–
Water on floor	DI3 [2]	–	–	–	–	–	–	–	–
Burglar alarm	DI4 [2]	–	–	–	–	–	–	–	–
Rainwater gauge	DI8 [2]	–	–	–	–	–	–	–	–
Resetting of motor protection	DO1 [2]	–	–	–	–	–	–	–	–
Start of drainage pump	DO3 [2]	–	–	–	–	–	–	–	–

[1] X is the value used in the CU 401/OD 401.

[2] Example when using the IO 403.

[3] Value from the SM 111.

[4] Value available, but is only used if the value is not available from other modules.

IO 403 inputs and outputs

The IO 403 inputs and outputs are connected according to the user's needs.

All input values are transferred directly to the SCADA system. The outputs are controlled by the SCADA system or via the inputs. See the table below.

The DI8 input can function as a counter input or as a digital input.

	Analog inputs				Digital inputs				Relay outputs			
Terminal designation	AI1	AI2	AI3	AI4	DI5	DI6	DI7	DI8	DO1	DO2	DO3	DO4

Further information about the individual inputs and outputs can be found in section *14.7.9 IO 403, I/O settings* and in the installation and operating instructions for the IO 403.

5.10 SM 111 and IO 111

The SM 111 and IO 111 modules enable the user to monitor more pumps and system data.

The SM 111 module is either mounted in the pump or in a control cabinet.

The IO 111 module is mounted in the control cabinet. The DIP switches on the IO 111 are used to configure the IO 111. All DIP switches are shown in position OFF. See fig. 2.

5.10.1 Communication between the SM 111 and IO 111

The power lines are used for the transfer of data between the SM 111 and IO 111. This means that only two cables connect the modules.

Note: For further information about the modules, see the relevant installation and operating instructions.

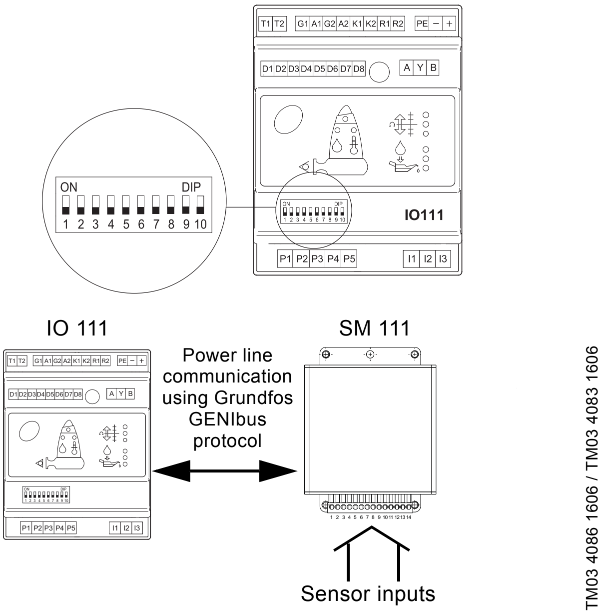

Fig. 2 Communication and power supply

5.10.2 Setting of pump variant

The user has to set which type of system the SM 111 and IO 111 modules are connected to.

DIP switches used for setting of pump variant

Pump variant	ON DIP 1 2 3 4 5 6 7 8 9 10	**Description**
Variant B	ON 1 2	System with SM 111: PTC sensors are not connected to the SM 111, but must be hardwired out of the pump.
Variant C	ON 1 2	System with SM 111: PTC sensors in the stator windings are connected to the SM 111.

5.10.3 Setting of GENI addresses

The IO 111 communicates with control systems via a GENIbus connection. In connection with bus communication, the IO 111 is a slave. As a control system must be able to identify the slave units it communicates with, the IO 111 modules must have unique addresses. Addresses between 40 and 45 are set by means of the switches 3, 4 and 5.

Note: The GENI address 40 is always used for pump 1.

DIP switches used for setting of pump number

ON DIP 1 2 3 4 5 6 7 8 9 10	**Description**
3 4 5	Pump 1
3 4 5	Pump 2
3 4 5	Pump 3
3 4 5	Pump 4
3 4 5	Pump 5
3 4 5	Pump 6

Note: For setting of GENI addresses, see installation and operating instructions for the IO 111.

5.10.4 Setting of analog outputs on the IO 111

The DIP switches 6 and 7 are used for setting the analog outputs. For further information, see installation and operating instructions for the IO 111.

5.10.5 Setting of communication between the IO 111 and CU 401

GENIbus communication is always used in wastewater applications. Set DIP switch 8 to OFF to enable GENIbus communication.

DIP switch used for setting of bus protocol

Bus protocol	ON DIP 1 2 3 4 5 6 7 8 9 10	**Description**
GENIbus	8	Grundfos standard protocol for communication between Grundfos products

5.10.6 Automatic sensor detection

The first time voltage is applied, the SM 111 will check all inputs to detect which sensors are connected. If a valid signal is measured, the SM 111 will register that a sensor is present and save the status.

If the IO 111 indicates a sensor fault or configuration fault, the sensor detection must be changed. Change DIP switch 9 on the IO 111 from OFF to ON.

After approx. 5 seconds, the SM 111 will reset, and the sensor detection is completed. Then set the DIP switch back to OFF.

DIP switch used for sensor detection

Sensor detection	ON DIP 1 2 3 4 5 6 7 8 9 10	Description
Activated.	9	Change DIP switch 9 on the IO 111 from OFF to ON. After approx. 5 seconds, the SM 111 will reset, and the sensor detection is completed.
Set back the DIP switch.	9	Set the DIP switch to OFF.

5.10.7 Connection

The IO 111 modules are connected in parallel with a daisy chain.

Note: The maximum length of the GENI cable is 1200 metres.

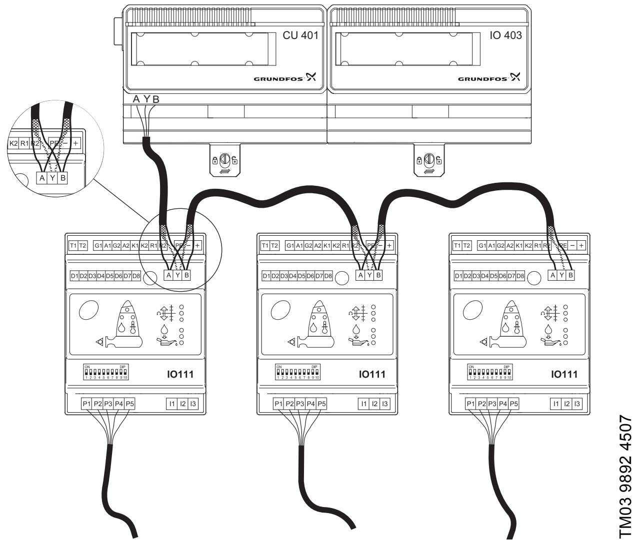

Fig. 3 Connecting several IO 111 modules with daisy chain

Measuring

The water-in-oil sensor is only active when the pump is running. The insulation resistance will only be measured when the pump is stopped.

5.10.8 Inputs on the SM 111

Analog inputs

The SM 111 module has seven analog inputs. One of the inputs is additional and can be configured by the user. It is possible to select the unit to be displayed when an analog sensor is connected to the additional input. See section *14.8.1 Units*.

All analog input signals are 4-20 mA.

Digital input and PTC input

The SM 111 has a digital input and a PTC sensor input.

For further information about electrical data, see installation and operating instructions for the SM 111.

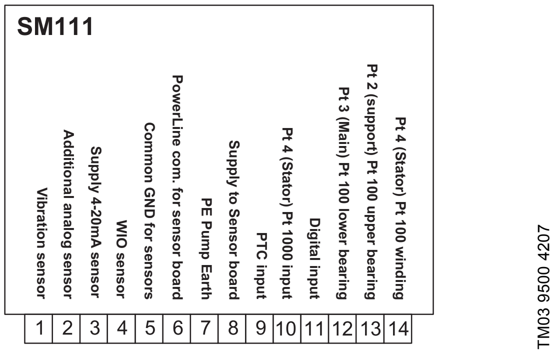

Fig. 4 SM 111 terminals

Terminal	Description
1	Vibration sensor
2	Additional analog sensor
3	Supply for 4-20 mA sensor
4	Water-in-oil sensor (WIO)
5	Common GND for sensors
6	Power line communication for sensor board
7	Earth connection for pump (PE)
8	Supply for sensor board
9	PTC input
10	Pt 4 (stator winding), Pt1000 input
11	Digital input
12	Pt 3 (main bearing), Pt100 input
13	Pt 2 (support bearing), Pt100 input
14	Pt 4 (stator winding), Pt100 input

Note: If both a Pt100 and a Pt1000 sensor are connected to the SM 111 (Pt 4), the signal from the Pt1000 sensor is used.

Note: When the user has made all connections, it must be ensured that all wires have been connected to the right terminals.

5.10.9 IO 111 and other modules

If both an IO 111 module and an MP 204 motor protector are connected to the system, the insulation resistance, including alarms and warnings, is taken from the MP 204. The Pt 2 temperature measurement, including alarms and warnings, is taken from the IO 111. See section *5.9 Use of inputs when connecting multiple units*.

5.10.10 Setting of alarms and warnings

The CU 401 can be configured to handle alarms and warnings detected by the IO 111 or SM 111. Changes of IO 111 and SM 111 alarm limits are to be made in the IO 111, possibly by means of the Grundfos PC Tool MP 204. Alarm handling is described in detail in section *14.2.6 Pump alarms and warnings - alarm handling*.

Each alarm and/or warning can be reset or acknowledged. See section *11.2.2 Alarm list - resetting/approving pump alarms and warnings*.

Note: The IO 111 module does not detect alarms or warnings from the additional analog input.

5.11 Combi alarms

Combi alarms enable the user to combine two alarms into one single alarm. Both alarms have to be active before the SCADA system is called. This means that combi alarms reduce the number of unscheduled service calls, and thereby cost savings are achieved.

5.11.1 Configuration

Combi alarms are only visible in the SCADA system and are set via the Grundfos PC Tool or the SCADA system. It is possible to set eight different combi alarms. See section *18.5.1 Setting of combi alarms*.

Configuration via Grundfos PC Tool

1. Put a check mark in the **Combi alarm x** check box to enable combi alarms.
2. Select an alarm from **Alarm source 1** and then an alarm from **Alarm source 2**.
3. Put a check mark in the **Call back** check box if the SCADA system should be called when the two selected alarms are active.
4. Select under **SMS action** whether an SMS should be sent or not.

Note: For further information about configuration of SMS, see section *14.10.1 SMS phone numbers*.

Configuration via SCADA system

See the functional profile for how to set combi alarms via the SCADA system.

Value	Alarm source
0	Not used
1	High level
2	Alarm level
3	Overflow
4	All pumps in alarm
5	Pump 1 in alarm
6	Pump 2 in alarm
7	Pump 3 in alarm
8	Pump 4 in alarm
9	Pump 5 in alarm
10	Pump 6 in alarm
11	GENI error, pump 1
12	GENI error, pump 2
13	GENI error, pump 3
14	GENI error, pump 4
15	GENI error, pump 5
16	GENI error, pump 6
17	Combi alarm 1
18	Combi alarm 2
19	Combi alarm 3
20	Combi alarm 4
21	Combi alarm 5
22	Combi alarm 6
23	Combi alarm 7
24	Combi alarm 8
25	User-defined I/O 1
26	User-defined I/O 2
27	User-defined I/O 3
28	User-defined I/O 4
29	User-defined I/O 5
30	User-defined I/O 6
31	User-defined I/O 7
32	User-defined I/O 8

Note: Only selected alarms will appear in the **Alarm source** list. This means that not all alarms can be combined.

Note: Combi alarms are only visible in the SCADA system.

Note: Combi alarms do not appear in the Modular Controls alarm log.

Note: Combi alarms do not appear in the OD 401.

Example:

Combi alarm 1, see fig. 34.

1. Select an alarm from **Alarm source 1** (High level) and then an alarm from **Alarm source 2** (Pump 1 in alarm).
2. Put a check mark in the **Call back** check box if the SCADA system should be called when the two selected alarms are active.
3. Select "Yes" under **SMS action** if an SMS should be sent to the service team.

Note: Only two combi alarms are enabled. The combi alarms 3 to 8 are disabled.

5.12 User-defined I/O

Via Grundfos PC Tool, the user can set the system inputs and outputs under **User IO**. Section *18.5.2 Example of user-defined I/O* shows the setting of a standard 2-pump wastewater station with stormwater retention basin, drain pump and mixers.

Logic block diagram for user-defined I/O

Channel 1	**Logic**	**Invert**	**Hold delay** (rising edge)	**Output** (internal flag)
Channel 2				

5.13 Radio modem

The Modular Controls control system is able to communicate with SCADA systems via a radio network. The radio network is used in areas without GSM/GPRS network facilities.

The radio network is a group of pumping stations (slaves) connected to a SCADA system (master).

The size of the radio network varies from 1 to 246 pumping stations. The structure depends on the use of frequencies and repeaters. The SCADA system must be configured to poll each pumping station or alarm.

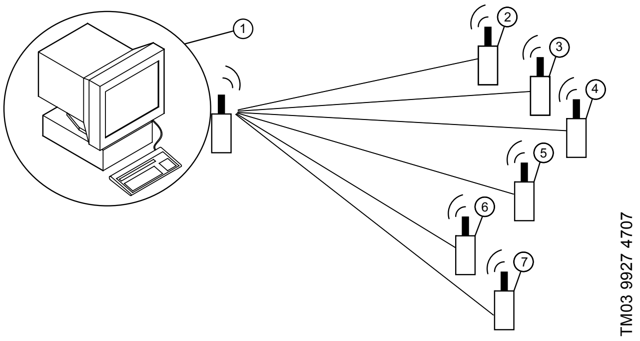

Fig. 5 SCADA system with radio network

Pos.	Description
1	SCADA system
2	Pumping station with radio modem
3	Pumping station with radio modem
4	Pumping station with radio modem
5	Pumping station with radio modem
6	Pumping station with radio modem
7	Pumping station with radio modem

5.13.1 Connection

The G 403 module is used as a gateway between the radio modem and the CU 401.

The radio modem must be connected to the G 403 with a RJ45 9-pin D-SUB cable plug.

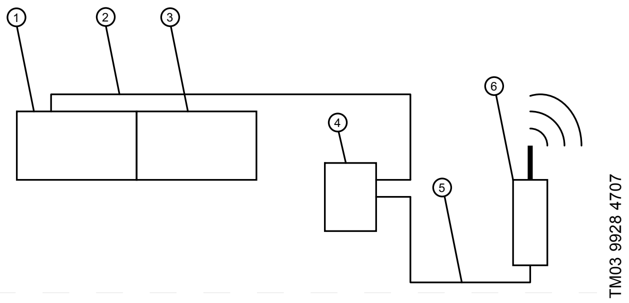

Fig. 6 Connection of radio modem

Pos.	Description
1	CU 401 control unit
2	Ethernet cable (crossed RJ45)
3	IO 403 module
4	G 403 module
5	Serial cable RJ45 (RS-232)
6	Radio modem

Note: Call-back and interlock functionality is not available when using radio network. Alarm monitoring is handled by the SCADA system by polling only. When using a radio network, the GSM/SMS/GPRS facilities are not available.

5.13.2 Power supply

The power supply and/or UPS system for the radio modem must be connected according to the supplier’s recommendation.

6. Program card

Switch off the power supply to the CU 401 control unit when replacing the program card or inserting the program card the first time.
Also switch off the PU 102.

6.1 General information

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The program card contains two programs and additional information. One program is run in the CU 401 control unit and the other in the OD 401 operator display. The OD 401 program is updated automatically when the system is started up; this means that the CU 401 control unit always has the program version matching the CU 401 program.

The additional information is for instance used for the configuration of the G 401 module.

The OD 401 operator display allows the system to be configured to control up to six pumps.

For a description of the OD 401, see sections *7. OD 401 operator display* and *8. Main menus*.

For pump, system, level and alarm settings, see section *14. Settings*.

6.2 Mounting

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Insert the program card into the control unit using the slot located at the opposite side of the terminals but in the same side as the indicator lights. See the picture to the left.

6.3 CU 401 software upgrade

Make sure that the latest version of the Grundfos PC Tool has been installed. It can be downloaded from http://www.grundfos.com/MR.

Note: A cable to connect the PC to the CU 401 is required. Consult the PC Tool online help for further instructions.

Procedure

1. Connect the PC to the CU 401 via the Grundfos PC Tool.
2. Transfer the settings from the CU 401 to the Grundfos PC Tool, and save the settings to the Grundfos PC Tool.
3. Disconnect the Grundfos PC Tool from the CU 401.
4. Switch off the CU 401.
5. Remove the old program card, and insert the new card.
6. Switch on the CU 401.
7. When the OD 401 is ready, re-establish the connection to the CU 401 via the Grundfos PC Tool.
8. Transfer the previously saved settings from the Grundfos PC Tool.
9. Select "Clone settings" in the "Save settings" menu, and save the settings to the CU 401. See fig. 7.
10. When the Grundfos PC Tool starts to save the settings, a caution message will appear saying that an older configuration will be saved. Click "Yes" to proceed. See fig. 8.
11. The old settings have been re-stored to the CU 401. The new settings now have the default values from the old configuration.

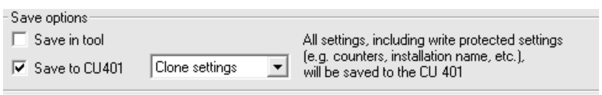

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Fig. 7 Save settings

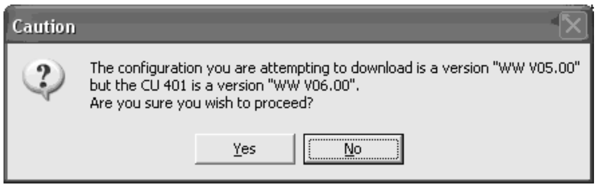

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Fig. 8 Caution message

7. OD 401 operator display

7.1 Buttons and indicator lights

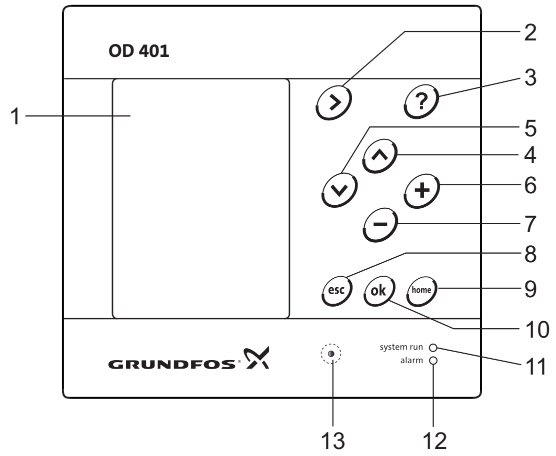


Fig. 9 OD 401 operator display

Pos.	Button/ indicator light	Description
1		LCD display
2		Changes to next column in menu structure
3		Changes to help text
4		Goes up in lists
5		Goes down in lists
6		Increases the value of a selected parameter
7		Reduces the value of a selected parameter
8		Goes one display back
9		Goes back to menu **Status**
10		Saves a value
11		Green indicator light (operation)
12		Red indicator light (alarm)
13		Changes the contrast of the display

Active buttons are illuminated from behind.

Arrow towards right (menu)

The "arrow towards right" button changes to the next column in the menu structure to the right. The last column changes to the first column to the left.

? (help)

When the help button is pressed, a help text for the actual display will appear. All elements of the display have a help text which can be called up with the help button.

The help text is closed by pressing the esc button.

Up and down

The up and down buttons are used to move up and down in lists.

If a text is marked and the up button is pressed, the text above will be marked instead. If the down button is pressed, the text below will be marked.

If the down button is pressed in the last line in the list, the first line in the list is activated.

If the up button is pressed in the first line in the list, the last line in the list is activated.

Plus and minus

The plus and minus buttons are used to increase and reduce values of a selected parameter.

The plus button increases the value and the minus button reduces the value.

Esc

The esc button is used to go one display back in the menu.

If a value has been changed and the esc button is pressed, the new value will not be activated.

Ok

The ok button is used as an enter button.

The ok button is also used to start the editing process for a value and to acknowledge the value.

Home

The home button is used to go back to the system overview.

Contrast

The contrast button is used to change the contrast in the display.

Press the plus or minus button to increase or reduce the contrast.

Indicator lights

The OD 401 incorporates two indicator lights.

The green indicator light is on when the power supply has been switched on.

The red indicator light is on if the system does not operate correctly due to a fault. The fault can be identified from the alarm list.

7.1.1 Display design

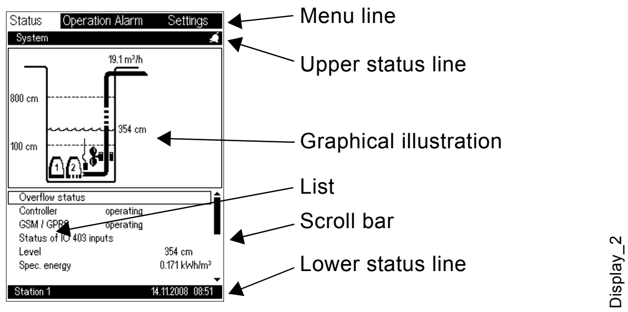


Fig. 10 Example of an application with two pumps and a mixer

Menu line

The display has four main menus:

Status:	Indication of system status
Operation:	Daily changes of operating parameters (access code option)
Alarm:	Alarm log for fault finding purposes
Settings:	Change of system configuration (access code option)

For use of the individual main menus, see section *8. Main menus*.

Upper status line

The upper status line shows

- the position in the menu structure (left side)
- status (actual operation, alarm) (right side).

Lower status line

The lower status line shows

- system name (left side)
- date and time (right side).

Graphical illustration

The graphical illustration may show the actual status, a historical indication or other graphics, depending on the position in the menu structure.

The illustration may show the entire system or parts of it as well as various settings.

When a graphical illustration is shown, a list will appear (see "List"). The list describes the elements of the illustration. It is possible to browse the illustration elements.

List

The list includes one or more lines with information grouped to the left and to the right.

The left side shows texts and the right side shows values.

Headlines and empty lines cannot be selected.

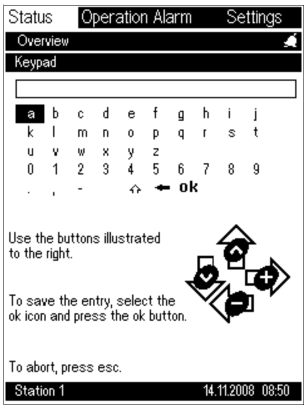
Scroll bar

If the list of illustration elements exceeds the display, the symbols "arrow up" and "arrow down" will appear in the scroll bar to the right. The up and down buttons can be used to move up and down in the list.

Note: Focus is marked with a frame.

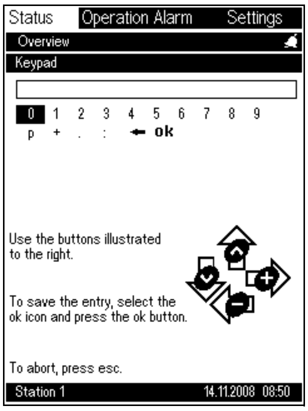
7.1.2 Keypad

The figures below show examples of keypads appearing when the user wants to make a change in the OD 401. Which of the three keypads will appear depends on whether text or digits are to be changed.



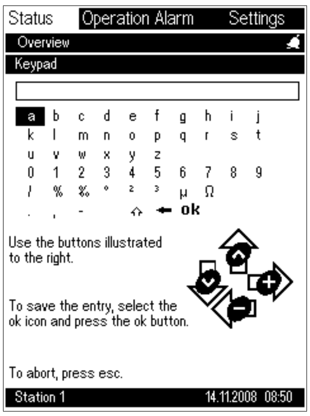
Display_40502

Fig. 11 Alphanumeric keypad



Display_40501

Fig. 12 Numeric keypad



Display_40504

Fig. 13 Alphanumeric keypad with units

7.2 Functions

7.2.1 Change of values

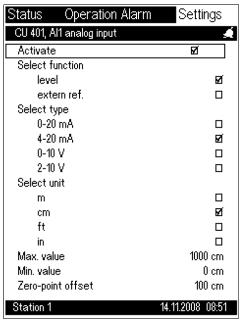


Fig. 14 Example of change of values

Values are changed as follows:

- Select the line to be changed.
- Press the ok button to activate the line (the focus frame starts flashing).
- Press the plus or minus button to change the value (the focus frame is still flashing).
- Abort the display by pressing the esc button.
  When the ok button is pressed, the value will be activated.
  When the esc button is pressed, the original value remains unchanged.

7.2.2 Lines with check mark

There are two kinds of check marks.

If the check mark is framed $\boxed{V}$, the value can be changed.

If the check mark is not framed V and is standing on a line, this is status information selected in another display. The value cannot be changed in this display.

Example:

In the example, there are for instance three subdisplays:

Level setting 1

Level setting 2 **V**

Level setting 3.

A check mark **V** (without frame) at the text "level setting 2" means that "level setting 2" was selected in a subdisplay.

If "level setting 1" is desired, mark the line and press the ok button.
"Level setting 1" can now be made in the next display.

Next time this display appears, it will look as follows:

Level setting 1 **V**

Level setting 2

Level setting 3.

7.3 Indications

7.3.1 Pop-up displays

There are four possible pop-up displays:

Help

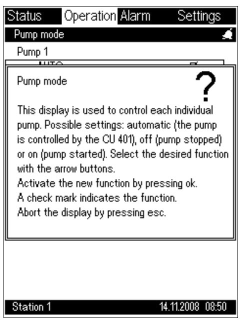


Fig. 15 Example of the help pop-up display

Press the help button to see explanation.

Access code

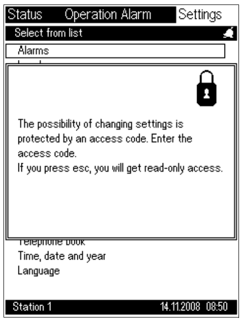


Fig. 16 Example of the access code pop-up display

The whole configuration or parts of it can be protected by means of an access code.

An access code is a combination of five button presses. The nine buttons on the operator display can be used, e.g. plus, minus, esc, minus, plus.

If an attempt is made to enter a value which is protected by an access code, the display in fig. 16 will appear. The display is not locked and it is possible to go on by pressing the esc button or by going to somewhere else in the menu.

Warning

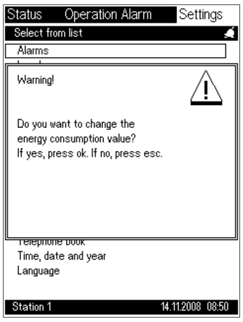


Fig. 17 Example of the warning pop-up display

The warning display appears if a value having consequence for the operation is being changed.

It is possible to regret the entry:

- Press the esc button to regret the entry.
- Press the ok button to activate the entered value.

Communication faults

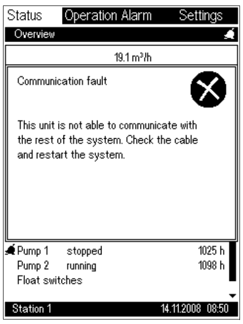


Fig. 18 Example of the communication faults pop-up display

This display appears if the communication between the OD 401 and the CU 401 has been interrupted.

Possible faults:

- The power supply has been switched off, but the OD 401 has its own power supply.
- Cable connection is defective.
- Module fault.

7.4 Communication with CU 401

The communication between the OD 401 and the CU 401 takes place via the COMLI (RS-485) protocol.

The OD 401 functions as "master" and the CU 401 as "slave".

The OD 401 must be connected to the CU 401 via a screened, twisted-pair standard cable (RS-485). The cable ends are fitted in terminals.

8. Main menus

When the hardware modules have been installed and all electrical connections made, the system must be configured. The configuration is made via the OD 401 operator display or by means of Grundfos PC Tool.

8.1 Use

8.1.1 Status

The status menu is used to obtain an overview of the system. Actual operating parameters can be shown as trend curves or figures. The status menu also displays actual alarms in the form of a small bell in the right side of the upper status line and an alarm line in the list. This provides a direct path to the alarm display.

See section *11. Status* for a detailed description.

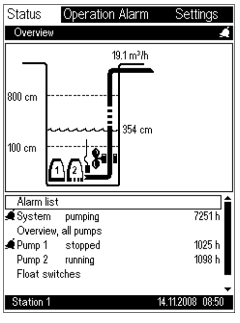


Fig. 19 Example of the status menu

8.1.2 Operation

The operation menu is intended for daily settings, such as start/stop and automatic/manual. Other changes are to be entered in the settings menu.

See section *12. Operation* for a detailed description.

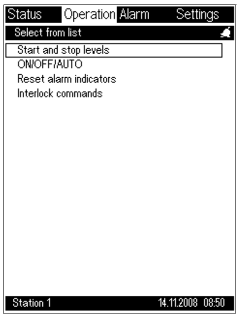


Fig. 20 Example of the operation menu

8.1.3 Alarm

The alarm menu is used as alarm log, storing a history log of up to 100 alarm activities. An alarm activity is stored when it occurs and when it disappears. Most alarms also have a SnapShot of the most important pump alarm parameters. See section *13.2 AlarmSnapShot*.

See section *13. Alarm* for a detailed description.

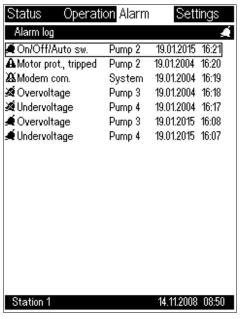


Fig. 21 Example of the alarm menu

8.1.4 Settings

The settings menu is used for the entry of reference parameters.

See section *14. Settings* for a detailed description.

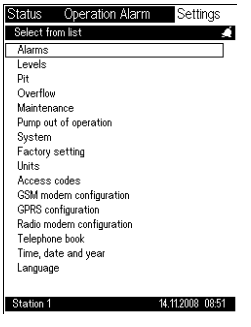


Fig. 22 Example of the settings menu

8.2 OD 401 menu overview

11. Status

- *11.2 Alarm list - status overview*
  - *11.2.1 Alarm list - resetting/approving system alarms and warnings*
  - *11.2.2 Alarm list - resetting/approving pump alarms and warnings*
- *11.3 Status - system*
  - *11.4 Status - overflow*
  - *11.5 Status - modules*
  - *11.7 GSM/GPRS modem*
  - *11.8 Status of IO 403 inputs*
- *11.1 Overview, all pumps*
- *11.6 Status - specific pump*
  - *11.11 Trend curves*
- *11.9 Status - position and location of float switches*
- *11.10 Mixer*

12. Operation

- *12.1.1 Operation - start and stop levels*
- *12.1.2 Direct operation of pumps - AUTO/ON/OFF mode*
- *12.1.3 Operation - resetting alarm indicators and alarm relays*
- *12.1.4 Interlock commands*

13. Alarm

- *13.1 Alarm log*
  - *13.2 AlarmSnapShot*

The menu overview continues on the next page

14. Settings

- 14.2 Alarms and warnings
  - 14.2.1 System alarms and warnings - enable/disable
  - 14.2.2 System alarms and warnings - limits
  - 14.2.3 System alarms and warnings - alarm handling
  - 14.2.4 Pump alarms and warnings - enable/disable
  - 14.2.5 Pump alarms and warnings - limits
  - 14.2.6 Pump alarms and warnings - alarm handling
- 14.3 Levels - float switches or level sensor
  - 14.3.1 Float switches
    - 14.3.2 One pump and two float switches
  - 14.3.8 Level sensor
    - 14.3.9 Variation of start level
- 14.3.10 Pit configuration and flow calculations
- 14.4 Overflow
  - 14.4.1 Overflow system
  - 14.4.10 Channel parameters
- 14.5 Maintenance
  - 14.5.1 Adjustment of counters
  - 14.5.2 Calibration
  - 14.5.3 Alarm log
  - 14.5.4 SMS counters
- 14.6 Pump out of operation
- 14.7 System
  - 14.7.2 I/O settings
    - 14.7.3 General settings
    - 14.7.4 CU 401, AI1 analog input
    - 14.7.5 CU 401, AI2 analog input
    - 14.7.6 CU 401, CNT counter input
    - 14.7.7 CU 401, DO7 relay output
    - 14.7.8 CU 401, DO8 relay output
    - 14.7.9 IO 403, I/O settings
    - 14.7.14 IO 401, AI analog inputs
    - 14.7.15 SM 111, analog input
  - 14.7.16 Pump - time settings
  - 14.7.17 Group configuration - alternation/advanced alternation
    - 14.7.18 Pump grouping
    - 14.7.19 Group properties
  - 14.7.20 Pump configuration (pump 1-6)
    - 14.7.21 VFD configuration
      - 14.7.22 PID user configuration
  - 14.7.25 Mixer configuration
  - 14.7.26 Userlog - selection of data to log
  - 14.7.27 Userlog intervals
  - 14.7.28 Installation name

The menu overview continues on the next page

9. Factory settings

9.1 System

Description	Factory setting	Own settings
IO 401 modules	One module installed (two pumps)	
IO 403, AI1 analog input	Not activated Name: IO 403, AI1 Name: IO 403, AI1 max. Name: IO 403, AI1 min. Signal type: 4-20 mA Max. value: 100 Min. value: 0 Unit: m	
IO 403, AI2 analog input	Not activated Name: IO 403, AI2 Name: IO 403, AI2 max. Name: IO 403, AI2 min. Signal type: 4-20 mA Max. value: 100 Min. value: 0 Unit: m	
IO 403, AI3 analog input	Not activated Name: IO 403, AI3 Name: IO 403, AI3 max. Name: IO 403, AI3 min. Signal type: 4-20 mA Max. value: 100 Min. value: 0 Unit: m	
IO 403, AI4 analog input	Not activated Name: IO 403, AI4 Name: IO 403, AI4 max. Name: IO 403, AI4 min. Signal type: 4-20 mA Max. value: 100 Min. value: 0 Unit: m	
IO 403, DI5 digital input	Name: IO 403, DI5 Name: IO 403, DI5 alarm Function: NO (normally open) Delay: 0 sec.	
IO 403, DI6 digital input	Name: IO 403, DI6 Name: IO 403, DI6 alarm Function: NO (normally open) Delay: 0 sec.	
IO 403, DI7 digital input	Name: IO 403, DI7 Name: IO 403, DI7 alarm Function: NO (normally open) Delay: 0 sec.	
IO 403, DI8 digital input	Name: IO 403, DI8 Name: IO 403, DI8 alarm Function: NO (normally open) Delay: 0 sec. Counter function selected Units per pulse selected Value: 10 Unit: m	
IO 403, DO1 relay output	Name: IO 403, DO1 Function: NO (normally open) Hold time: 0 sec. Controlled by DI5 not selected Controlled by SCADA selected	
IO 403, DO2 relay output	Name: IO 403, DO2 Function: NO (normally open) Hold time: 0 sec. Controlled by DI6 not selected Controlled by SCADA selected	
IO 403, DO3 relay output	Name: IO 403, DO3 Function: NO (normally open) Hold time: 0 sec. Controlled by DI7 not selected Controlled by SCADA selected	

IO 403, DO4 relay output	Name: IO 403, DO4 Function: NO (normally open) Hold time: 0 sec. Controlled by DI8 not selected Controlled by SCADA selected	
Number of pumps	2	
Analog level sensor connected to AI1 analog input	Level, 4-20 mA, 0-500 cm	
Analog sensor connected to AI2 analog input	4-20 mA	
Counter connected	None	
Level control	Level sensor	
Float switches	Not installed	
Frequency	50 Hz	
CU 401, DI1 digital input	NO (normally open)	
CU 401, DI2 digital input	NO (normally open)	
CU 401, DI3 digital input	NO (normally open)	
CU 401, DI4 digital input	NO (normally open)	
CU 401, DI5 digital input	NO (normally open)	
CU 401, DI6 digital input	NO (normally open)	
PU 102	Not installed	
Max. start-up delay	10 seconds	
After-run time	0 seconds	
Anti-seizing, start interval	24 hours	
Anti-seizing, runtime	2 seconds	
Daily emptying time	Not active	
Min. start/start delay	2 seconds	
Min. stop/stop delay	2 seconds	
Start/stop, stop/start delay	2 seconds	
Foam drain, start interval	Not active	
Foam drain, stop delay	10 seconds	
High-water after-run time	30 seconds	
Group 1 to 2 by level	400 cm	
Group 2 to 1, max. runtime	180 seconds	
Pump 1	Group 1	
Pump 2	Group 1	
Pump 3	Group 1	
Pump 4	Group 1	
Pump 5	Group 1	
Pump 6	Group 1	
Group 1	Alternation; max. number of started pumps: 6	
Group 2	Alternation; max. number of started pumps: 6	
Group 3	Alternation; max. number of started pumps: 6	
Cut-out other groups	Yes	
CU 401, DO7 relay output	All alarms; manual resetting: Yes	
CU 401, DO8 relay output	High level; manual resetting: Yes	
IO 401, AI1 analog input	Not active; Signal type: 4-20 mA; Max. value: 20; Min. value: 0; Unit: %	
SM 111, analog output	Not active; Max. value: 255; Min. value: 0; Unit: mm	

Overflow	Overflow system: 1; Fault switch: Not installed; Channel form: Rectangular; Diameter/width (b): 50 cm; Wall angle (α): 90; Manning's coefficient: Code 0.001; Channel slope (S): 100 %	

9.2 Mixer

Description	Factory setting	Own settings
Mixer	Not installed; Start level 1 <-> Start level, mixer: 5 cm; Stop level, mixer: 100 cm; Mixer ratio 1: 10; Mix while pumping: Not active; Contactor feedback: Not active; Max. starts/hour: 30; Max. runtime: 30 minutes	

9.3 Pump

Description	Factory setting	Own settings
Pump 1	Installed; out of operation; without contactor feedback; MP 204: Not installed; IO 111: Not installed	
Pump 2	Installed; out of operation; without contactor feedback; MP 204: Not installed; IO 111: Not installed	
Pump 3	Installed; out of operation; without contactor feedback; MP 204: Not installed; IO 111: Not installed	
Pump 4	Installed; out of operation; without contactor feedback; MP 204: Not installed; IO 111: Not installed	
Pump 5	Installed; out of operation; without contactor feedback; MP 204: Not installed; IO 111: Not installed	
Pump 6	Installed; out of operation; without contactor feedback; MP 204: Not installed; IO 111: Not installed	

9.4 Frequency converter

Description	Factory setting	Own settings					
		1	2	3	4	5	6
VFD type	CUE						
Min. frequency	30 Hz						
Max. frequency	50 Hz						
Economy frequency	30 Hz						
Economy level, setpoint	1.75 m						
Economy level, max.	2.00 m						
Operating mode	Linear control						
2 operating mode	Max. speed						
Activate reversing at start	Active						
Reversing time	10 sec.						
Reversing interval	240 min.						
Flush at start	Active						
Flush time at start	40 sec.						
Flush at intervals	Active						
Flush time at intervals	20 sec.						
Flush interval	120 min.						
Activate flush before stop	Not active						
Flush time before stop	20 sec.						
Gain	2.00						
Integral time	1.00						
Differential time	0.00						
Pv filter time	1.00						
Der filter time	2.00						
Offset	60.00						
Dead zone	0.00						
Ff gain	0.00						
Direct	Active						
Controller type	PID						
Out inc. limit	1.00						
Out dec. limit	1.00						
Max. relay amplitude	10.00						
External ref.	Not active						
Ff ref.	Not active						
Secure stop	Stop in case of dry running						
Use special setpoint	Not active						
Special setpoint	50 %						

9.5 Grundfos PC Tool

Description	Factory setting	Own settings
IO 403 Digital inputs	DI5 – Name – Selection – Delay DI6 – Name – Selection – Delay DI7 – Name – Selection – Delay DI8 – Name – Selection – Delay Counter function – Counter function – Counting unit – Counter scaling – Log interval – Preset counter	
IO 403 Analog inputs	AI1 – Name – Measuring unit – max. range – min. range – input signal type – Log interval AI2 – Name – Measuring unit – max. range – min. range – input signal type – Log interval AI3 – Name – Measuring unit – max. range – min. range – input signal type – Log interval AI4 – Name – Measuring unit – max. range – min. range – input signal type – Log interval	

IO 403 Relay outputs	DO1 – Name – Selection – Hold time – Controlled by DO2 – Name – Selection – Hold time – Controlled by DO3 – Name – Selection – Hold time – Controlled by DO4 – Name – Selection – Hold time – Controlled by	
User IO 1	Input 1 – type – IO source – Input – Trigger level – Invert – Trigger delay Input 2 – type – IO source – Input – Trigger level – Invert – Trigger delay Output – Logic – Invert – Hold delay – Output	
User IO 2	Input 1 – type – IO source – Input – Trigger level – Invert – Trigger delay Input 2 – type – IO source – Input – Trigger level – Invert – Trigger delay Output – Logic – Invert – Hold delay – Output	

User IO 3	Input 1 – type – IO source – Input – Trigger level – Invert – Trigger delay Input 2 – type – IO source – Input – Trigger level – Invert – Trigger delay Output – Logic – Invert – Hold delay Output	
User IO 5	Input 1 – type – IO source – Input – Trigger level – Invert – Trigger delay Input 2 – type – IO source – Input – Trigger level – Invert – Trigger delay Output – Logic – Invert – Hold delay – Output	
User IO 6	Input 1 – type – IO source – Input – Trigger level – Invert – Trigger delay Input 2 – type – IO source – Input – Trigger level – Invert – Trigger delay Output – Logic – Invert – Hold delay – Output	

9.6 Userlog

Description	Factory setting	Own settings
System, level	Active	
System, average flow	Active	
System, power	Active	
System, specific energy	Active	
Pump, temperature, Pt 1	Active	
Pump, temperature, Pt 2	Active	
Pump, water in oil	Active	
Pump, average flow	Active	
Pump, average mains voltage	Active	
Pump, average current	Active	
Pump, power	Active	
Pump, cos φ	Active	
Level	30 seconds	
Average flow	3600 seconds (cannot be changed)	
Temperature	120 seconds	
Current	30 seconds	
Voltage	30 seconds	
Power	30 seconds	
cos φ	30 seconds	
Specific energy	3600 seconds (cannot be changed)	
Water in oil	3600 seconds (cannot be changed)	

9.7 Pit

Description	Factory setting	Own settings
Pit depth	500 cm	
Upper measurement level	160 cm	
Lower measurement level	55 cm	
Flow min. multiply	5	
Flow max. multiply	7	
Volume	--- l	
Max. measuring time	3600 seconds	

9.8 Level

Description	Factory setting	Own settings
Overflow level	500 cm	
High level	475 cm	
Max. height above start level 1	10 cm	
Alarm level	350 cm	
Start level 6	--- cm	
Start level 5	--- cm	
Start level 4	--- cm	
Start level 3	--- cm	
Start level 2	200 cm	
Start level 1	175 cm	
Stop level 6	--- cm	
Stop level 5	--- cm	
Stop level 4	--- cm	
Stop level 3	--- cm	
Stop level 2	50 cm	
Stop level 1	50 cm	
Dry-running level	25 cm	
Foam drain level	10 cm	
Variation of start level	Active	

9.9 OD 401 configuration

Description	Factory setting	Own settings
Units	SI	
Access code, operation	Not active	
Access code, settings	Not active	
Language	Local language	

9.10 SCADA

Description	Factory setting	Own settings
Dialup to SCADA	Not active	
Redial delay	180 seconds	
Priority 1 SCADA	SCADA ID: Station name Telephone number: 1234 Max. number of dials: 3 PIN code: 0	
Priority 2 SCADA	SCADA ID: Station name Telephone number: 1234 Max. number of dials: 0 PIN code: 0	
Priority 3 SCADA	SCADA ID: Station name Telephone number: 1234 Max. number of dials: 0 PIN code: 0	
Priority 4 SCADA	SCADA ID: Station name Telephone number: 1234 Max. number of dials: 0 PIN code: 0	
Priority 5 SCADA	SCADA ID: Station name Telephone number: 1234 Max. number of dials: 0 PIN code: 0	
No. 1 installation	Installation name: Station name Telephone number: 1234 PIN code: 0 Interlock mode: Transmit Interlock message: Stop	
No. 2 installation	Installation name: Station name Telephone number: 1234 PIN code: 0 Interlock mode: Not active Interlock message: Stop	
No. 3 installation	Installation name: Station name Telephone number: 1234 PIN code: 0 Interlock mode: Not active Interlock message: Stop	
No. 4 installation	Installation name: Station name Telephone number: 1234 PIN code: 0 Interlock mode: Not active Interlock message: Stop	
No. 5 installation	Installation name: Station name Telephone number: 1234 PIN code: 0 Interlock mode: Not active Interlock message: Stop	

9.11 Interlock

Description	Factory setting	Own settings
Interlock	Not active	
Variation of start level	Not active	

9.12 Advanced SMS

Description	Factory setting	Own settings
SMS	Not active	
SMS No 1	-	
SMS No 2	-	
SMS No 3	-	
Send SMS to	Primary number	
Acknowledge time	30 minutes	
Period, work (Monday-Sunday)	08:00	
Period, off (Monday-Sunday)	16:00	
Period, sleep (Monday-Sunday)	22:00	
Change 1	08:00	
Primary number	-	
Secondary number	-	
Change 2	16:00	
Primary number	-	
Secondary number	-	
Change 1	22:00	
Primary number	-	
Secondary number	-	
Heartbeat (Monday-Sunday)	Active	
Heartbeat time	00:00	
Heartbeat text	I am alive	
Protection	Protection, both	
Password	-	

9.13 Simple SMS

Description	Factory setting	Own settings
SMS	Use simple SMS: Deactivated Delay: 30 seconds	
Telephone number	0000	
Alarm/warning message	Not active	
Message	Alarm message	
Heartbeat (Monday-Sunday)	Not active	
Heartbeat time	12:30	
Heartbeat text	I am alive	

9.14 System alarms

Description	Factory setting	Own settings
Program card	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
UPS	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Module	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	

Description	Factory setting	Own settings
Mains voltage	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Modem	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Modem communication	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No.	
Float switch	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Flowmeter	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No	
Power meter	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Level sensor	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
High level	Not active; limit: 475 cm; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Level	Not active; limit: 300 cm; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Dry-running level	Not active; limit: 25 cm; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Conflicting levels	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Forced relay output	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Overflow	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Level sensor, overflow	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	

Description	Factory setting	Own settings
Flowmeter, overflow	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Fault switch, overflow	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	

9.15 Mixer alarms

Description	Factory setting	Own settings
Mixer contactor	Active; autom. resetting time: 1 second; manual resetting: Yes; SCADA message: No; SMS message: No	
Max. starts/hour, mixer	Active; warning limit: 15; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Time for service, mixer	Active; warning limit: 1000 hours; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	

9.16 Pump alarms

Description	Factory setting	Own settings
On/Off/Auto switch	Active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Motor protection, tripped	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Mains supply, off	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No	
Missing phase	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Wrong phase sequence	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Overtemperature, sensor 1	Not active; autom. resetting time: 15 seconds; manual resetting: No; SCADA message: No; SMS message: No	
Overtemperature, sensor 2	Not active; autom. resetting time: --- seconds; manual resetting: Yes; SCADA message: No; SMS message: No	

Description	Factory setting	Own settings
Overtemperature, Pt sensor	Not active; alarm limit: 145 degrees; warning limit: 130 degrees; autom. resetting time: 120 seconds; manual resetting: No; SCADA message: No; SMS message: No	
Water in oil	Not active; warning limit: 15 %; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Max. starts/hour	Not active; warning limit: 40; autom. resetting time: 1 second; manual resetting: Yes; SCADA message: No; SMS message: No	
Moisture in motor	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Low flow	Not active; warning limit: --- m^3/h; autom. resetting time: 1 second; manual resetting: Yes; SCADA message: No; SMS message: No	
Time for service	Not active; warning limit: 1000 hours; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Latest runtime	Not active; alarm limit: --- seconds; warning limit: --- seconds; autom. resetting time: 300 seconds; manual resetting: No; SCADA message: No; SMS message: No	
Overvoltage	Not active; alarm limit: --- V; warning limit: --- V; autom. resetting time: 10 seconds; manual resetting: No; SCADA message: No; SMS message: No	
Undervoltage	Not active; alarm limit: --- V; warning limit: --- V; autom. resetting time: 10 seconds; manual resetting: No; SCADA message: No; SMS message: No	
Overload	Not active; alarm limit: --- A; warning limit: --- A; autom. resetting time: 10 seconds; manual resetting: No; SCADA message: No; SMS message: No	
Underload	Not active; alarm limit: --- A; warning limit: --- A; autom. resetting time: 10 seconds; manual resetting: No; SCADA message: No; SMS message: No	

Description	Factory setting	Own settings
Insulation resistance, low	Not active; alarm limit: 20 kΩ; warning limit: 50 kΩ; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Current unbalance	Not active; alarm limit: 12 %; warning limit: 10 %; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
cos φ min.	Not active; alarm limit: 70; warning limit: 75; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
cos φ max.	Not active; alarm limit: 99; warning limit: 95; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Overtemperature, Pt 2	Not active; alarm limit: 145 degrees (applies to IO 111 only); warning limit: 130 degrees (applies to IO 111 only); autom. resetting time: 10 seconds; manual resetting: No; SCADA message: No; SMS message: No	
Overtemperature, MP 204 PTC	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Start capacitor, low	Not active; alarm limit: ---; warning limit: ---; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Run capacitor, low	Not active; alarm limit: ---; warning limit: ---; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Contactor	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
GENIbus communication fault	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Auxiliary winding fault	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	

Description	Factory setting	Own settings
Temperature sensor, Pt 3 (main bearing)	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No; warning limit: 135 °C; alarm limit: 145 °C	
Temperature sensor, Pt 4 (stator winding)	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No; warning limit: 135 °C; alarm limit: 145 °C	
Power line communication	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Vibration sensor	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No; warning limit: --- mm/s; alarm limit: --- mm/s	
Temperature sensor fault, Pt 3	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Config. conflict, fault	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Temperature sensor fault, Pt 4	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
Vibration sensor fault	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
SM 111, analog sensor, fault	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	
VFD alarm	Not active; autom. resetting time: 1 second; manual resetting: No; SCADA message: No; SMS message: No	

See also section *14.8 Factory settings*.

10. Quick start

When the system is started up for the first time, the pumps are set to be out of operation. This prevents the pumps from starting until all settings have been made.

Proceed as follows:

1. Insert the button cell battery into the CU 401 and make sure that the PU 102 is disconnected.
2. Switch on the CU 401.
3. Set the CU 401 clock. See section *14.11 Time and date settings*.
4. Change the userlog intervals, if required. See section *14.7.27 Userlog intervals*.
5. Switch off the CU 401 and connect the PU 102.
6. Switch on the CU 401.
7. Check that the clock is correct after start-up.
8. Set the system frequency (if other than 50 Hz). See section *14.7.3 General settings*.
9. Select the number of pumps in the system. See section *14.7.1 System configuration*.
10. Set the sensors. See section *14.3 Levels - float switches or level sensor*.
11. Set pit parameters. See section *14.3.10 Pit configuration and flow calculations*.
12. Select and set the desired alarms. See section *14.2.1 System alarms and warnings - enable/disable*.
13. Select and set the start and stop levels, etc. See section *12.1.1 Operation - start and stop levels*.
14. Start the pumps in accordance with section *14.6 Pump out of operation*.

When all settings have been made, the pumps are started.

The program card incorporates a number of factory settings for quick configuration of a two-pump system. An overview of all factory settings is given in section *9. Factory settings*.

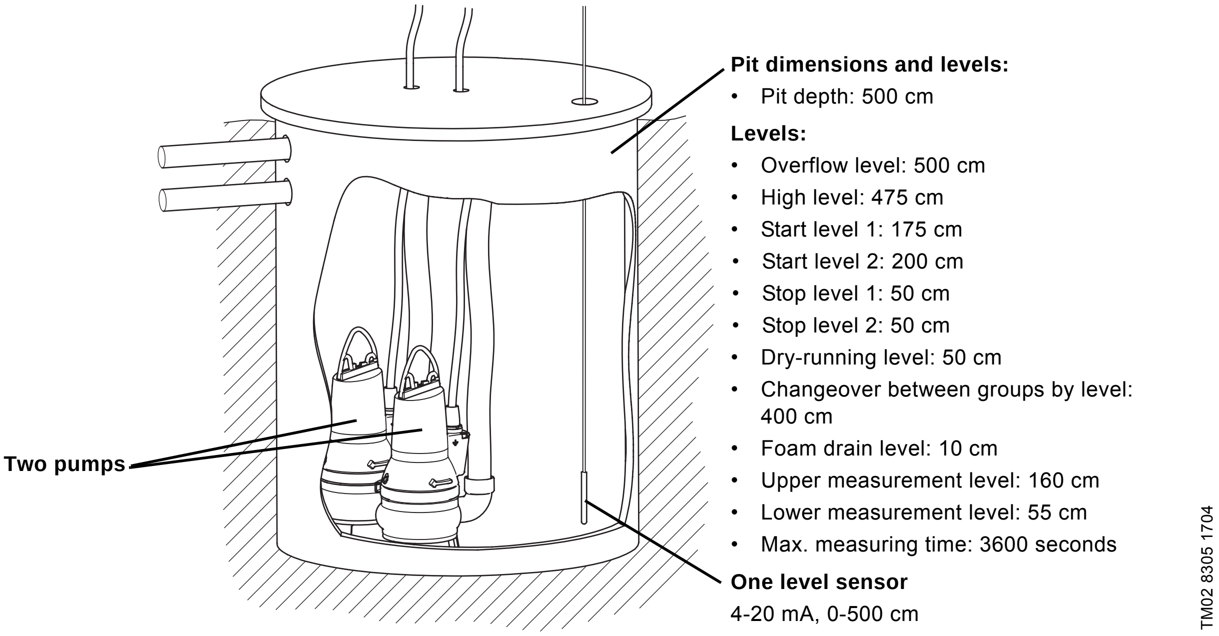


Fig. 23 Example of two-pump system (factory settings)

11. Status

The status display is the main opening display of the Grundfos Modular Controls system.

Note: If the buttons have not been activated for a couple of minutes, the operator display automatically reverts to the status display.

The condition of pumps and components can be monitored from the status display.

The alarm condition is shown by a bell against the units causing the alarm.

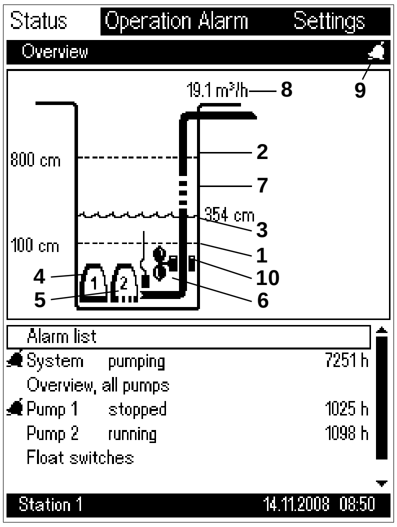
When the system registers an alarm, the following will take place:

- A bell is shown in the right side of the upper status line.
- The red indicator light on the operator display is on.
- Alarm list" is displayed as the first line below the pit graphics.
- Alarm relay operates.
- High-level alarm relay operates (only in the event of high-level alarm and if high level is enabled).
- The indication is maintained as long as the system has an active alarm. An alarm is active until it is approved, either automatically after a preset time (see section *14.2 Alarms and warnings*) or manually via the status display (see section *11.2.1 Alarm list - resetting/approving system alarms and warnings*).
- In order to reset an alarm, the situation causing the alarm must be corrected. For example, in case of alarm due to overtemperature the pump must be cool before the alarm can be approved/reset.

See section *11.11 Trend curves*.

Note: The display shown below should be considered as an example.

Path: Status**>**



Display_1

Description

Press **ok** to move to the "Alarm status" display of the individual alarm groups. In case of active alarms, focus will be on the "Alarm list" line.

Possible alarm groups:

- **Alarm list**
  - See actual alarms for system or pumps.
- **System**
  - See operating parameters for the system and move to trend curves.
- **Overview, all pumps**
- **Pump 1**
  - See operating parameters for pump 1.
- **Pump 2**
- **Pump 3**
- **Pump 4**
- **Pump 5**
- **Pump 6**
- **Float switches**
  - See actual location and status of float switches.
- **Mixer**
  - See the actual status of the mixer.

Note: Float switches and mixer are displayed only if incorporated in the system.

Key to display

Pos.	Description
1	Lowest stop level: If the water level falls below this level, the pumps will stop.
2	Highest start level: If the water level rises above this level, the pumps will start.
3	This value and the wave line indicate the actual water level in the pit.
4	Displays pump 1 in stopped state. The bottom is a thick, unbroken line.
5	Displays pump 2 running. The broken bottom line is moving rightward.
6	Pressure sensor symbol. Symbolising an ordinary pressure sensor, the sensor displayed is located at the bottom of the pit.
7	The broken line is moving upwards, symbolising a flow. The line is displayed when one or more pumps are running.
8	The actual flow is measured with a flowmeter or by means of level measurement and pit data. See section *14.3.10 Pit configuration and flow calculations*.
9	Alarm bell: Displayed as long as there are active alarms. The red indicator light on the operator display indicates the same function.
10	Mixer: The mixer propeller will rotate when the mixer is running.

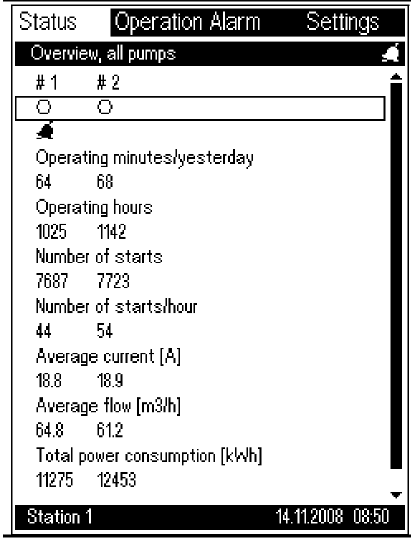
11.1 Overview, all pumps

This display shows an overview of all pumps in the pit.

The display appears automatically when the display has been inactive for more than eight minutes. The OD 401 then alternately shows the main display for 20 seconds and the overview display for 10 seconds until a button on the OD 401 is touched.

Note: The display shown below should be considered as an example.

Path: Status>Overview, all pumps>



Display_20

Description

- #: Pump number.
- A mark in the circle shows that the pump is running.
- Operating minutes/yesterday (number of pump operating hours). The number of operating minutes is set zero at 00:00 (factory setting). The time can be changed via Grundfos PC Tool.
- Operating hours (accumulated number of pump operating hours).
- Number of starts (number of pump starts).
- Number of starts/hour (average number of starts per hour).
- Average current (actual current consumption of three phases).
- Average flow (flow delivered by the pumps when running).
- Total power consumption (pump consumption in kWh since installation).

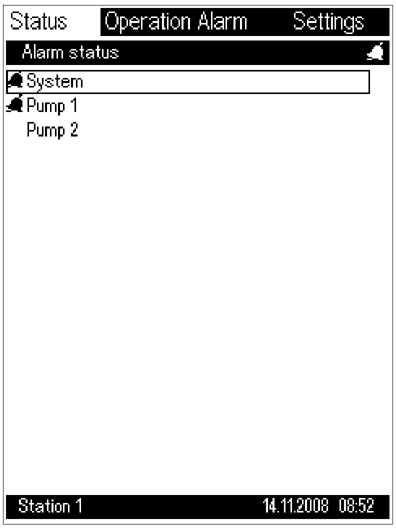
11.2 Alarm list - status overview

This display can be used to get an overview of the actual alarms.
See section *13.1 Alarm log*.

Symbol	Description
	Alarm
	Warning

Note: The display shown below should be considered as an example.

Path: Status>Alarm status>



Display_600

Description

Press **ok** to move to "System" or "Pump" to reset or approve the alarms as required.

- **System**
  - Select the desired alarm list line.
- **Pump 1**
  - Select the desired alarm list line.
- **Pump 2**
- **Pump 3**
- **Pump 4**
- **Pump 5**
- **Pump 6.**

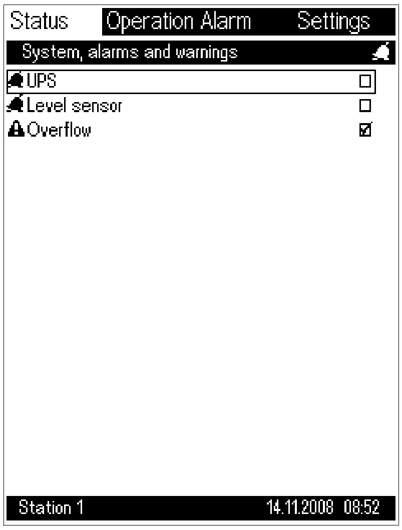
11.2.1 Alarm list - resetting/approving system alarms and warnings

This display shows all active system alarms. The alarms can be inspected and reset as required. See section *13.1 Alarm log*.

Symbol	Description
	Alarm
	Warning

Note: The display shown below should be considered as an example.

Path: Status>Alarm status>System>



Display_601

Description

- A check mark **V** can be put in the frame to the right on the line. Press **ok**, then **plus** and finally **ok** again to reset the alarm in question.
- This procedure is repeated for all active alarms to be reset manually. Press **esc** to return to the "Alarm status" display, or **home** if there are no more alarms to reset.
- Once all alarms have been approved, the "Alarm list" line and bell in the upper status line disappear, and the red indicator light on the operator display is off.

Note: The alarm relay cannot be reset until the reset button is pressed (if installed), or the alarm relay is reset on the OD 401 via the "Reset alarm indicators" menu. See section *12.1.3 Operation - resetting alarm indicators and alarm relays*.

The alarm log shows the alarms of the system since the last resetting. Thus it is possible to see whether the system has registered an alarm since the latest reset, despite automatic resetting. A maximum of 100 alarm activities are stored in the log. See also section *13.1 Alarm log*.

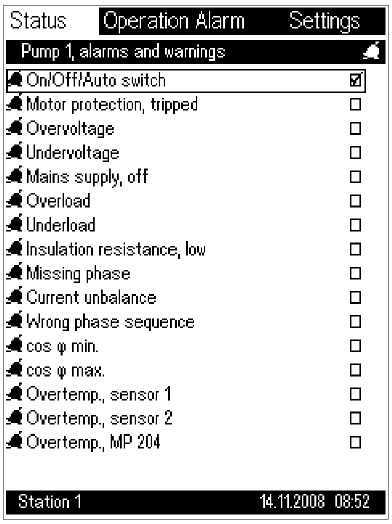
11.2.2 Alarm list - resetting/approving pump alarms and warnings

This display shows all active pump alarms. The alarms can be inspected and reset as required. See section *13.1 Alarm log*.

Symbol	Description
	Alarm
	Warning

Note: The display shown below should be considered as an example.

Path: Status>Alarm status>Pump 1>



Display_602_1

Description

- A check mark **V** can be put in the frame to the right on the line. Press **ok**, then **plus** and finally **ok** again to reset the alarm in question.
- This procedure is repeated for all active alarms to be reset manually. Press **esc** to return to the "Alarm status" display, or **home** if there are no more alarms to reset.
- Once all alarms have been approved, the "Alarm list" line and bell in the upper status line disappear, and the red indicator light on the operator display is off.

Note: The alarm relay cannot be reset until the reset button is pressed (if installed), or if the alarm relay is reset on the OD 401 via the "Reset alarm indicators" menu. See section *12.1.3 Operation - resetting alarm indicators and alarm relays*.

The alarm log shows the alarms of the pump since the last resetting. Thus it is possible to see whether the system has registered an alarm since the latest reset, despite automatic resetting. A maximum of 100 alarm activities are stored in the log. See also section *13.1 Alarm log*.

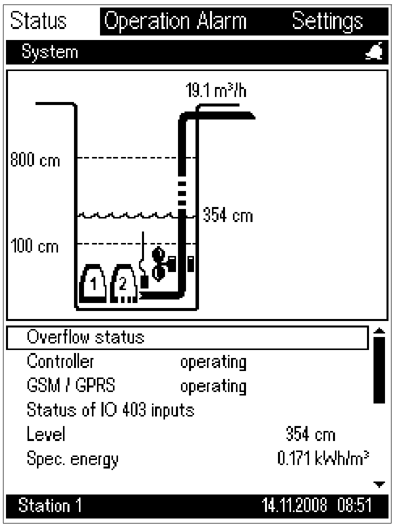
11.3 Status - system

This display shows actual system operating parameters.

Symbol	Description
	Alarm
	Warning

Note: The display shown below should be considered as an example.

Path: Status>System>



Display_2

Description

The display shows operating overview and power consumption.

Possible operating parameters:

- Overflow status
- Controller
- GSM / GPRS
- Level
- Specific energy
- Power
- Energy consumption
- Average flow
- Volume
- Overflow volume
- Overflow time
- Parallel-operation time
- Status of IO 403 inputs.

Note: Selecting "Controller" displays the modules installed with indication as to whether they function correctly.

Note: If *Variation of start level* is active, start level 1 (here 800 cm) will change after each pump start.

Function of operating parameters

Operating parameter	Function
Overflow status	Entry to overflow display.
Controller	Displays the operational state of the system: - standby - start delay - pumping - pumping max. - off - foam drain - daily emptying - mains fault - manual - alarm on all pumps - anti-seizing - level sensor fault - interlocked. **Note:** Some of the above operational states only occur if the system incorporates extra equipment, such as FB 101, MP 204, etc.
GSM / GPRS	Displays the status of the GSM/GPRS modem: - operating - not connected - SIM card failure - no GSM signal - battery failure.
Level	Displays the actual level measured by the analog level sensor. **Note:** "Level" is only displayed if an analog level sensor is connected.
Specific energy	Displays the specific energy, indicating the efficiency of the pump to convert the electrical energy (measured in kWh) to pumped volumes (measured in m^3). The specific energy is indicated in kWh/m^3. To be able to make a satisfactory average measurement, the measuring interval is 24 hours. Thus 24 hours will pass before a value is displayed. **Note:** To utilise this option, an energy meter, e.g. an MP 204, must be fitted in the supply cable to the pump.

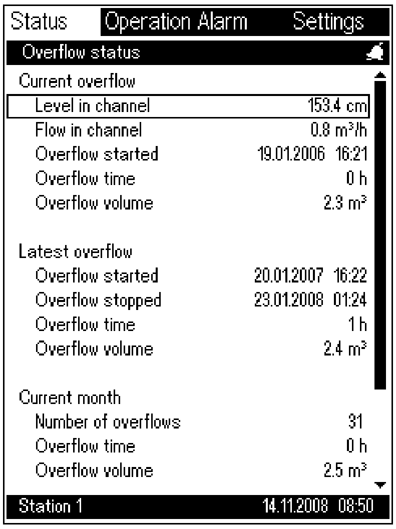
Operating parameter	Function
Power	Displays the actual power, measured in kW, supplied to the pumping station. The figure is updated once a second, if the measurement is made on the analog input. If the measurement is made on the counter input, the figure is updated every 15 seconds. **Note:** To utilise this option, an energy meter, e.g. an MP 204, must be fitted in the supply cable to the pump.
Energy	Displays the accumulated energy measured in kWh. The figure is updated once a second. **Note:** To utilise this option, an energy meter, e.g. an MP 204, must be fitted in the supply cable to the pump.
Average flow	Displays the average flow for the pumping station. The unit can be selected. The figure is updated once a second, if the measurement is made on the analog input. If the measurement is made on the counter input, the figure is updated every 15 seconds. **Note:** For this value to be displayed, a flowmeter must be fitted. Alternatively, the flow must be calculated by means of the analog sensor, on the basis of the knowledge about the pit geometry. See the pit configuration, section *14.3.10 Pit configuration and flow calculations*.
Volume	Displays the accumulated value for the volume of liquid removed. The unit can be displayed in litre/gallons or m^3. **Note:** For this value to be displayed, a flowmeter must be fitted. Alternatively, the flow must be calculated by means of the analog sensor, on the basis of the knowledge about the pit geometry. See the pit configuration, section *14.3.10 Pit configuration and flow calculations*.
Overflow volume	Displays the estimated overflow volume, based on the last flow calculation.
Overflow time	Displays the period during which there has been an overflow. See also section *14.4 Overflow*.
Parallel-operation time	Displays the accumulated period during which more than one pump has been running.
Status of IO 403 inputs	Displays the actual input values.

11.4 Status - overflow

This display shows when there was an overflow, the overflow volume and the number of overflows.

Note: The display shown below should be considered as an example.

Path: Status**>**System**>**Overflow status**>**



Display_40

Description

This display gives an overview of previous and current overflows, if any.

Current overflow:

- Level in channel
- Flow in channel
- Overflow started (date and time)
- Overflow time (period during which there has been an overflow)
- Overflow volume (number of m^3 that ran out).

The current overflow is added to the values for the current month when the overflow has stopped.

Latest overflow:

- Overflow started (date and time)
- Overflow stopped (date and time)
- Overflow time (period during which there was an overflow)
- Overflow volume (number of m^3 that ran out).

Current month (status for current month, i.e. from the 1st until today)

- Number of overflows (total number of overflows)
- Overflow time (accumulated overflow time)
- Overflow volume (accumulated overflow volume).

Previous month (status for previous month, i.e. from the 1st until the 31st)

- Number of overflows (total number of overflows)
- Overflow time (accumulated overflow time)
- Overflow volume (accumulated overflow volume).

11.5 Status - modules

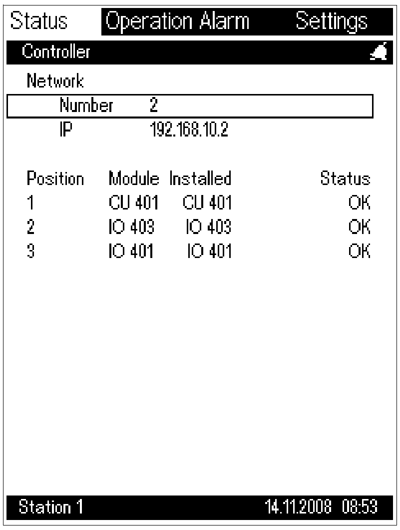
This display shows the CU 401 network number and the modules connected in the system.

If the number of modules shown is lower than the actual number of modules, one or more modules are defective or not connected correctly. In that case, check the module bus plug.

Note: The network number only appears if a G 40X module is connected.

Note: The display shown below should be considered as an example.

Path: Status>System>Controller>



Display_90_IO403

Description

Network:

- Network number (CU 401)
- IP number (CU 401).

Note: The network and IP numbers are automatically allocated to the CU 401.

Modules installed in the system and status of each module:

- CU 401
- IO 403
- IO 401.

Status:

- OK
- Fault.

Note: If a WW1 program card is inserted by accident, the module positions in this display will be wrong.

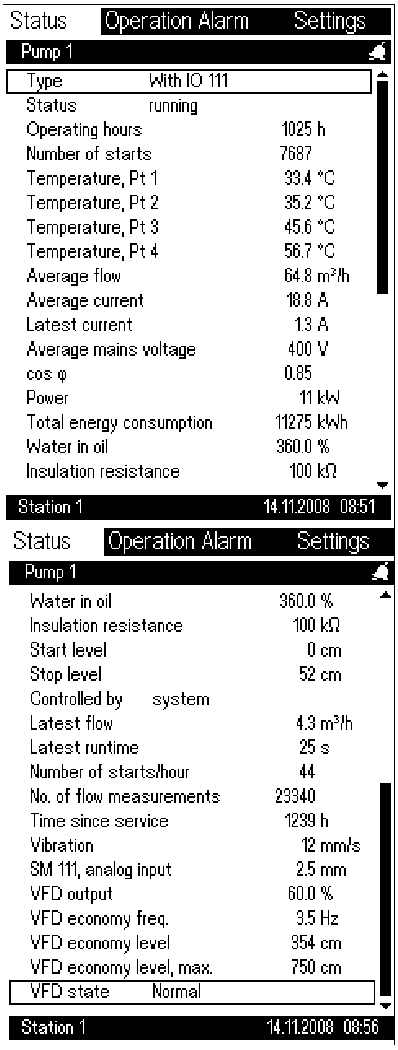
11.6 Status - specific pump

This display shows the actual operating parameters of pump 1.

Many parameters only show a value when the sensor or module (e.g. MP 204) is connected.

Note: The display shown below should be considered as an example.

Path: Status>Pump 1>



Display_3

Description

The display shows these operating parameters:

- Operating hours (number of pump operating hours. Can be set to zero, if another pump is installed).
- Number starts (number of pump starts since installation/connection. Can be changed, if another pump is installed).
- Temperature, Pt 1 (measured by means of an IO 401. Requires a Pt sensor in the stator housing).
- Temperature, Pt 2 (requires an MP 204 or IO 111 and a Pt sensor for measurement).
- Temperature, Pt 3 (requires an IO 111 or an SM 111 with IO 111 communication module and a Pt sensor for measurement).
- Temperature, Pt 4 (requires an IO 111 or an SM 111 with IO 111 communication module and a Pt sensor for measurement).
- Average flow (requires an analog level sensor or flow sensor). See section *14.3.10 Pit configuration and flow calculations*).
- Actual current (actual average current consumption of three phases. Requires an MP 204 or IO 401 with current sensor).
- Latest current (the current value at the time when the pump stopped last time. Requires an MP 204 or IO 401 with current sensor).
- Actual mains voltage (average mains voltage of three phases).
- cos φ (requires an MP 204).
- Power (requires an MP 204).
- Energy consumption (requires an MP 204).
- Water in oil (water in oil can be detected in three different ways):
  – IO 401 and an analog water-in-oil sensor.
  – IO 111 and an analog water-in-oil sensor.
  – SM 111 with IO 111 communication module and an analog water-in-oil sensor.
- Insulation resistance (the insulation resistance between the stator coils, cable and earth is measured. Requires an MP 204 or IO 111).
- Start level (requires an analog level sensor).
- Stop level (requires an analog level sensor).
- Controlled by (CU 401 (system), manually via On/Off/Auto switch or via SCADA system).
- Latest flow (requires an analog level sensor or flow sensor).
- Latest runtime (the latest operating period of the pump).
- Number of starts/hour (number of pump starts per hour). The value is a calculated average value which is stable after approximately 10 pump starts).
- Number of flow measurements (requires an analog level sensor). See section *14.3.10 Pit configuration and flow calculations*).
- Time since service (time since last service on the pump. Can be reset by Grundfos Service).
- SM 111, analog sensor (requires an SM 111 and an IO 111).
- Vibration sensor (requires an SM 111, an IO 111 and a vibration sensor for measurement).
- VFD output (setpoint of the frequency converter. Requires a CUE/frequency converter).
- VFD economy freq. (requires a CUE/frequency converter).
- VFD economy level (requires a CUE/frequency converter).
- VFD economy level, max. (requires a CUE/frequency converter).
- VFD state (status of a CUE/frequency converter).

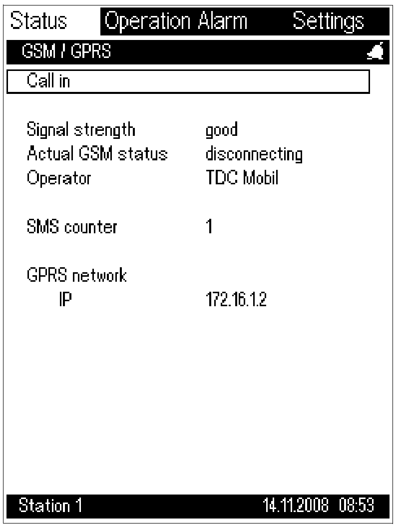
11.7 GSM/GPRS modem

This display shows the status of the GSM modem.

The display can be used in connection with fault finding and checking of antenna conditions.

Note: The display shown below should be considered as an example.

Path: Status>GSM / GPRS>



Display_9

Description

Display example:

GSM status:

- Not active
- Call out
- Call in.

- Signal strength:
  - weak
  - good
  - very good.

Note: The G 401 modem will always use the strongest GSM signal/antenna.

- Actual GSM status:
  - connecting
  - connected
  - disconnecting
  - not active.

If GPRS is used, the GSM status will always be "not active".

- Operator: GSM operator name (comes up automatically).
- SMS counter: Number of sent SMS messages.
- IP number of GPRS network: IP address of the G 401 on the GPRS network. The IP number appears when the connection to the GPRS network has been established. The number is also used for the setting of the interlock function.

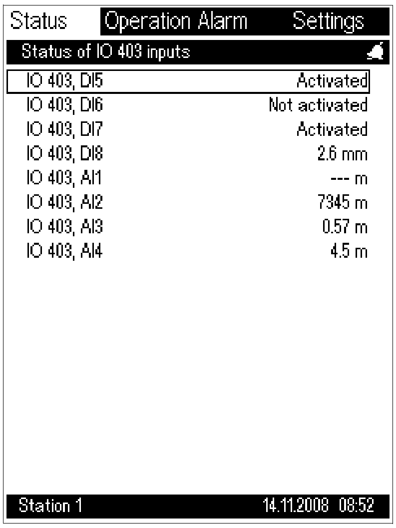
Note: The IP address is provided by the SIM card in the G 401 and cannot be changed.
See also section *11.5 Status - modules*.

11.8 Status of IO 403 inputs

This display shows the status of the IO 403 inputs.

Note: The display shown below should be considered as an example.

Path: Status>Status of IO 403 inputs>



Display_577

Description

Display example:

Status of the DI5, DI6 and DI7 inputs:

- Activated
- Not activated.

Status of the DI8 input:

- Activated (used as digital input)
- Not activated (used as digital input)
- xxx.x m (used as counter input).

Status of the AI1, AI2, AI3 and AI4 inputs:

- Activated (xxx m)
- Not activated (--- m).

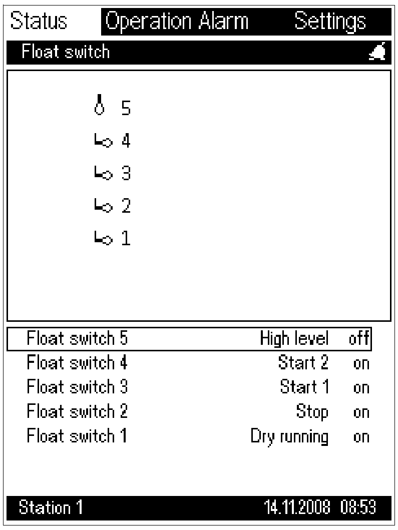
11.9 Status - position and location of float switches

This display shows the actual position and location of the float switches.

The display can be used in connection with fault finding and testing of functions.

Note: The display shown below should be considered as an example.

Path: Status>Float switch>



Display_98

Description

See the actual position and location of the float switches.

Display example:

- Float switch 5: High level.
- Float switch 4: Pump 2 starts.
- Float switch 3: Pump 1 starts.
- Float switch 2: The system stops.
- Float switch 1: Dry running.

11.10 Mixer

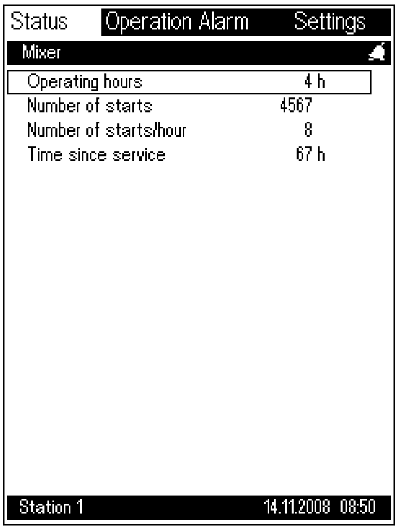
This display shows the counters for the mixer.

The display can be used in connection with fault finding and service.

Note: The display will only appear when the mixer is enabled. See section *14.7.7 CU 401, DO7 relay output*.

Note: The display shown below should be considered as an example.

Path: Status>Mixer>



Display_30

Description

See the actual number of operating hours and number of starts.

Display example:

- Operating hours
- Number of starts
- Number of starts/hour
- Time since service.

11.11 Trend curves

Generally the following types of measurements can be logged:

- level
- flow
- voltage
- current
- power
- cos φ
- specific energy
- temperature
- water in oil.

A logging interval can be selected for each of the above types of measurements. Subsequently, a logging interval related to the physical nature of the measurement can be selected, for example water in oil can be logged with a much longer interval than voltage. The logging interval can be set in seconds from 30 to 3600 seconds.

For the system, the following four measurements can be logged:

- level
- average flow
- power
- specific energy.

For each pump, the following eight measurements can be logged:

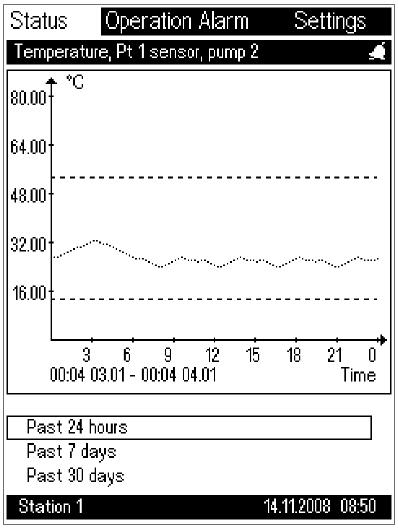
- temperature, Pt 1 (IO 401)
- temperature, Pt 2 (from MP 204 or IO 111. If both units are connected, only IO 111 data will be logged.)
- water in oil
- average flow
- average mains voltage
- average current
- power
- cos φ.

For the setting of data to log, see sections *14.7.26 Userlog - selection of data to log* and *14.7.27 Userlog intervals*.

If the logging of a given pump parameter is initiated, the logging is made for all the pumps in the system up to a maximum of six pumps.

Note: The display shown below should be considered as an example.

Path: Status>System>Power>30 days>



Display_10

Description

To see trend curves, select "System" in the "Alarm status" display and press **ok**. The subsequent display shows a list of the values which can be displayed as trend curves. Trend curves can be shown for 24 hours, 7 days or 30 days back in time.

Select the desired value and press **ok**.

Possible settings:

- Latest 24 hours
- Latest 7 days
- Latest 30 days.

Display of trend curves

The logged data can be displayed as curves on the OD 401. Alternatively, log data can be retrieved and displayed on a SCADA system. These installation and operating instructions only describe the display of curves on the OD 401.

The OD 401 graph can contain 200 x 150 points, the time and value axis, respectively.

Full scale on the time axis can be 24 hours, 7 days or 30 days. The factory setting is a full scale of 24 hours.

When a curve is selected, the time is frozen and the program retrieves a number of measurements, depending on the full scale and the interval chosen.

As the time axis solution is restricted to 200 points, the program calculates an average value of a number of measurements, depending on the logging interval and time scale chosen. If for example a full scale of 24 hours is selected with a logging interval of 30 seconds, 14 measurements are averaged into one measurement.

For the first 200 measurements, the program makes no averaging in the time dimension. Therefore the time axis is not correct until after a minimum of 200 measurements times the logging interval of the measurement in question, e.g. 30 second interval = 1 hour + 40 min.

In the y-axis dimension, the program first makes an offset adjustment, so that the zero point of the y-axis is the smallest value measured in the logging period. Then a compression of the dynamics of the signal is made, so that the value can be represented on 150 points.

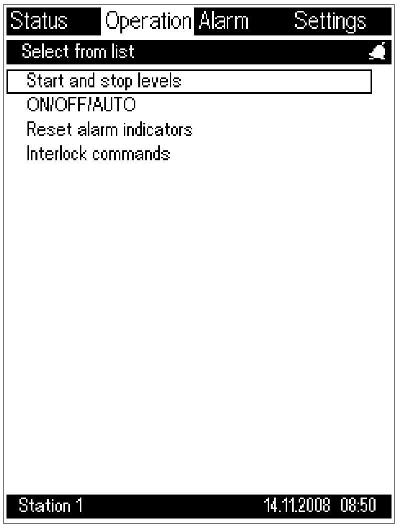
12. Operation

12.1 Overview

This menu item contains the most common settings of the pumping station such as start and stop levels, direct control of the pumps (ON/OFF/AUTO) and resetting of alarm indicators.

Note: The display shown below should be considered as an example.

Path: Operation>



Display_101

Description

Select from the list and press **ok**.

Possible settings:

- Start and stop levels, see section *12.1.1 Operation - start and stop levels*.
- ON/OFF/AUTO, see section *12.1.2 Direct operation of pumps - AUTO/ON/OFF mode*.
- Reset alarm indicators, see section *12.1.3 Operation - resetting alarm indicators and alarm relays*.
- Interlock commands, see section *12.1.4 Interlock commands*.

12.1.1 Operation - start and stop levels

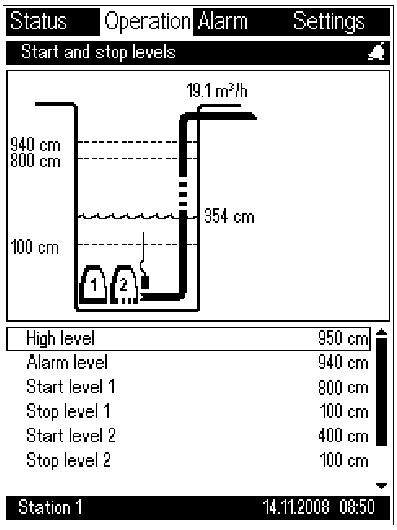
This display can be used to control or change the values for level control.

Note: If variation of start level has been activated, the display will show start level 1 (here 800 cm) and the set maximum variation (here 140 cm) as dotted lines.

Note: This is only possible if an analog sensor is fitted in the system. See validation rules concerning float switches, section *14.3.8 Level sensor*.

Note: The display shown below should be considered as an example.

Path: Operation**>**Start and stop levels**>**



Display_103

Description

Select the desired level to be changed and press **ok** (the focus frame starts flashing); use the **plus** and **minus** buttons to enter a new value. Press **ok** to enter the new value.

Possible settings:

- Overflow level
- High level
- Alarm level (enables interlock)
- Start level 1
- Start level 2
- Start level 3
- Start level 4
- Start level 5
- Start level 6
- Stop level 1
- Stop level 2
- Stop level 3
- Stop level 4
- Stop level 5
- Stop level 6.

Note: At the highest stop level, interlock is disabled. See also section *14.9.3 Interlock*.

- Dry-running level.

Description of display texts

Display text	Description
Overflow level	When this level is reached, the water runs over the edge of the pit or into an overflow channel. The overflow level triggers an alarm.
High level	This is the highest alarm level, indicating high water level. When this level is reached, the system will attempt to start or restart all pumps (the number of pumps can be limited in the group configuration display. See section *14.7.17 Group configuration - alternation/advanced alternation*).
Alarm level	The alarm level triggers an alarm, if desired. See section *14.2 Alarms and warnings*. The alarm level can be set between the lowest stop level and high level. The alarm level also triggers interlock, if interlock has been selected in the display named "Interlocking". See section *14.9.3 Interlock*.
Start level 1	This is the lowest start level. At start level 1, the first pump starts (not necessarily pump 1. See section *14.7.19 Group properties*).
Start level 2	This is the next start level. It is recommended to set start level 2 equal to or higher than start level 1.
Start level 3	This is the next start level. It is recommended to set start level 3 equal to or higher than start level 2.
Start level 4	This is the next start level. It is recommended to set start level 4 equal to or higher than start level 3.
Start level 5	This is the next start level. It is recommended to set start level 5 equal to or higher than start level 4.
Start level 6	This is the next start level. It is recommended to set start level 6 equal to or higher than start level 5.
Stop level 1	This is the lowest stop level. At stop level 1, the last pump stops. Stop level 1 can be set between the dry-running level and start level 1.
Stop level 2	This is the second stop level from the bottom. At stop level 2, the second-last pump stops. It is recommended to set stop level 2 higher than or equal to stop level 1.
Stop level 3	This is the third stop level from the bottom. At stop level 3, the third-last pump stops. It is recommended to set stop level 3 higher than or equal to stop level 2.
Stop level 4	This is the fourth stop level from the bottom. At stop level 4, the fourth-last pump stops. It is recommended to set stop level 4 higher than or equal to stop level 3.
Stop level 5	This is the fifth stop level from the bottom. At stop level 5, the fifth-last pump stops. It is recommended to set stop level 5 higher than or equal to stop level 4.
Stop level 6	This is the sixth stop level from the bottom. At stop level 6 the sixth-last pump stops. It is recommended to set stop level 6 higher than or equal to stop level 5.
Dry-running level	When this level is reached, the system will attempt to stop all pumps (again). The dry-running level triggers an alarm, if desired. See section *14.2 Alarms and warnings*.

12.1.2 Direct operation of pumps - AUTO/ON/OFF mode

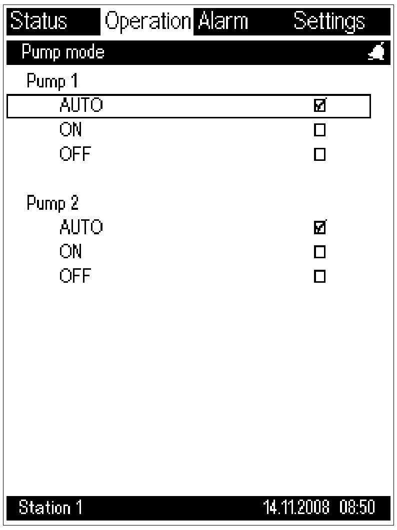
This display is used to switch between different pump modes. ON and OFF are used for direct operation of the pump.

The display is used for example for testing of pumps or for forced pumping/emptying.

Note: The ON/OFF/AUTO input on the IO 401 has highest priority. The OD 401 can only control the ON and OFF mode if it is in AUTO mode or not used.

Note: The display shown below should be considered as an example.

Path: Operation>Pump mode>



Display_102

Description

Select the function to be changed, press **ok** and then the **plus** or **minus** button.

Possible settings:

Pump 1

- AUTO (the pump is controlled automatically by the controller).
- ON (the pump is running).
- OFF (the pump has stopped).

Pump 2

- AUTO (the pump is controlled automatically by the controller).
- ON (the pump is running).
- OFF (the pump has stopped).

If the pump is set to ON, all pump protection and safety settings will be disabled, except the settings for motor-protective circuit breaker, overtemperature sensor 2 and missing phase.

If the pump is set to OFF, the float switch and other system functions will be disabled.

12.1.3 Operation - resetting alarm indicators and alarm relays

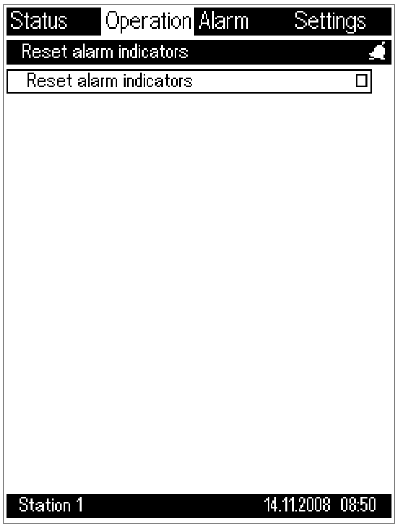
This display is used for resetting alarm indicators and alarm relays.

The alarm relays can be reset by pressing the reset button (if installed) or by using this OD 401 display.

Symbol	Description
	Alarm
	Warning

Note: The display shown below should be considered as an example.

Path: Operation>Reset alarm indicators>



Display_106

Description

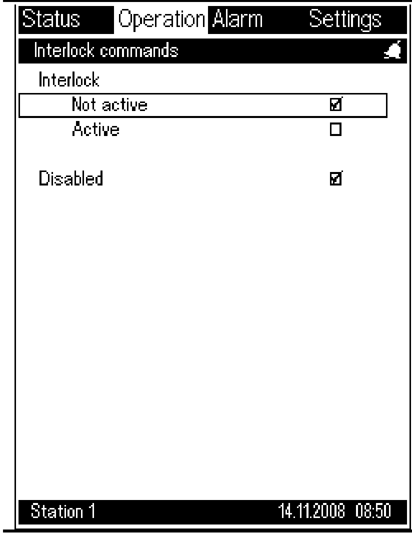
"Reset alarm indicators" is marked with a focus frame.

Reset alarm indicators in the following way:

1. Press **ok** (the focus frame is flashing).
2. Press **plus** (a check mark **V** appears).
3. Press **ok** to approve.

All alarm indicators are now reset.

12.1.4 Interlock commands



Display_101

Description

Select from the list and press **ok**.

- Not active (the interlock function is not active).
- Active (the interlock function is active).
- Disabled (the interlock function is disabled).
  The warning "Interlock disabled" is shown when interlock is disabled to remind the user of this setting.

13. Alarm

13.1 Alarm log

This display shows a log of the latest alarm activities (max. 112). The alarm log is used to see a historical overview of the alarms that have occurred in the pumping station. It is possible to trace the alarms chronologically back in time. If the maximum number of alarm activities is exceeded, the oldest alarm activity will be deleted.

The alarm log can be reset. See section *14.5.3 Alarm log*.

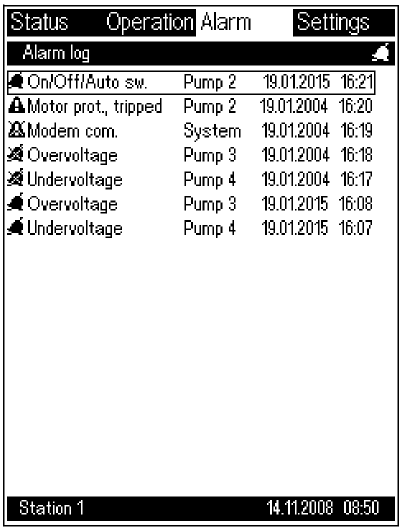
Note: The alarms **cannot** be reset from this display.
Actual alarms are reset in the alarm list. See section *11.2.1 Alarm list - resetting/approving system alarms and warnings*.

Most alarms also have SnapShot. See section *13.2 AlarmSnapShot*. This makes it possible to see whether a fault has more than one cause.

Symbol	Description
	Alarm
	Warning
	Non-active alarm
	Non-active warning

Note: The display shown below should be considered as an example.

Path: Alarm**>**Alarm log**>**



Display_201

Description

To show a SnapShot of an alarm, select the desired alarm and press **ok**.

An AlarmSnapShot of the desired alarm is displayed.

Note: The alarm log can contain up to 112 items. When the alarm log is full, the last item will be removed from the log if there is a new alarm.

Possible fault indications

System
Program card
UPS
Module
Mains
Modem
Modem com.
Float switch
Level sensor
Flowmeter
Overflow
Level sens., overflow
Flowmeter, overflow
Fault switch, overflow
Power meter
High level
Level
Dry running
Conflicting levels
Forc. rel.output act. (forced relay output enabled)
Mixer contactor
Time f. service, mix.
Max. starts/h, mixer

IO 403
IO 403, AI1 max.
IO 403, AI1 min.
IO 403, AI2 max.
IO 403, AI2 min.
IO 403, AI3 max.
IO 403, AI3 min.
IO 403, AI4 max.
IO 403, AI4 min.
IO 403, DI5
IO 403, DI6
IO 403, DI7
IO 403, DI8
IO 403, AI 1 sensor fault
IO 403, AI 2 sensor fault
IO 403, AI 3 sensor fault
IO 403, AI 4 sensor fault

Note: All the mentioned fault indications can be displayed both as a warning and an alarm. All alarm texts are displayed as the user defined them during the configuration. See section *14.7.9 IO 403, I/O settings*.

Pumps
Power meter
High level
Level
Dry running
Conflicting levels
Forc. rel.output act. (forced relay output enabled)
Mixer contactor
Time f. service, mix.
Max. starts/h, mixer
On/Off/Auto sw.
Motor protect., tripped
Mains supply, off
Overload
Underload
Overtemp., sensor 1
Overtemp., sensor 2
Overtemp., Pt 1
Pt 1 sensor
Water in oil
Water-in-oil sensor
Max. starts/hour
Moisture in motor
Low flow
Time for service
Latest runtime
Contactor

System with FB 101
Missing phase
Wrong phase seq.

System with CUE or frequency converter
VFD alarm

System with MP 204
Overvoltage
Undervoltage
Overload
Underload
Low insul. resist.
Current unbalance
cos φ min.
cos φ max.
Overt., MP 204 PTC
Start capacitor, low
Run capacitor, low
Aux. winding fault
Overtemp., Pt 2
Pt 2 sensor
Wrong phase seq.
GENIbus
Protection warning

System with IO 111
Low insul. resist.
Overtemp., Pt 2
Pt 2 sensor
Water in oil
Moisture in motor
GENIbus
IO 111, alarm
IO 111, warning

System with IO 111 and SM 111
Temperature sensor, Pt 2 (support bearing)
Temperature sensor, Pt 3 (main bearing)
Temperature sensor, Pt 4 (stator winding)
Insulation sensor
Water-in-oil sensor
Power line communication
Vibration sensor
Moisture sensor
PTC sensor
Temperature sensor fault, Pt 2
Temperature sensor fault, Pt 3
Temperature sensor fault, Pt 4
Water-in-oil sensor fault
Vibration sensor fault
PTC sensor fault
Config. conflict, fault
SM 111, analog sensor, fault
GENIbus
IO 111, alarm
IO 111, warning

13.2 AlarmSnapShot

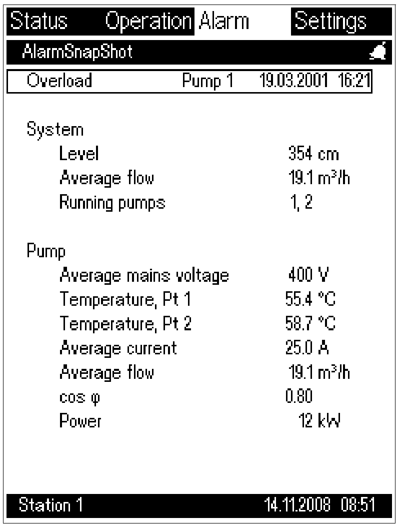
The AlarmSnapShot is mainly used for fault finding of the pumping station. The display shows a number of pumps and system values as they were measured when the alarm was tripped. This may indicate where the fault has occurred.

Most alarms have AlarmSnapShot. See section *13.1 Alarm log*.

Symbol	Description
	Alarm
	Warning

Note: The display shown below should be considered as an example.

Path: Alarm>



Display_202

Description

Displays date, time and fault indication for the alarm as well as operating parameters.

- System
  - Level
  - Average flow
  - Running pumps (shows running pumps with comma separation).
- Pump
  - Average mains voltage
  - Temperature, Pt 1 (IO 401)
  - Temperature, Pt 2 (from MP 204 or IO 111. If both units are connected, the IO 111 will be used)
  - Average current consumption
  - Average flow
  - cos φ
  - Power.

Note: AlarmSnapShot values are only measured in connection with pump alarms, not in connection with pump warnings. A pump alarm must be reset before a new SnapShot is taken.

14. Settings

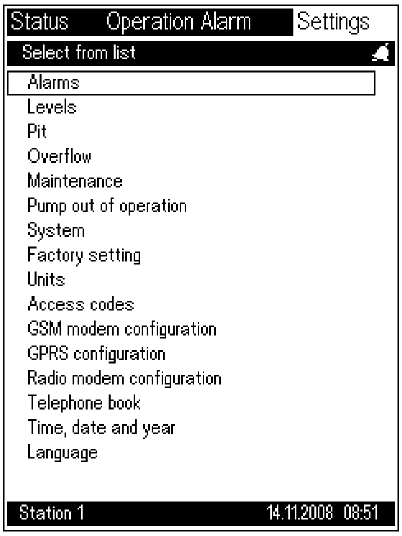
14.1 System overview

The following minimum settings must be made in the order shown below.

Note: Setting of alarms is especially important when using an analog sensor due to the fact that for example the levels of alternating pump operation vary with the system.

1. **Pump(s)**,
   path: Settings>System>Number of pumps>
2. **System**,
   path: Settings>System>
3. **Levels** (float switch or analog sensors),
   path: Settings>Levels>
4. **Alarms**,
   path: Settings>Alarms>

Path: Settings>



Display_301

Description

Possible settings:

- **Alarms**
  - For alarm settings, see section *14.2 Alarms and warnings*.
  - Select the alarms and warnings to be monitored by the system. Enable alarms and warnings, set limits for alarms and warnings and select whether a SCADA message or an SMS is to be sent in the event of an alarm/ warning.
    Note: These functions do not work until the alarm is active. See section *14.2 Alarms and warnings*.
- **Levels**
  - For level settings and variation of start level, see section *14.3.8 Level sensor*.
  - Select level sensor system, control method and setting of levels for the control of pump functions, alarm levels, etc.
- **Pit**
  - For pit configuration, see section *14.3.10 Pit configuration and flow calculations*.
  - Enter pit geometry data for the purpose of graphical display and flow calculations.
- **Overflow**
  - For configuration of overflow system, see section *14.4.1 Overflow system*.
  - Select overflow system. If overflow system 3, 4 or 5 is selected, enter the data for the overflow channel. See section *14.4.10 Channel parameters*.

Description

- Maintenance
  - Adjustment of counters, see section *14.5.1 Adjustment of counters*.
  - Calibration of sensors, see section *14.5.2 Calibration*.
  - Alarm log, see section *14.5.3 Alarm log*.
  - SMS counters, see section *14.5.4 SMS counters*.
- Pump out of operation
  - To enable/disable pumps, see section *14.6 Pump out of operation*.
  - The individual pumps can be taken out of operation in connection with service, etc.
- System
  - Number of pumps
  - I/O settings
  - PU 102 installed
  - Pump, time settings
  - Pump groups
  - Pump 1, 2, 3, 4, 5, 6 (MP 204, FB 101, IO 111, CUE, contactor feedback, VFD)
  - Mixer settings
  - Userlog selection of data
  - Userlog intervals
  - Installation name.
- Factory setting
  - Return to factory settings for the system
  - Return to factory settings for each pump.
- Units
  - For configuration of units, see section *14.8.1 Units*.
  - Select units to be used for presentation of data.
    Note: The configuration of units for analog inputs is made in the display in section *14.7.2 I/O settings*.
- Access codes
  - Operation
  - Setting.
- GSM modem configuration
  - PUK code
  - GSM protocol
    - ModBus
    - COMLI.
- GPRS configuration
  - APN
  - User name
  - Access code.
- Radio modem configuration
  - Address
  - Protocol
  - Baud rate.
- Telephone book
  - SCADA network
  - Interlock network
  - SMS.
- Time, date and year
  - Time setting.
- Language
  - Select the language to be used by the OD 401 (local language or English).

14.2 Alarms and warnings

This display is used for setting alarms and warnings. The display is divided into system alarms and pump alarms. The display allows the user to

- enable/disable alarms and warnings
- set alarm and warning limits
- set manual or automatic alarm approval, including automatic alarm-approval delay.

These settings can be made for each data point measured or calculated by the system, allowing adjustment of alarm and warning settings to the user's requirements.

In general, the system reacts on alarms by typically starting or stopping pumps as a result of an alarm state.

After a warning, the pumps continue to run. The warning indicates that the system is proceeding towards an alarm state. Service is not yet required.

Pump alarms and warnings contain alarms and warnings applying to the individual pump.

Alarms and warnings not relating to pump(s) have been gathered under "System alarms".

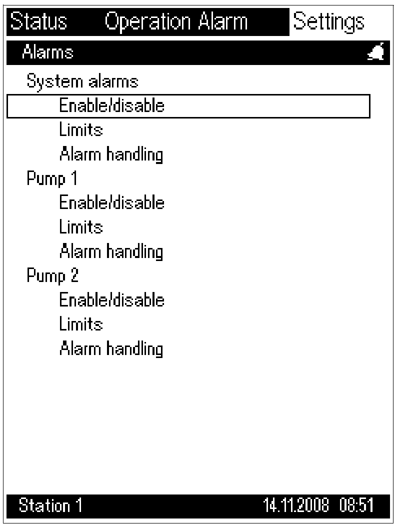
Note: The individual sensors must be set before this menu point can be used.

Symbol	Description
	Alarm
	Warning

Note: Enabling/disabling of alarms also enables/disables the sensor function. If, for instance, ON/OFF/AUTO is disabled, the switch input cannot be used to control the pump.

Note: The display shown below should be considered as an example.

Path: Settings>Alarms>



Display_302

Description

- Enable/disable
  - Select "Enable/disable" to display a list of alarms and warnings that can be enabled. **Note:** "Enable/disable" must be selected first as only enabled alarms and warnings will be displayed in the "Limits" and "Alarm handling" displays.
- Limits
  - Select "Limits" to set actual alarm and warning limits for enabled alarms/warnings.
  - It is possible to get an indication, if a value falls below or exceeds the alarm/warning limits set.
  - No changes will take place in the operation of the pumping station if a warning limit is exceeded. However, exceeding an alarm limit will result in a change in the operation of the pumping station, typically a pump stoppage or pump start.
- Alarm handling
  - Select "Alarm handling" to approve an alarm or delete it automatically from the system after a specified time without alarm, or to require manual resetting before operations can be resumed.

14.2.1 System alarms and warnings - enable/disable

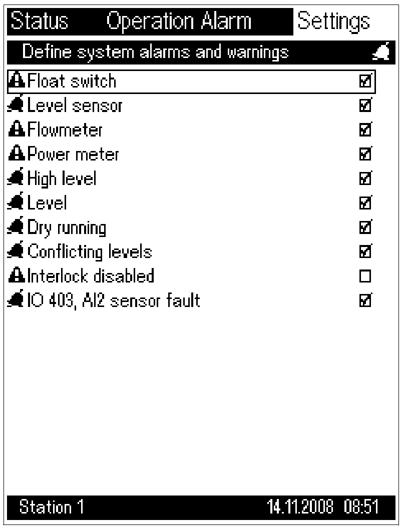
This display shows the system units that may cause an alarm or a warning. Select the unit(s) to be monitored.

After having selected the units in this display, the limits can be set.

Symbol	Description
🔔	Alarm
⚠	Warning

Note: The display shown below should be considered as an example.

Path: Settings**>**Alarms**>**System alarms - Enable/disable**>**



Display_303

Description

Setting of system alarms and system warnings. Select the alarms and warnings to be monitored by the system. Alarms or warning limits are displayed in the alarm list and the alarm log. The IO 403 alarm texts are displayed as the user defined them during the configuration. See section *14.7.9 IO 403, I/O settings*.

See the possible alarms and warnings and their functions in the list below.

Note: For external modules (MP 204 and other motor protectors as well as IO 111), alarms and warnings are not suppressed when they are deselected. This means that they will be shown in the alarm list and the alarm log.

Description of system alarms and system warnings

Alarms and warnings	Function
Program card	Program-card warning is displayed in the event of misreading/miswriting of the program card. If this fault occurs, contact Grundfos for possible replacement of the program card.
UPS	(Uninterruptible Power Supply). An UPS alarm is displayed if the UPS module PU 102 no longer provides power to the system. Possible cause: Exhausted battery, broken cable, etc.
Module	Module alarm is displayed if a module is not positioned correctly in the row, or if a fault occurs on the module during operation.
Mains voltage	System mains voltage supply alarm can be displayed if a UPS module is connected. There is no current supply to the system.
Modem	The alarm is displayed if the modem is defective.
Modem communication	The alarm is displayed if a communication fault has occurred in connection with dialup to or from the modem.
Float switch	The warning is displayed in the event of a discrepancy between the input signals from the various float switches if the float switch "start" is enabled and the float switch "stop" is disabled in a wastewater pit.
Level sensor	The alarm is displayed if the level sensor input is outside the measuring range.
Flowmeter	The warning is displayed if the flowmeter input is outside the measuring range.
Power meter	The warning is displayed if the power meter input is outside the measuring range.
High level	High level can be set to activate the DO8 relay output. When the high level is reached, the system attempts to restart all pumps. High level must always be the highest level in the system.
Level	Level can be selected at random with the lowest stop level as the lower limit and high level as the upper limit. Enables interlock. See also section *14.9.3 Interlock*.
Dry running	The dry-running level can be set at random as the lowest level in the system. All pumps are stopped, if this limit is reached. Foam drain level overrules the dry-running alarm and pumps down to foam drain level.

Conflicting levels	The alarm is displayed if the signal from the level sensor does not correspond with the float switches. This situation may occur if the level sensor is damaged or clogged. If the dry-running float switch displays dry running without the level sensor having displayed dry running, or if the high-level float switch is enabled without the level sensor also displaying high level, the system considers the level sensor to have failed and disregards it. If this situation occurs, it is possible to continue with only a high-level float switch and a dry-running float switch. When the high-level float switch is enabled, the pumps that are allowed to start will start pumping for a preset period of time or until the dry-running float switch displays dry running. **Note:** If the analog sensor is damaged, "Dry running" and "High level" will appear in the display even if the alarms have not been set to "active". **Note:** If the analog sensor is damaged, "Conflicting levels" will appear in the display even if the alarms "High level" and "Dry running" have not been set to "active".
Forced relay output enabled	The alarm is displayed if the DO7 or DO8 output is manually forced.
Mixer contactor	Contactor feedback. Contactor welded or no feedback from contactor.
Max. starts/hour, mixer	The maximum number of starts per hour. The number is specified in the manual for the mixer.
Time for service, mixer	Gives a warning when the counter has exceeded the number of operating hours set in section *14.2.2 System alarms and warnings - limits*.
Overflow	The alarm is displayed if an overflow has been registered by an analog level sensor or an overflow float switch.
Level sensor, overflow	The alarm is displayed if the level sensor connected for overflow measurement is outside the measuring range.
Flowmeter, overflow	The alarm is displayed if the flowmeter connected for overflow measurement is outside the measuring range.
Fault switch, overflow	The alarm is displayed if the overflow measurement must stop because the overflow tank is full and there is a risk of backflow.
Interlock	The alarm will be shown if the user disables interlock.
IO 403	The alarm is displayed if the current measurements of AI1 to AI4 lie outside the limits (minimum and maximum) and if the DI5 to DI8 inputs are activated. If the DI8 input is used as a counter input, a warning/alarm is displayed when the maximum limit is exceeded. Faults on one of the four analog inputs are shown in the alarm log. The alarm is displayed as the user defined it during the configuration. See section *14.7.9 IO 403, I/O settings*.

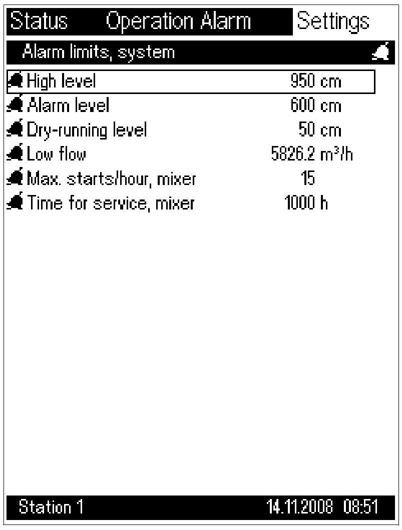
14.2.2 System alarms and warnings - limits

This display allows the setting of alarm and warning limits for the elements selected in the display. Alarm and warning limits can be set for each analog input. The digital inputs have two positions (alarm active or no alarm).

Symbol	Description
	Alarm
	Warning

Note: The display shown below should be considered as an example.

Path: Settings>Alarms>System alarms - Limits>



Display_304

Description

Setting of system alarms and warnings.

1. Select alarm and warning limits for the level functions.
2. Select the value to be changed and press **ok**.
3. Then use the **plus** and **minus** buttons to change the value.
4. Press **ok** again to approve the value.

Possible settings:

- High level
- Alarm level (enables interlock, if configured. See also section *14.9.3 Interlock*.)
- Dry-running level
- Low flow
- Max. starts/hour, mixer
- Time for service, mixer
- IO 403 warning and alarm limits.

14.2.3 System alarms and warnings - alarm handling

This display gives a number of setting options for handling of system alarms and warnings.

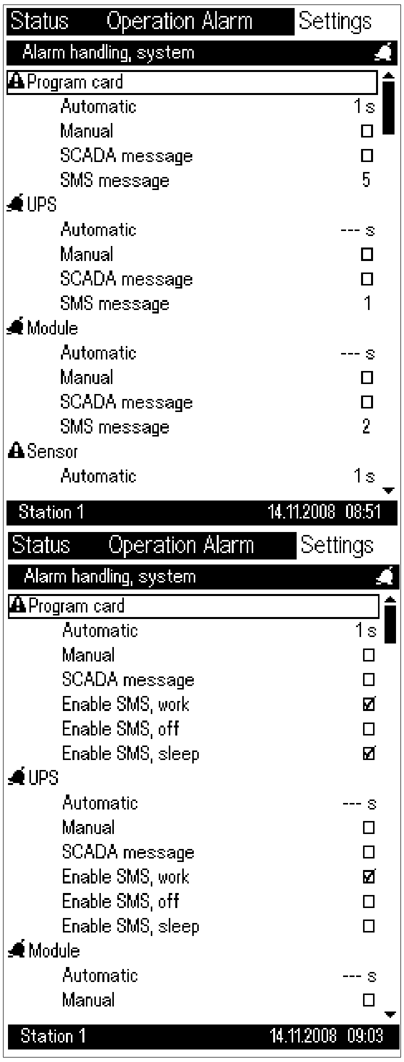
Possible settings:

- Automatic resetting
  Select "Automatic" to set the time period during which the value must be within the normal range before the alarm is reset.
- Manual resetting
  Note: Manual resetting means that a pump does not resume operation until the alarm has been reset in the alarm list.
- Message to the SCADA system
  The SCADA system is informed about the event.
  Note: If GPRS communication is used, this setting has no effect, as the SCADA system constantly communicates with the CU 401.
  For setting of the SCADA system, see sections *14.9.1 SCADA network* and *14.9.2 SCADA*.
- SMS message
  An SMS message is sent, informing the recipient about the event.
  Note: For setting of SMS message, see section *14.10 SMS*.

Symbol	Description
	Alarm
	Warning

Note: The display shown below should be considered as an example.

Path: Settings>Alarms>System alarms - Alarm handling>



Display_305

Description

Select the method to be used for approving the alarms.

1. Select the desired function and press **ok** (the focus frame is flashing).
2. Use the **plus** and **minus** buttons to change the value.
3. Press **ok** to approve.

Possible settings:

- Automatic resetting after a preset time.
- Manual resetting approved by user.
- SCADA message: An alarm telegram is sent to the SCADA system.
- SMS message
  – Advanced SMS
  – Simple SMS.

Note: The indication depends on whether advanced or simple SMS has been selected.

14.2.4 Pump alarms and warnings - enable/disable

This display is used to indicate the subsystems that trigger alarms and warnings. Several of these are only relevant if external units such as On/Off/Auto switch, MP 204 motor protector, IO 111, SM 111, temperature sensor, etc. are connected.

In general, alarms result in an action, such as stop or start of pumps, whereas warnings result in an indication and possibly a feedback to the SCADA system.

Symbol	Description
🔔	Alarm
⚠	Warning

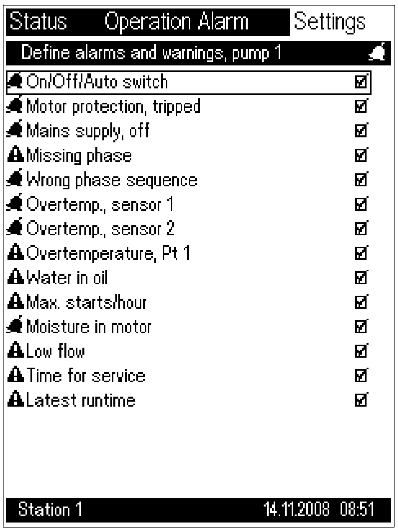
Note: The analog sensors have an alarm limit as well as a warning limit. If both limits have been chosen, a bell will be displayed in the list. If a warning has been chosen, a triangle will be displayed.

Note: The following alarm/warning functions will not be enabled until the alarm/warning has been set:

- On/Off/Auto switch
- Overtemperature, Pt sensor
- Water in oil.

Note: The display shown below should be considered as an example.

Path: Settings>Alarms>Pump 1 - Enable/disable>



Display_411-416

Description

Setting of pump alarms and pump warnings. Select the alarm/warning functions to be monitored by the system. Alarm or warning limits are displayed in the alarm list and in the alarm log.

See the possible alarms/warnings and functions in the table below.

Possible pump alarms and warnings	Function	Alarm	Warning
System with standard pumps			
On/Off/Auto switch	The alarm must be set to enable the On/Off/Auto switch function. If the switch fails, this alarm will be displayed. To prevent personal injury in the event of failure, the pump must be stopped.	🔔	
Motor protection, tripped	Must be connected to motor-protective circuit breaker terminals 97 and 98 allowing the system to detect a release. **Note:** The motor-protective circuit breaker provides the motor protection.	🔔	
Overtemp., sensor 1	The pump stops when this fault occurs. When the fault has been eliminated and the pump has cooled, the pump can be restarted automatically.	🔔	
Overtemp., sensor 2	The pump stops at this temperature, but it **cannot** be restarted automatically.	🔔	
Overtemperature, Pt 1	When the alarm limit is exceeded, the pump will be switched off. Automatic restarting is allowed.	🔔	⚠
Water in oil	Typical limit values are between 0 % and 20 %. This value is measured either with the IO 401 or the IO 111. It is also possible to get a Pt sensor warning if the sensor is defective.		⚠
Max. starts/hour	The desired maximum number of starts per hour can be set. The warning is displayed if the number of pump starts per hour exceeds the limit set.		⚠
Moisture in motor	Feedback from some of the pumps.	🔔	
Low flow	The low-flow limit can be set to receive an indication that a decrease in pump performance has occurred.		⚠
Time for service	The warning is displayed if the recommended service interval has been exceeded or if the total pump operating time exceeds the limit set.		⚠

Possible pump alarms and warnings	Function	Alarm	Warning
Latest runtime	The maximum time the pump is allowed to operate without interruptions. At the end of the alarm time set, the pump stops and another pump starts, provided the conditions of pump operation continue to be fulfilled. This setting is intended for a system with pump alternation, characterised by an almost identical inflow and pump capacity. This results in forced alternation, when a pump reaches its maximum operating time.	🔔	⚠
GENIbus	The communication between the system and the MP 204 is via a GENIbus. An alarm is displayed if communication to the MP 204 is interrupted.	🔔	
Contactor	An "NO" auxiliary contact is used for feedback from the main contactor to check that the contact set is not welded or hanging.	🔔	
System with FB 101			
Missing phase	One or more phases missing in a three-phase system is detected by the IO 401 (with FB 101). The pumps will stop.	🔔	
Wrong phase sequence	If the phase sequence is not correct, the pump rotates in the wrong direction. If this event occurs during operation, the pump either stops or does not start. **Note:** The pump can still be started by setting On/Off/Auto to "On". This is intended for use in connection with service.	🔔	
Mains supply, off	All phases are missing.	🔔	
System with CUE or frequency converter			
VFD alarm	If there is a fault on the frequency converter, there will be a VFD alarm. The alarm is general and only says that there is a fault on the frequency converter.	🔔	
System with IO 111			
Low insul. resist.	The IO 111 can do an average measurement of the discharging resistor. As standard the warning limit is set at 50 kΩ and the alarm limit at 20 kΩ.	🔔	⚠
Overtemp., Pt 2	Overtemperature, from IO 111 Pt100/Pt1000 or Tempcon sensor. When the alarm limit is exceeded, the pump will be switched off. Automatic restarting is allowed.	🔔	⚠
Water in oil	Typical limit values are between 0 % and 20 %. This value is measured either with the IO 401, IO 111 or the SM 111. It is also possible to get a Pt sensor warning if the sensor is defective.		⚠
Moisture in motor	Feedback from some of the pumps.	🔔	
Overtemp., sensor 1	Overtemperature, from IO 111 sensor/thermal switch. The pump stops when this fault occurs. When the fault has been eliminated and the pump has cooled, the pump can be restarted automatically.	🔔	
IO 111, alarm	The alarm is displayed when an alarm is active on the IO 111.	🔔	
IO 111, warning	The warning is displayed when a warning is active on the IO 111.		⚠
System with SM 111 and IO 111 communication module			
Vibration	Too high vibration level from the SM 111 vibration sensor. The pump needs an unscheduled service overhaul.	🔔	⚠
Overtemp., Pt 3	Overtemperature in the pump main bearing. When the alarm limit is exceeded, the pump will be switched off. Automatic restarting is allowed.	🔔	⚠
Overtemp., Pt 4	Overtemperature in the stator winding (from Pt100, Pt1000 or IO 111 klixon). When the alarm limit is exceeded, the pump will be switched off. Automatic restarting is allowed.	🔔	⚠
Overtemp., PTC	Overtemperature from the SM 111 PTC sensor. When the alarm limit is exceeded, the pump will be switched off. Automatic restarting is allowed.	🔔	⚠
Power line communication	No communication between the SM 111 and IO 111. The sensors monitored by the SM 111 will no longer be visible in the SCADA system or OD 401.	🔔	
Config. conflict, fault	Fault in the configuration of the DIP switch.	🔔	
GENIbus	The system and the IO 111 communicate via a GENIbus. An alarm is displayed if the communication to the IO 111 is interrupted.	🔔	
Water-in-oil sensor fault	The water-in-oil value will be shown with the last measured value.	🔔	
Vibration sensor fault	The vibration value will be shown with the last measured value.	🔔	
IO 111, alarm	The alarm is displayed when an alarm is active on the IO 111.	🔔	
IO 111, warning	The warning is displayed when a warning is active on the IO 111.		⚠

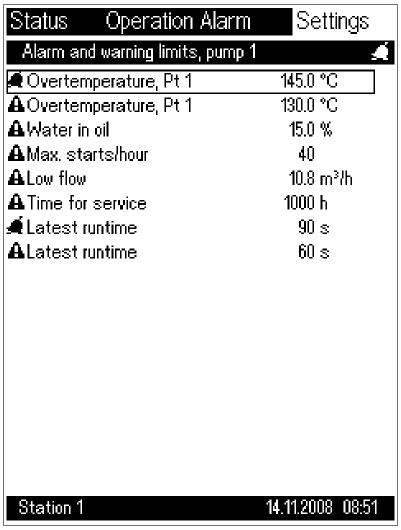
14.2.5 Pump alarms and warnings - limits

This display allows the setting of alarm and warning limits for measurements selected in the display. Alarm and warning limits can be set for each analog input. Limits for digital alarms cannot be set, as they are either active or not active.

Symbol	Description
	Alarm
	Warning

Note: The display shown below should be considered as an example.

Path: Settings>Alarms>Pump 1 - Limits>



Display_421

Description

Setting of pump alarms and warnings. Select alarm and warning limits for level functions:

1. Select the value to be changed and press **ok**.
2. Use the **plus** and **minus** buttons to change the value.
3. Press **ok** again to approve the value.

Possible settings:

- Overload
- Overload
- Underload
- Underload
- Insulation resistance
- Insulation resistance
- Overtemperature, Pt 1
  (Overtemperature, Pt 2,
  if an IO 111 is connected and in use)
- Overtemperature, Pt 1
  (Overtemperature, Pt 2,
  if an IO 111 is connected and in use)
- Water in oil
- Water in oil
- Max. starts/hour
- Max. starts/hour
- Time for service
- Time for service
- Latest runtime
- Latest runtime.

14.2.6 Pump alarms and warnings - alarm handling

This display gives a number of setting options for handling of alarms and warnings.

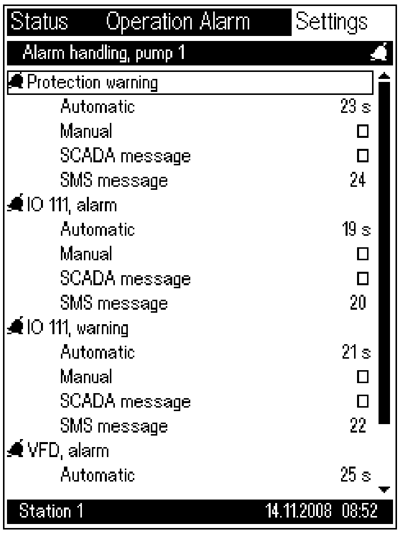
Possible settings:

- Automatic resetting
  Select "Automatic" to set the time period during which the value must be within the normal range before the alarm is reset.
- Manual resetting
  Note: Manual resetting means that a pump does not resume operation until the alarm has been reset in the alarm list.
- Message to SCADA system
  The SCADA system is informed about the event.
  Note: If GPRS communication is used, this setting has no effect, as the SCADA system constantly communicates with the CU 401.
  For setting of the SCADA system, see sections *14.9.1 SCADA network* and *14.9.2 SCADA*.
- SMS message
  An SMS message is sent, informing the recipient about the event.
  Note: For setting of SMS message, see section *14.10 SMS*.

Symbol	Description
	Alarm
	Warning

Note: The display shown below should be considered as an example.

Path: Settings>Alarms>Pump 1 - Alarm handling>



Display_431_436-13

Description

Select the method to be used for approving the alarms and warnings:

1. Select the desired function and press **ok** (the focus frame is flashing).
2. Use the **plus** and **minus** buttons to change the value.
3. Press **ok** to approve.

Possible settings:

- Automatic resetting after a preset time
- Manual resetting approved by user
- SCADA message: An alarm telegram is sent to the SCADA system.
- SMS message
  – Advanced SMS
  – Simple SMS.

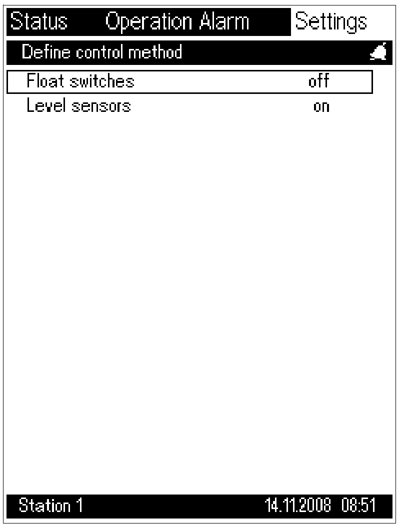
Note: The indication depends on whether advanced or simple SMS has been selected.

14.3 Levels - float switches or level sensor

In this display, the method for controlling the pumps is selected. If the system comprises more than two pumps, an analog sensor is required.

Note: The display shown below should be considered as an example.

Path: Settings>Levels>



Display_325

Description

First select the control method and then the individual start/stop settings.

- Float switches
  This selection can be made for one or two pumps. Pump start and stop are controlled by float switches located in the pit. This setting requires that a minimum of two float switches are connected for one pump and three float switches for two pumps.
  Press **ok** to display the configuration display for float switches. See section *14.3.1 Float switches*.
- Level sensors
  This selection can be made for an arbitrary number of pumps up to a maximum of six pumps. The setting requires an analog sensor.
  Press **ok** to display the setting display for the individual start/stop levels.

Note: Additional float switches in connection with level sensors (high level and dry running) are to be set at the same time as the level sensor. See section *14.3.8 Level sensor*.

Use the below links to find the desired empty or analog-sensor function:

Empty function with one pump:

14.3.2 One pump and two float switches

14.3.3 One pump and three float switches

14.3.4 One pump and four float switches

Empty function with two pumps:

14.3.5 Two pumps and three float switches

14.3.6 Two pumps and four float switches

14.3.7 Two pumps and five float switches

Analog-sensor function with more than two pumps:

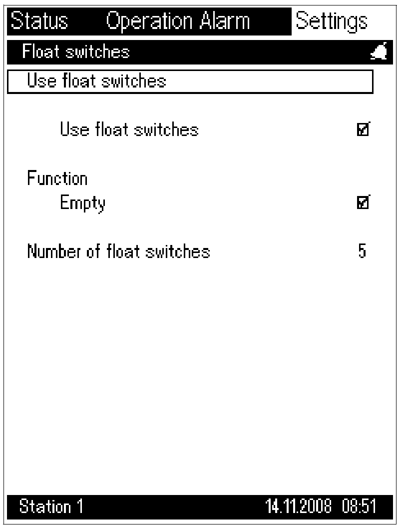
14.3.8 Level sensor

14.3.1 Float switches

This display allows the user to enter the number of float switches.

Note: The display shown below should be considered as an example.

Path: Settings>Levels>Float switches>



Display_316

Description

Procedure:

First set the number of float switches:

1. Select "Number of float switches" and press **ok** (the focus frame is flashing).
2. Use the **plus** and **minus** buttons to change the number.
3. Press **ok** to approve.

Then select the float switch function:

1. Select "Use float switches" and press **ok**.
2. In the new display that appears, select the function. See sections *14.3.2* to *14.3.7*.

14.3.2 One pump and two float switches

This display allows the user to select the function of the float switches connected.

The left side of the display shows the existing, saved function.

To see other options,

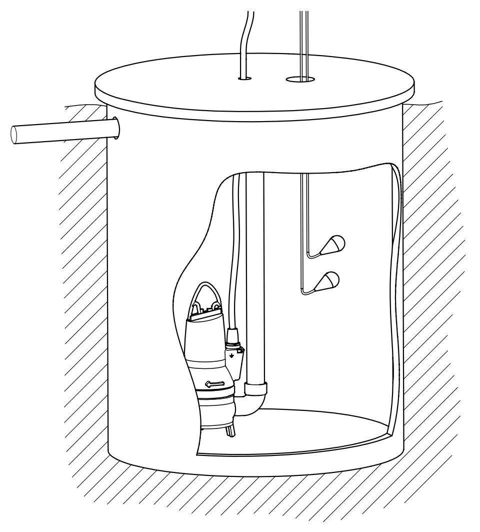
1. select a configuration and press **ok**.

When the desired configuration has been found,

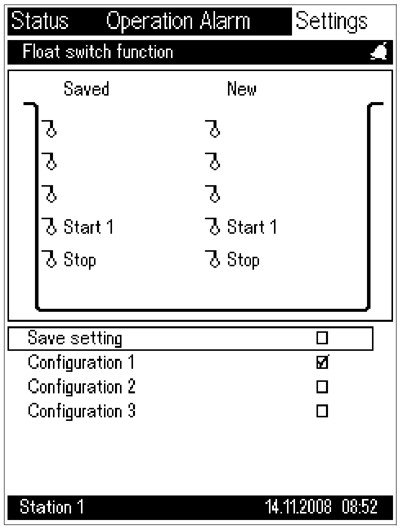
2. select "Save setting" and put a check mark **V** in the frame to the right.

Note: The display shown below should be considered as an example.

Path: Settings**>**Levels**>**Float switches**>**Use float switches**>**



TM02 8114 4703



Display_650_1_2

Configurations

Float switch	Configuration		
	1	**2**	**3**
2	Start	High level	Start/Stop
1	Stop	Start/Stop	Dry running

14.3.3 One pump and three float switches

This display allows the user to select the function of the float switches connected.

The left side of the display shows the existing, saved function.

To see other options,

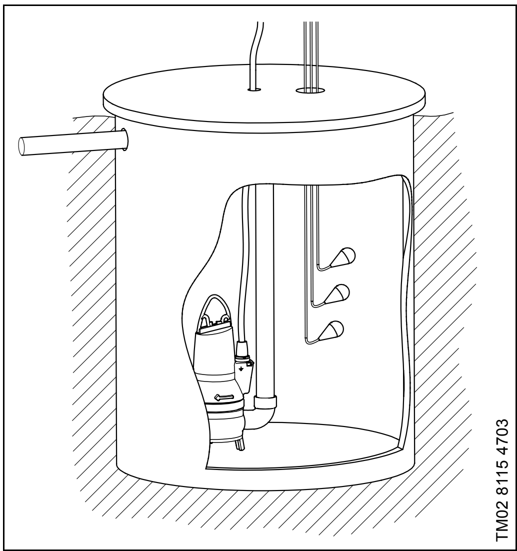
1. select a configuration and press **ok**.

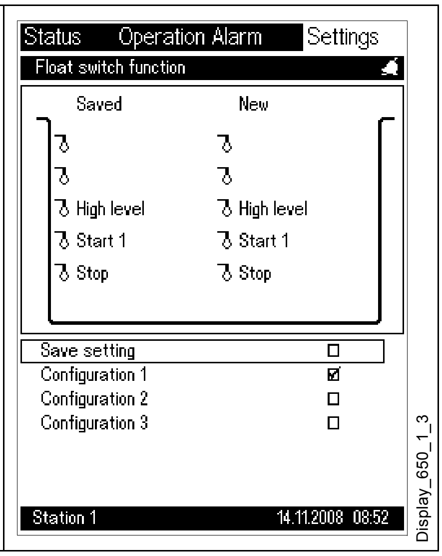
When the desired configuration has been found,

2. select "Save setting" and put a check mark **V** in the frame to the right.

Note: The display shown below should be considered as an example.

Path: Settings**>**Levels**>**Float switches**>**Use float switches**>**





Configurations

Float switch	Configuration		
	1	**2**	**3**
3	High level	High level	Start
2	Start	Start/Stop	Stop
1	Stop	Dry running	Dry running

14.3.4 One pump and four float switches

This display allows the user to select the function of the float switches connected.

The left side of the display shows the existing, saved function.

To see other options,

1. select a configuration and press **ok**.

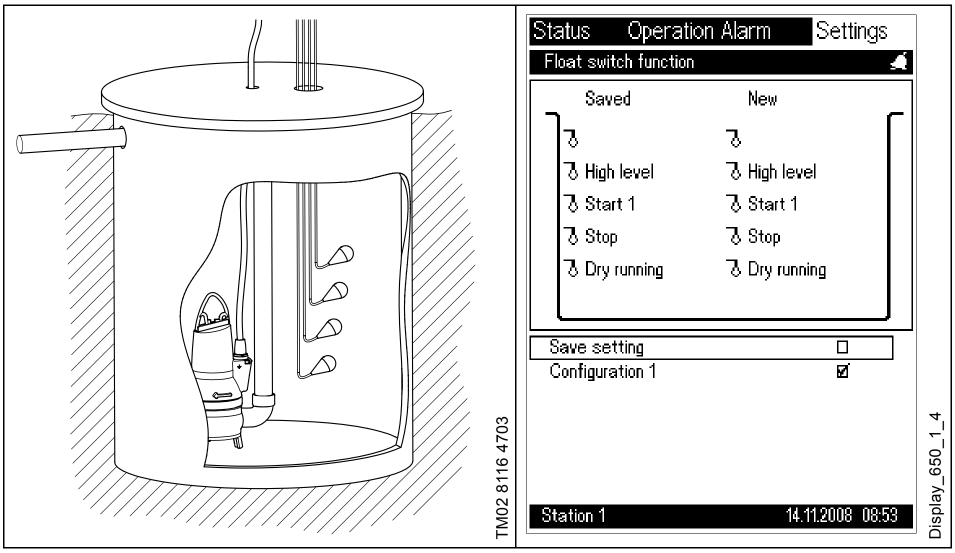
When the desired configuration has been found,

2. select "Save setting" and put a check mark **V** in the frame to the right.

There is only one configuration option for one pump and four float switches.

Note: The display shown below should be considered as an example.

Path: Settings**>**Levels**>**Float switches**>**Use float switches**>**



Configurations

Float switch	Configuration
	1
4	High level
3	Start
2	Stop
1	Dry running

14.3.5 Two pumps and three float switches

This display allows the user to select the function of the float switches connected.

The left side of the display shows the existing, saved function.

To see other options,

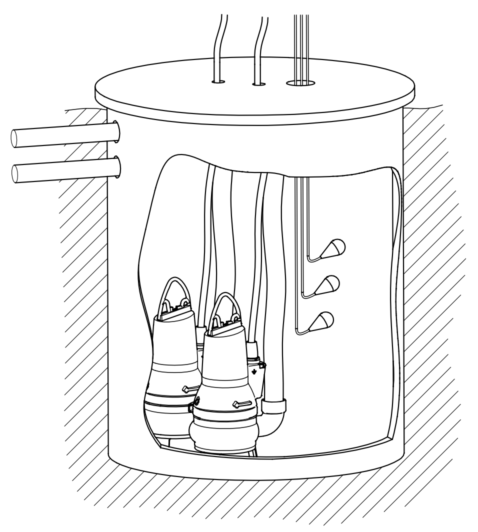
1. select a configuration and press **ok**.

When the desired configuration has been found,

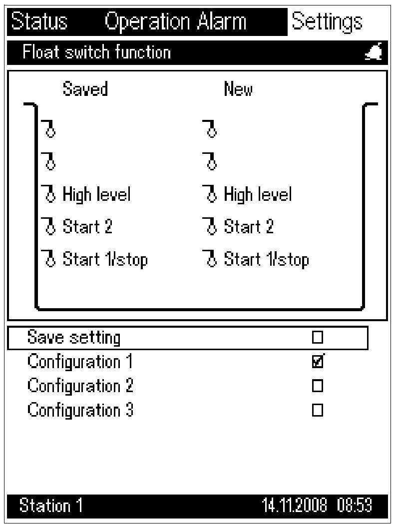
2. select "Save setting" and put a check mark **V** in the frame to the right.

Note: The display shown below should be considered as an example.

Path: Settings**>**Levels**>**Float switches**>**Use float switches**>**



TM02 8299 4903



Display_650_2_3

Configurations

Float switch	Configuration			
	1	**2**	**3**	**4**
3	High level	Start 2	Start 2	Start 2
2	Start 2	Start 1/ Stop	Alarm	Start 1
1	Start 1/ Stop	Dry run- ning	Start 1/ Stop	Stop

14.3.6 Two pumps and four float switches

This display allows the user to select the function of the float switches connected.

The left side of the display shows the existing, saved function.

To see other options,

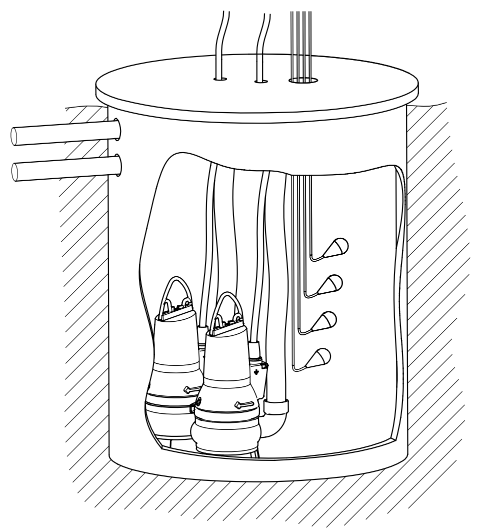
1. select a configuration and press **ok**.

When the desired configuration has been found,

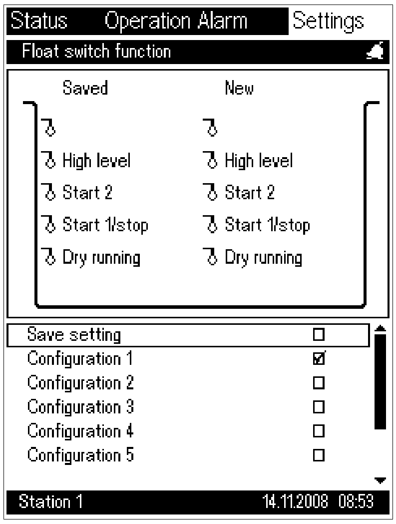
2. select "Save setting" and put a check mark **V** in the frame to the right.

Note: The display shown below should be considered as an example.

Path: Settings>Levels>Float switches>Use float switches>



TM02 8300 4903



Display_650_2_4

Configurations

Float switch	Configuration							
	1	2	3	4	5	6	7	8
4	High level	High level	Start 2	Start 2	Start 2	Start 2	Start 2	Start 2
3	Start 2	Start 2	Alarm	Alarm	Start 1	Start 1	Start 1	Stop 2
2	Start 1/Stop	Start 1	Start 1	Start 1/Stop	Stop	Stop 2	Stop 1	Start 1/Stop
1	Dry running	Stop	Stop	Dry running	Dry running	Stop 1	Stop 2	Dry running

14.3.7 Two pumps and five float switches

This display allows the user to select the function of the float switches connected.

The left side of the display shows the existing, saved function.

To see other options,

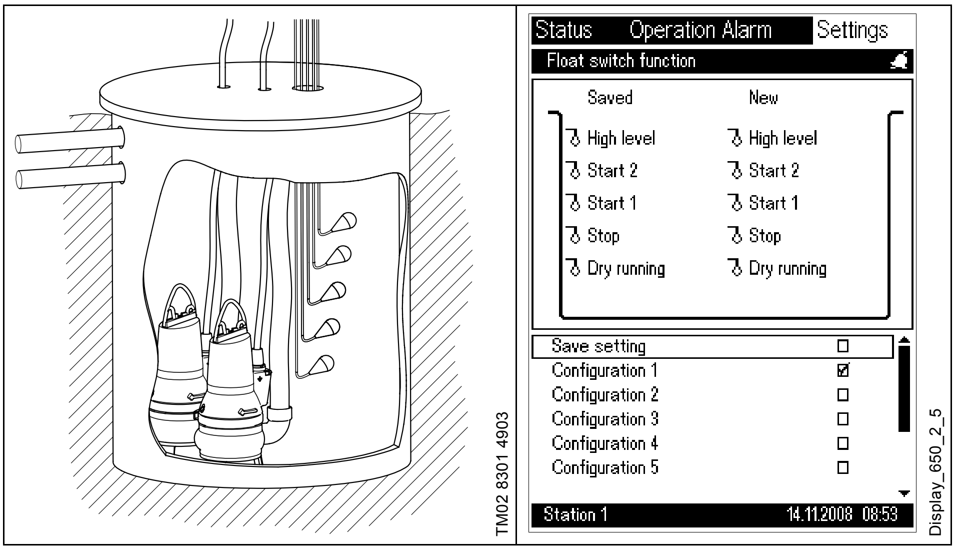
1. select a configuration and press **ok**.

When the desired configuration has been found,

2. select "Save setting" and put a check mark **V** in the frame to the right.

Note: The display shown below should be considered as an example.

Path: Settings>Levels>Float switches>Use float switches>



Configurations

Float switch	Configuration												
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**
5	High level	High level	High level	Start 2	Start 2	High level	Start 2	Start 2	Start 2	High level	Start 2	High level	Start 2
4	Start 2	Start 2	Start 2	Alarm	Start 1	Start 2	Alarm	Start 1	Stop 2	Start 2	Alarm	Start 2	Alarm
3	Start 1	Alarm	Alarm	Start 1	Stop 2	Start 1	Start 1	Stop 1	Start 1	Start 1	Start 1	Stop 2	Stop 2
2	Stop	Start 1/ Stop	Start 1	Stop	Stop 1	Stop 2	Stop 2	Stop 2	Stop 1	Stop 1	Stop 1	Start 1	Start 1
1	Dry running	Dry running	Stop	Dry running	Dry running	Stop 1	Stop 1	Dry running	Dry running	Stop 2	Stop 2	Stop 1	Stop 1

14.3.8 Level sensor

This display allows the user to set the stop and start levels for the number of pumps in the system as well as the high level, dry-running level, foam-drain level and overflow.

If alternation has been deselected, the pumps are numbered according to their position. See section *4. Positioning*. Start 1 and Stop 1 apply to the pump connected as number 1 in the system.

In connection with alternation, this one-to-one rule does not apply. This means that the lowest level always starts one pump and the next level two pumps, etc.

In connection with alternation, the number of operating hours are distributed equally between the pumps in the system.

The system ensures that the below rules are observed by automatically adjusting the other **levels** accordingly to meet the rules:

- The dry-running level is always lower than the lowest stop level.
- The high level is always the highest level.
- The start level must always be higher than the stop level of the same pump.
- The alarm level can be set higher than the lowest start level and lower than the high level.

At the overflow level, the liquid runs over the edge of the pit or into an overflow channel. The overflow level typically lies between the highest start level and the edge of the pit.

If, in addition to the level sensor, the system incorporates a high-level float switch and/or a dry-running float switch, these float switches must be selected in this display.

In the pit, the high-level float switch must be physically installed above the level indicated as high level, otherwise a "conflicting level" alarm will occur and the sensor is considered defective.

The dry-running float switch must be physically installed below the level indicated as dry-running level, otherwise a "conflicting level" alarm will occur and the sensor is considered defective.

When the high-level float switch is enabled, the high-level alarm is set. At the same time, all pumps will start.

For the purpose of emergency operation in the event of a defective sensor, the time from the deactivation of the high-level float switch to the stopping of the pumps can be set. The length of this time is best found by trial and error, as it depends on the actual amount of water the pumps are capable of moving. See section *14.7.16 Pump - time settings*.

If there is more than one group, the user selects the group that each pump is to belong to. If alternation has been deselected, the start level of each pump determines the group in which a pump is to start. For example, if pump 1 belongs to group 2, start 1 means that a pump from group 2 is to start. See section *14.7.18 Pump grouping*.

For the connection between levels and alternation, see section *14.7.17 Group configuration - alternation/advanced alternation*.

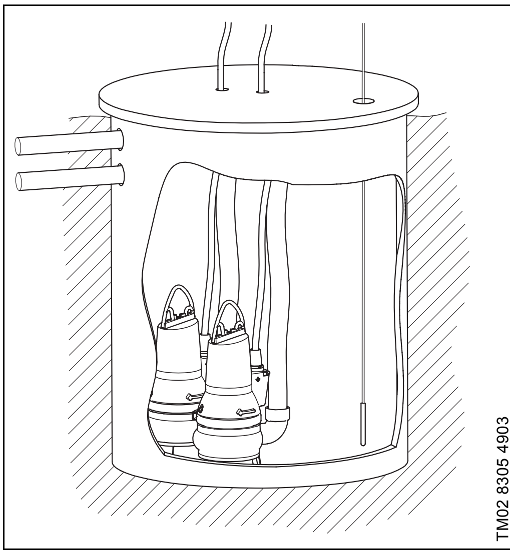
For further information about configuration, see section *14.7.18 Pump grouping*.

The foam-drain level function means that water is drained to this level. See also section *14.7.16 Pump - time settings*.

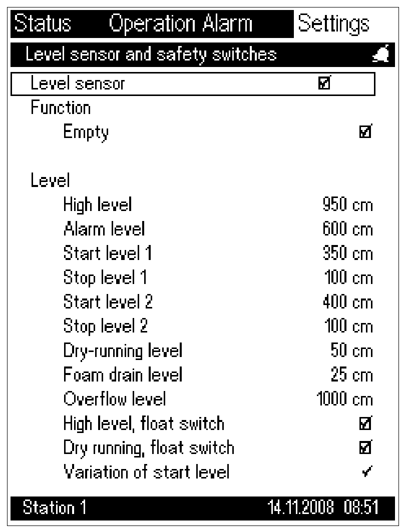
Note: The display shown below should be considered as an example.

Note: The overflow float switch is selected in section *14.4.1 Overflow system*.

Path: Settings>Levels>Level sensors>



TM02 8305 4903



Display_317

14.3.9 Variation of start level

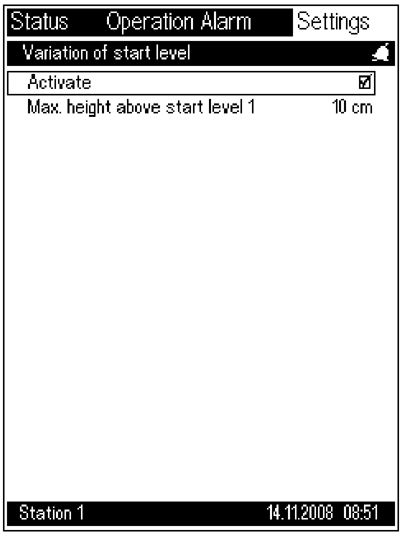
In order to prevent deposits of grease and other organic material at the same level, variation of start level 1 can be set.

Start level 1 is changed to a random height after each start. A maximum variation must be set.

Note: The function requires an analog sensor in the system.

Note: The display shown below should be considered as an example.

Path: Settings>Levels>Level sensor>Variation of start level>



Display_318

Description

- Activate (the check mark shows that variation of the start level has been activated. The variation can only be positive).
- Max. height above start level 1.

14.3.10 Pit configuration and flow calculations

For a useful pit display, the pit depth entered should be at a suitable level. Then enter the measurement data to calculate the flow.

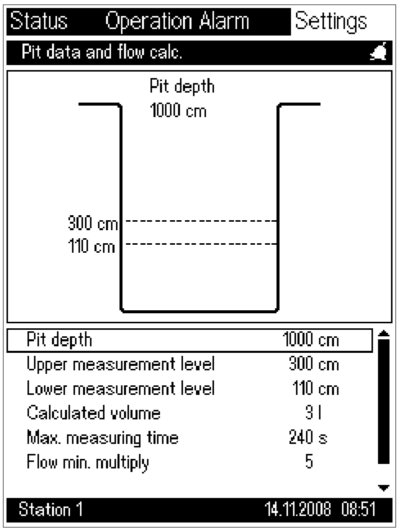
When the pumps have stopped, the time it takes to fill the volume is measured, thus calculating the inlet flow. The calculations are based on a constant flow during the pumping time.

The pit volume between the lower measurement level (the height "h_1") and the upper measurement level (the height "h_2") should be stated as accurately as possible to enable the system to calculate the correct flow. A flow calculated in this way has an empirical accuracy of ± 10 %, provided the inlet flow is constant during the pumping time and that the height/volume values selected provide for a suitable pumping time - pit ratio.

When pumping down a pit, the time it takes a pump to remove this volume is measured, thus calculating the pump performance of the pump.

Note: The display shown below should be considered as an example.

Path: Settings**>**Pit**>**



Display_355

Description

Enter the following pit data:

- Pit depth
- Upper measurement level (the level must be lower than the "Start 1" level)
- Lower measurement level (the level must be higher than the "Stop 1" level)
- Calculated volume
- Max. measuring time
- Flow min. multiply
- Flow max. multiply.

Description of display texts

Display text	Function
Pit depth	Enter the actual pit depth. If the levels on the display in section *11. Status* overlap, reduce the pit depth in this display; this will increase the distance between the actual start and stop levels displayed. Pit depth settings are only used for the graphical presentation of the pit.
Upper measurement level	Enter the upper measurement level for the flow calculation. The level must be lower than the "Start 1" level. See fig. 23, page 37.
Lower measurement level	Enter the lower measurement level for the flow calculation. The level must be higher than the "Stop 1" level. See fig. 23, page 37.
Max. measuring time	Enter the maximum permissible time for filling the volume between the lower measurement level and the upper measurement level. This time is found by measuring the time it normally takes to fill the volume between the lower and the upper measurement level. The time entered should be 1.2 times longer. **Example:** It takes 20 minutes to fill the pit, including 15 minutes for filling the volume between the lower and the upper measurement level. The time is set to 15 x 60 x 1.2 = 1080 seconds. The time is set in seconds.
Flow min. multiply	The calculation of the flow min. multiply value is described below. Factory setting: 5.
Flow max. multiply	The calculation of the flow max. multiply value is described below. Factory setting: 7.
Calculated volume	Enter the pit volume between the lower and the upper measurement level.

A flow calculation is expected to be made in 80-100 % of the pump starts. If a flow calculation is not made in at least 70 % of the pump starts in the actual pit, start by checking the time it takes to fill the volume between the lower and upper measurement level. If the time measured is higher than the max. measuring time set, the latter time must be changed. See the example in the table above.
If flow calculations are still not made, the empty time is also measured.

Example 1: It takes approx. 3 minutes to fill the pit and approx. 1 minute to empty it. The ratio between filling and emptying the pit is 3:1 = approx. 3. Set the flow min. multiply value to 2.

Example 2: It takes approx. 20 minutes to fill the pit and approx. 2 minutes to empty it. The ratio between filling and emptying the pit is 20:2 = approx. 10. Set the flow max. multiply value to 11.

Note: Changes in the flow min. multiply and flow max. multiply values affect the flow calculation. The best accuracy is obtained with values between 5 and 7.

Flow calculation theory

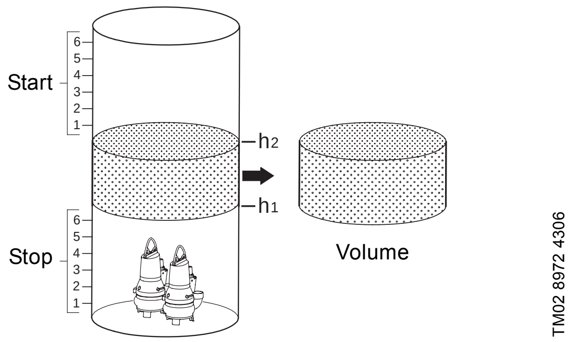


Fig. 24 Example of pit

Note: Fig. 24 shows an ideal pit.

To obtain the optimum flow calculation, the following situations must be taken into account:

- the pit is not cylindrical.
- the pumps are included in the calculated volume; in that case, the volume of the pumps must be deducted from the calculated volume.
- any other physical factors in the pit affecting the calculated volume.

The inlet flow is measured when the pumps have stopped and the pit is filled. The time it takes to fill the volume from h_1 to h_2 is called t_2.

The time it takes one pump to remove the volume is called t_1.

See fig. 25.

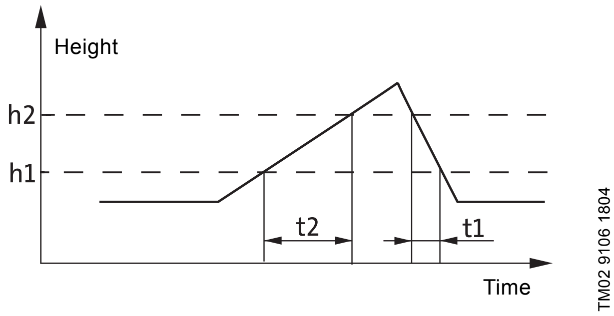


Fig. 25 Height in pit as a function of time

The optimum measurement is obtained if $5 \times t_1 \leq t_2 \leq 7 \times t_1$. See the table on the previous page. If t_2 is outside this interval, the calculation is ignored and the actual pump flow is not updated.

If the time between useful measurements is too long, or if no useful measurements are obtained, the flow min. multiply value can be reduced and the flow max. multiply value increased.

The inlet flow is expected to be constant during the time period t_1. If the volume between the lower measurement level h_1 and the upper measurement level h_2 is called V, the pump flow Q_p is calculated as follows:

$$Q_p = V \frac{t_1 + t_2}{t_1 \times t_2}$$

Flow calculation for two pump sizes:

The following applies to two different pump sizes:

"flow min. factor" x t_{1small} < t_2 < "flow max. factor" x t_{1large}.

Explanation:

t_{1small} = pumping down time for a small pump

t_{1large} = pumping down time for a large pump

t_2 = average filling time (e.g. not right after a large quantity).

Example:

A small pump needs 90 seconds to pump from "upper measurement level" to "lower measurement level".

A large pump needs 30 seconds to pump from "upper measurement level" to "lower measurement level".

The average filling time from "lower measurement level" to "upper measurement level" has been measured to 320 seconds.

flow min. factor = 320/90 = 3.56 ~ **3** (small factor rounded down).

flow max. factor = 320/30 = 10.7 ~ **11** (large factor rounded up).

Check:

The following condition must be fulfilled:

"flow min. factor" x t_{1small} < t_2 < "flow max. factor" x t_{1large}.

3 x 90 < 320 < 11 x 30.

270 < 320 < 330.

If "flow min. factor" x t_{1small} is **larger than** t_2, the "flow min. factor" must be reduced until the above condition is fulfilled.

If "flow max. factor" x t_{1large} is **smaller than** t_2, the "flow max. factor" must be increased until the above condition is fulfilled.

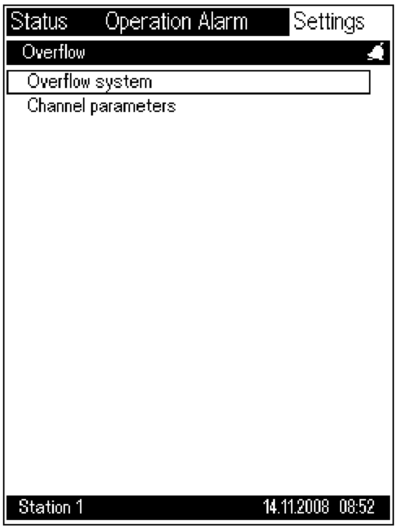
14.4 Overflow

This display is the opening display for overflow.

The display allows the user to select various subdisplays for configuration of overflow system and channel parameters.

Note: The display shown below should be considered as an example.

Path: Settings>Overflow>



Display_560

Description

Select from the list:

- Overflow system
- Channel parameters.

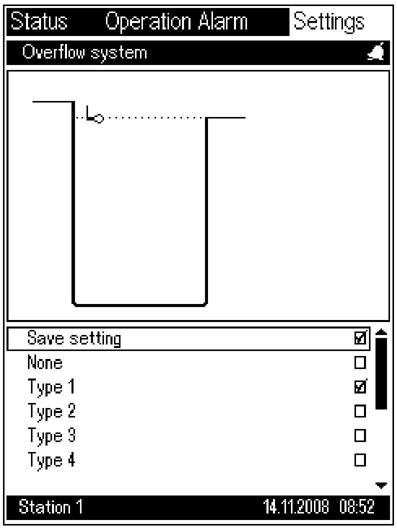
14.4.1 Overflow system

This display is used to select the type of overflow system.

Important: The selected overflow system will not operate until "Save setting" has been selected.

Note: The display shown below should be considered as an example.

Path: Settings>Overflow>Overflow system>



Display_561

Description

Select overflow system from the list:

- None (no overflow will be calculated).
- Type 1, see section *14.4.2 Overflow system 1*.
- Type 2, see section *14.4.3 Overflow system 2*.
- Type 3, see section *14.4.4 Overflow system 3*.
- Type 4, see section *14.4.5 Overflow system 4*.
- Type 5, see section *14.4.6 Overflow system 5*.
- Type 6, see section *14.4.7 Overflow system 6*.
- Type 7, see section *14.4.8 Overflow system 7*.
- Fault switch (optional), see section *14.4.9 Fault switch*.
  This function can be used for all overflow systems. If the fault switch is activated, the overflow calculation will be stopped because the overflow tank is full and there is a risk of backflow.

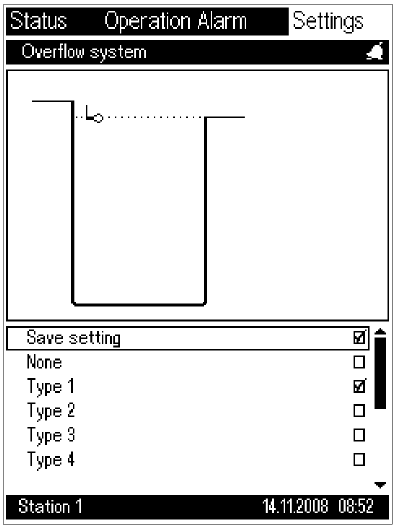
If overflow system 3, 4 or 5 is selected, the channel parameters must also be set, see section *14.4.10 Channel parameters*.

14.4.2 Overflow system 1

This display shows overflow system 1.

Note: The display shown below should be considered as an example.

Path: Settings**>**Overflow**>**Overflow system**>**



Display_561_1

Description

Overflow system 1 calculates:

- Number of overflows
- Overflow time.

Unit connected to the CU 401:

- High-level float switch (overflow float switch).

This overflow system is used if all levels are controlled

- by float switches or
- by an analog sensor and one overflow float switch.

If an analog sensor is used, the system will register an overflow as soon as the overflow float switch is activated. The overflow float switch can be placed at the level required. The activation of the overflow float switch does not result in a fault indication for the analog sensor.

Position of float switches

Contrary to the overflow float switch, the high-level float switch must be placed above the high level. See section *14.3.8 Level sensor*.

Example

Wastewater pit data	
Pit height	500 cm
High level	450 cm
Overflow level	400 cm

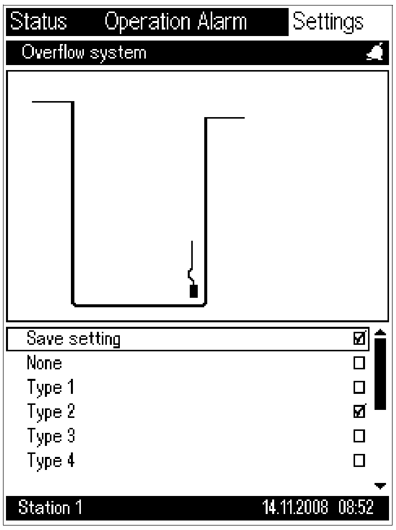
As shown in the table, the overflow level is lower than the high level. This means that the overflow pipe is placed in a position lower than the physical top of the wastewater pit. To ensure that the system works as intended, the high-level float switch must be placed above 450 cm.

14.4.3 Overflow system 2

This display shows overflow system 2.

Note: The display shown below should be considered as an example.

Path: Settings>Overflow>Overflow system>



Display_561_2

Description

Overflow system 2 calculates:

- Number of overflows
- Overflow time
- Overflow volume.

Units connected to the CU 401:

- Analog level sensor (pit) to AI1 input.
- Safety float switches (optional) to DI1 or DI2 input.

The overflow calculation is based on the inflow calculation and a signal from the analog level sensor which registers the overflow level. The overflow calculation must be activated when the level reaches the overflow level which is measured by the analog sensor.

The inflow must be constant in the overflow situation. The "Overflow time" is then used to calculate an overflow. This is a rough calculation. If a more accurate calculation is necessary, another overflow system should be chosen.

Formula for the volume calculation:

- latest measured inflow x time for overflow level measured by the analog sensor.

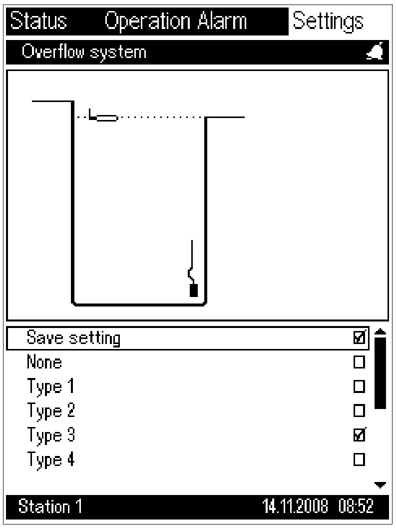
If a safety float switch is installed, this switch has no influence on the overflow calculation.

14.4.4 Overflow system 3

This display shows overflow system 3.

Note: The display shown below should be considered as an example.

Path: Settings>Overflow>Overflow system>



Display_561_3

Description

Overflow system 3 calculates:

- Number of overflows
- Overflow time
- Overflow volume.

Units connected to the CU 401:

- Analog level sensor (pit) to AI1 input
- Overflow float switch to DI3 input.

This overflow system has an overflow channel and indirectly measures the level in the channel via a signal from the analog level sensor which is installed in the pit.

The overflow float switch registers when there is an overflow and when the overflow measurement must stop. The signal from the overflow float switch activates the overflow alarm and starts the calculation of the level in the channel.

The overflow float switch must be installed in the right position to ensure that the level in the pit is calculated correctly.

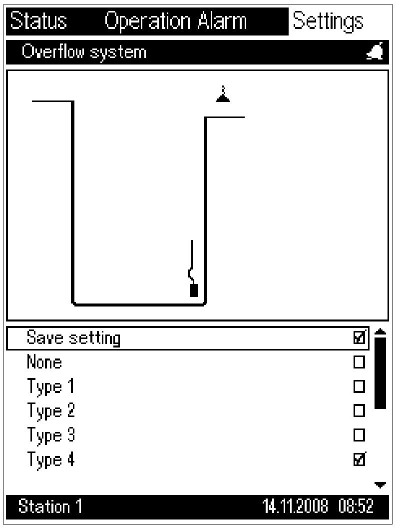
Note: The channel parameters must be set for this overflow system. See section *14.4.10 Channel parameters*.

14.4.5 Overflow system 4

This display shows overflow system 4.

Note: The display shown below should be considered as an example.

Path: Settings>Overflow>Overflow system>



Display_561_4

Description

Overflow system 4 calculates:

- Number of overflows
- Overflow time
- Overflow volume.

Units connected to the CU 401:

- Analog level sensor (pit) to AI1 input
- Analog level sensor (overflow) to AI2 input (level).

This overflow system has an overflow channel and measures the level directly in the channel via a separate analog level sensor.

The level sensor in the pit registers that there is an overflow. The level sensor in the channel registers the level in the channel.

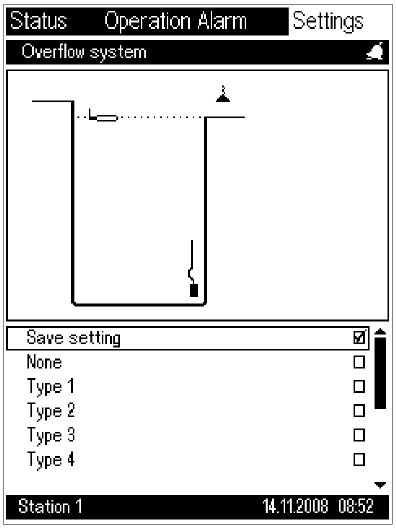
Note: The channel parameters must be set for this overflow system. See section *14.4.10 Channel parameters*.

14.4.6 Overflow system 5

This display shows overflow system 5.

Note: The display shown below should be considered as an example.

Path: Settings>Overflow>Overflow system>



Display_561_5

Description

Overflow system 5 calculates:

- Number of overflows
- Overflow time
- Overflow volume.

Units connected to the CU 401:

- Analog level sensor (pit) to AI1 input
- Analog level sensor (overflow) to AI2 input (level)
- Overflow float switch to DI3 input.

This overflow system has an overflow channel and measures the level directly in the channel via a separate analog level sensor.

The overflow float switch registers when there is an overflow and when the overflow measurement must stop. The signal from the overflow float switch activates the overflow alarm and starts the calculation of the level in the channel.

The overflow float switch must be installed in the right position to ensure that the level in the pit is calculated correctly.

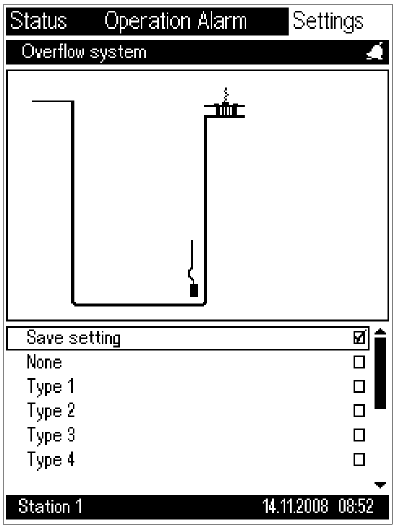
Note: The channel parameters must be set for this overflow system. See section *14.4.10 Channel parameters*.

14.4.7 Overflow system 6

This display shows overflow system 6.

Note: The display shown below should be considered as an example.

Path: Settings>Overflow>Overflow system>



Display_561_6

Description

Overflow system 6 calculates:

- Number of overflows
- Overflow time
- Overflow volume.

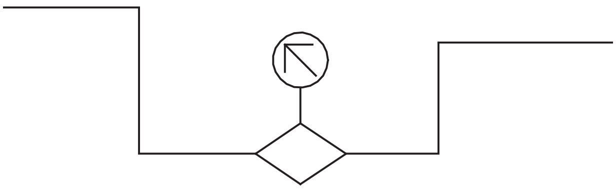
Units connected to the CU 401:

- Analog level sensor (pit) to AI1 input
- Analog flow sensor (overflow) to AI2 input (flow).

This overflow system measures the flow in the channel via a flow sensor which is installed in the channel. The level sensor registers that there is an overflow.

The flow sensor measures the current overflow.

The flowmeter must be installed so that it is always filled with water. See fig. 26.



TM03 6050 4306

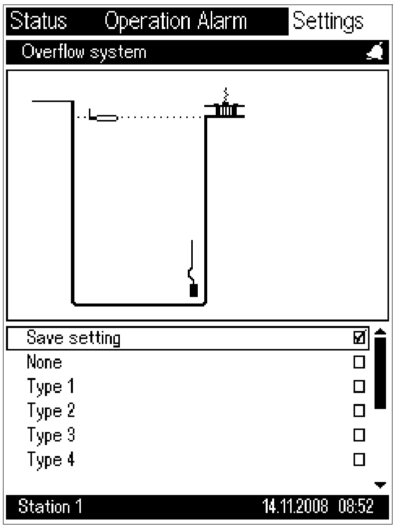
Fig. 26 Position of flowmeter

14.4.8 Overflow system 7

This display shows overflow system 7.

Note: The display shown below should be considered as an example.

Path: Settings>Overflow>Overflow system>



Display_561_4

Description

Overflow system 7 calculates:

- Number of overflows
- Overflow time
- Overflow volume.

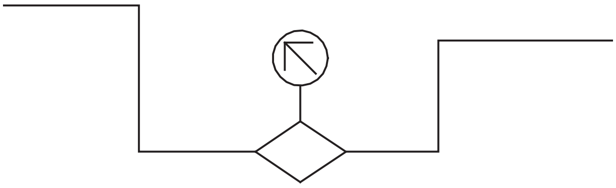
Units connected to the CU 401:

- Analog level sensor (pit) to AI1 input
- Analog flow sensor (overflow) to AI2 input (flow)
- Overflow float switch to DI3 input.

This overflow system measures the flow in the channel via a flow sensor which is installed in the channel.

The overflow float switch registers when there is an overflow and when the overflow measurement must stop. The signal from the overflow float switch activates the overflow alarm and starts the calculation of the level in the channel.

The overflow float switch must be installed in the right position to ensure that the level in the pit is calculated correctly.



TM03 6050 4306

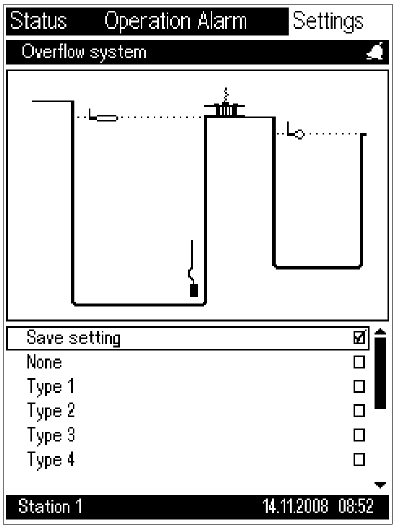
Fig. 27 Position of flowmeter

14.4.9 Fault switch

This display shows the fault switch (optional) in the overflow tank.

Note: The display shown below should be considered as an example.

Path: Settings>Overflow>Overflow system>



Display_561_9

Description

Unit connected to the CU 401:

- Fault switch to DI4 input (optional).

This function can be used for all overflow systems. If the fault switch is activated, the overflow calculation will stop.

The overflow channel is connected to a tank in which an additional sensor is installed. When the sensor is activated, the overflow measurement must stop because the overflow tank is full and there is a risk of backflow.

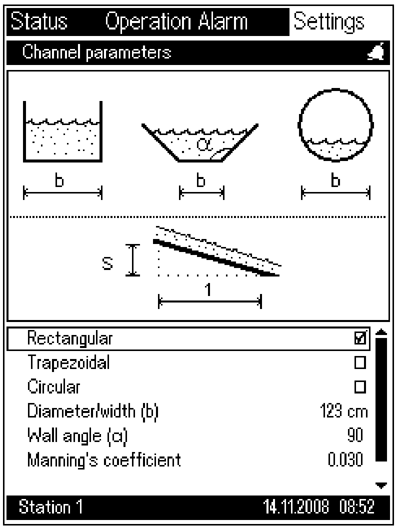
The overflow tank is full when the fault switch is activated. Then the calculation of the flow in the overflow channel will not be correct.

14.4.10 Channel parameters

This display is used for the configuration of the overflow parameters for the overflow systems 3, 4 and 5.

Note: The display shown below should be considered as an example.

Path: Settings>Overflow>Channel parameters>



Display_562

Description

Select the form of the overflow channel:

- Rectangular
- Trapezoidal
- Circular (pipe).

Enter the other channels parameters:

- Diameter/width (b): Diameter/width in cm.
- Wall angle (α): Angle in degrees (90 ° corresponds to a rectangular form).
- Manning's coefficient: Coefficient for the surface finish of the overflow channel, see table below.
- Channel slope (S): Slope in percent (100 % corresponds to a slope of 45 °).

Material	Manning's coefficient
Natural streams	
Clean and straight	0.030
Major rivers	0.035
Sluggish with deep pools	0.040
Excavated earth channels	
Clean	0.022
Gravelly	0.025
Weedy	0.030
Stony, cobbles	0.035
Flood plains	
Pasture, farmland	0.035
Light brush	0.050
Heavy brush	0.075
Trees	0.15
Metals	
Brass	0.011
Cast iron	0.013
Smooth steel	0.012
Corrugated metal	0.022
Non-metals	
Glass	0.010
Clay tile	0.014
Brickwork	0.015
Asphalt	0.016
Masonry	0.025
Finished concrete	0.012
Unfinished concrete	0.014
Gravel	0.029
Earth	0.025
Planed wood	0.012
Unplaned wood	0.013

Material	Manning's coefficient
PE and PVC	
Corrugated polyethylene (PE) with smooth inner walls	0.009-0.015
Corrugated polyethylene (PE) with corrugated inner walls	0.018-0.025
Polyvinyl chloride (PVC) with smooth inner walls	0.009-0.011

Note: The values of the Manning's coefficient should be used as a guide only. For more exact values, the actual flow should be verified against other measuring methods, such as a controlled overflow of an exact amount.

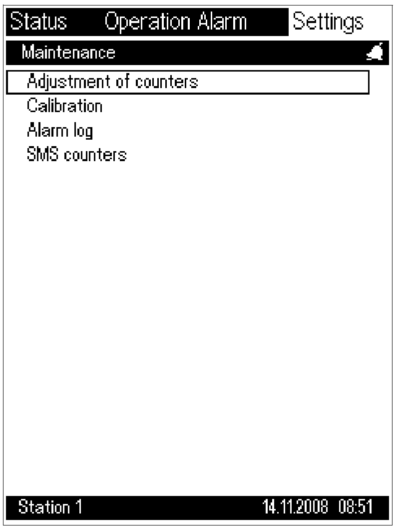
14.5 Maintenance

This is the opening display for maintenance.

This display allows the user to select various subdisplays for maintenance.

Note: The display shown below should be considered as an example.

Path: Settings**>**Maintenance**>**



Display_335

Description

Select from the list:

- Adjustment of counters
- Calibration
- Alarm log
- SMS counters.

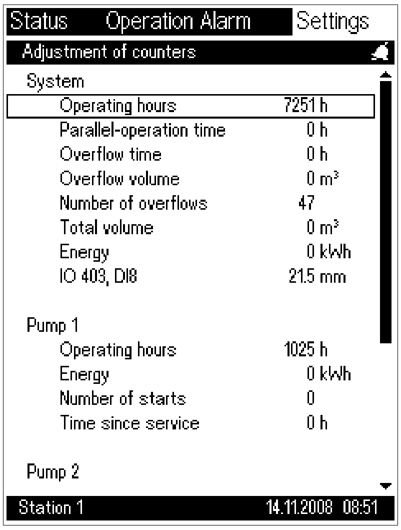
14.5.1 Adjustment of counters

This display allows the user to set the start- and energy-counter values. These values are useful for pump replacement purposes.

Example: Pump "A" is taken out of operation at 350 operating hours and 700 starts.
These values are stored in the service log. The pump is replaced by a repaired pump with 250 operating hours and 800 starts; after the user has entered these values, the system continues to count from the new values.

Note: The display shown below should be considered as an example.

Path: Settings>Maintenance>Adjustment of counters>



Display_350

Description

Select from the list:

- System:
  - Operating hours
  - Parallel-operation time
  - Overflow time
  - Overflow volume
  - Number of overflows
  - Total volume
  - Energy
  - IO 403, DI8.
- Pump:
  - Operating hours
  - Energy
  - Number of starts
  - Time since service.
- Mixer (appears only when the mixer is enabled. See section *14.7.7 CU 401, DO7 relay output*):
  - Operating hours
  - Number of starts
  - Time since service.

Description of display texts

Display text	Function
System:	
Operating hours	Number of hours the CU 401 has been in operation (can be modified if another CU 401 control unit is installed).
Parallel-operation time	Accumulated hours during which more than one pump has been running.
Overflow time	Period during which there has been an overflow.
Overflow volume	Number of m^3 which ran out.
Number of over-flows	Total number of overflows.
Total volume	Total volume removed/pumped by all the pumps.
Energy	Accumulated energy measured in kWh. **Note:** This option requires the fitting of an energy meter in the supply cable to the pump.
IO 403, DI8	If, for example, a rainwater gauge is connected, the amount of rainwater measured will be displayed. **Note:** The user has to select the unit to be used. In this example, mm was selected. The text "IO 403, DI8" can be changed into "Rainwater gauge". See section *14.7.9 IO 403, I/O settings*.
Pumps:	
Operating hours	Number of hours the pump has been running (can be modified if another pump is installed).
Energy	Accumulated energy measured in kWh. **Note:** This option requires the fitting of an energy meter in the supply cable to the pump, e.g. an MP 204.
Number of starts	Number of pump starts since installation/connection (can be modified if another pump is installed).
Time since service	Number of hours since last service of the pump (can be reset by Grundfos Service).
Mixer:	
Operating hours	Number of hours the mixer has been running (can be modified if another mixer is installed).
Number of starts	Number of mixer starts since installation/connection (can be modified if another mixer is installed).
Time since service	Number of hours since last service of the mixer (can be reset by Grundfos Service).

14.5.2 Calibration

This display is used for calibrating the water-in-oil sensor of each individual pump. Indicate the relevant sensor by a check mark, and the sensor will be calibrated.

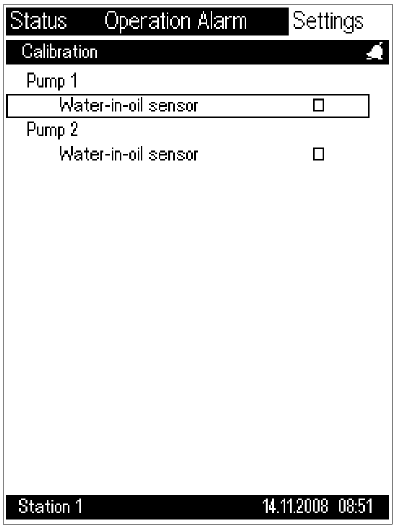
The signal will be calibrated so that the measured mA gives 0 % water in oil (offset). This calibration should be made when a new pump is installed or a pump is replaced.

Note: 20 mA does not correspond to 20 % water in oil after calibration.

The IO 111 signal converter box for SE pumps has the same function.

Note: The display shown below should be considered as an example.

Path: Settings>Maintenance>Calibration>



Display_351

Description

Select from the list:

- Pump number:
  - Water-in-oil sensor.

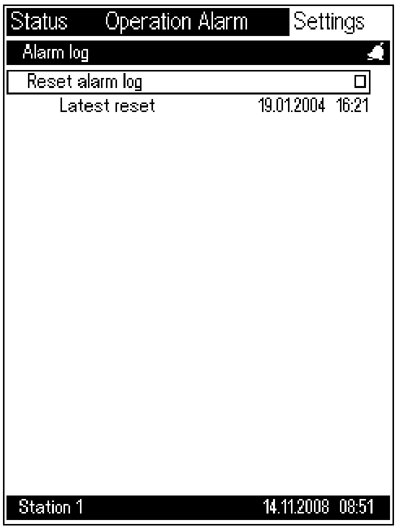
14.5.3 Alarm log

This display is used for resetting the entire alarm log.

Date and time for last reset is shown in the display.

Note: The display shown below should be considered as an example.

Path: Settings>Maintenance>Alarm log>



Display_352

Description

Select from the list:

- Reset alarm log.

Note: If the alarm log is reset, the warranty will be invalidated.

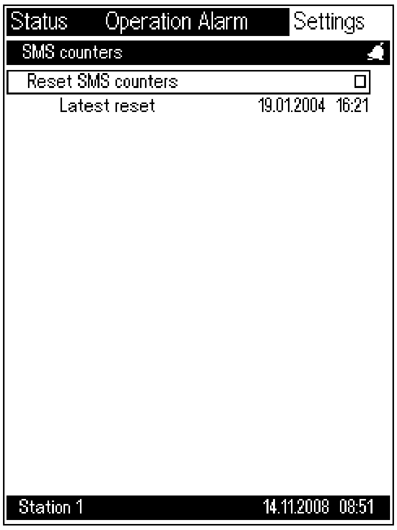
14.5.4 SMS counters

This display is used for resetting the number of sent SMSes.

Date and time for last reset is shown in the display.

Note: The display shown below should be considered as an example.

Path: Settings>Maintenance>SMS counters>



Display_353

Description

Select from the list.

- Reset SMS counter.

14.6 Pump out of operation

This display allows the user to take pumps temporarily out of operation for service overhaul or on account of disturbances of operation.

When taken out of operation, the pump is cancelled from the list of pumps capable of starting. However, the system will attempt to continue operation with the remaining pumps.

Example: If pump 3 is taken out of operation in a three-pump system using alternation, the alternation between the remaining two pumps continues. No matter which of the three pumps is taken out of operation, the remaining pumps will run according to the start/stop levels of pumps 1 and 2. This process will continue until alternation is deselected.

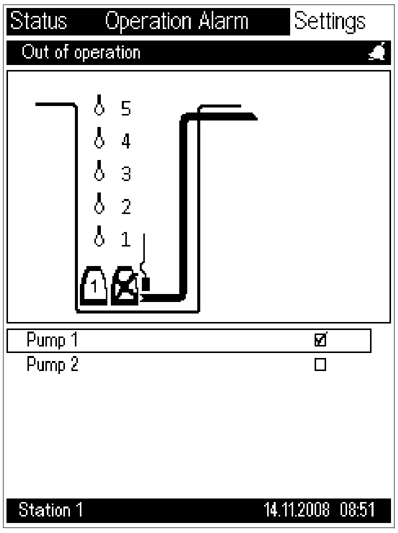
If alternation is deselected, the start/stop values of the pumps will become active. See section *14.3.8 Level sensor*.

When a pump is taken out of operation, none of the alarms selected for the pump will be indicated. Thus, it is an advantage to take a (defective) pump out of operation to avoid superfluous fault indications on the OD 401/SCADA system.

Note: For safety reasons, all pumps are out of operation when a new program card is installed.

Note: The display shown below should be considered as an example.

Path: Settings>Pump out of operation>



Display_345

Description

The display example to the left displays the following operational state:

- Pump 1 running
- Pump 2 out of operation.

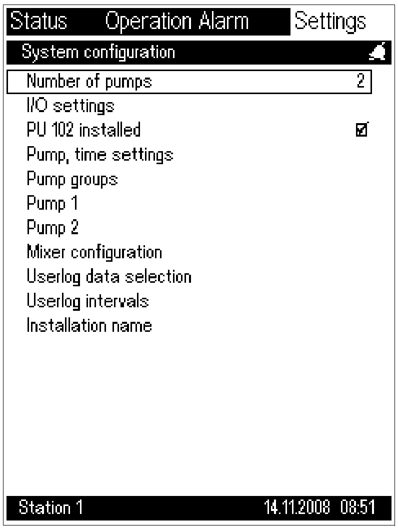
14.7 System

14.7.1 System configuration

This display allows the user to define the main components of the pit installation, such as number of pumps, pump type, frequency converter and operating mode, e.g. alternating operation. For further information, see the various subdisplays.

Note: The display shown below should be considered as an example.

Path: Settings>System>



Display_370

Description

Select:

- "Number of pumps", and enter the number of pumps in the system.*
- "I/O settings" for setting of module inputs and outputs.
- "PU 102 installed" to automatically detect or insert the PU 102.**
- "Pump, time settings" to move to the "Pump, time settings" display.
- "Pump groups" to move to the "Group configuration" display.
- "Pump 1", "Pump 2", etc. to move to the display "Pump 1, configuration", "Pump 2, configuration", etc.
- "Mixer configuration" to move to the "Mixer configuration" display.
- "Userlog data selection" to move to the "Data to log" display.
- "Userlog intervals" to move to the "Userlog intervals" display.
- "Installation name" to move to the "Installation name" display.

* The number of pumps parameter affects a number of other setting options and must be set as one of the first parameters. Control by means of float switches can only be made with up to two pumps.
Systems with three or more pumps require a analog level sensor, e.g. a pressure sensor.

** The PU 102 (battery back-up/UPS) is deselected in the factory settings. However, the system automatically detects whether it is connected the first time the battery is fully loaded. Consequently, a modification of this setting is only required if the PU 102 is removed again from the system. This must be done to prevent incorrect alarms.

14.7.2 I/O settings

This display is used to select the input or output to be set.

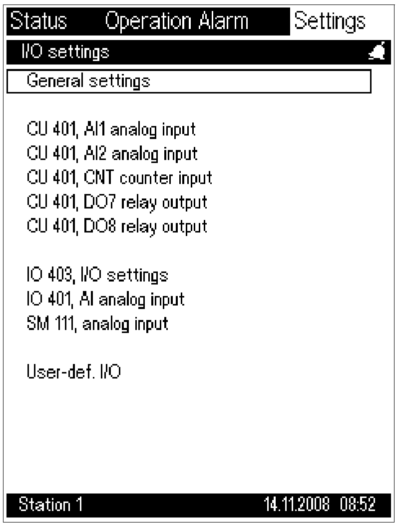
Note: A pressure sensor or other level sensor must be connected to the AI1 analog input on the CU 401. A volume measuring device or energy meter can be connected to

- the CU 401, AI2 analog input or
- the CU 401, CNT counter input.

Note: A possible rainwater gauge or other additional equipment can be connected to the IO 403 inputs and outputs.

Note: The display shown below should be considered as an example.

Path: Settings>System>I/O settings>



Display_550

Description

Select the input or output to be set:

- General settings
- CU 401, AI1 analog input
- CU 401, AI2 analog input
- CU 401, CNT counter input
- CU 401, DO7 relay output
- CU 401, DO8 relay output
- IO 403, I/O settings
- IO 401, AI analog input
- User-defined I/O.

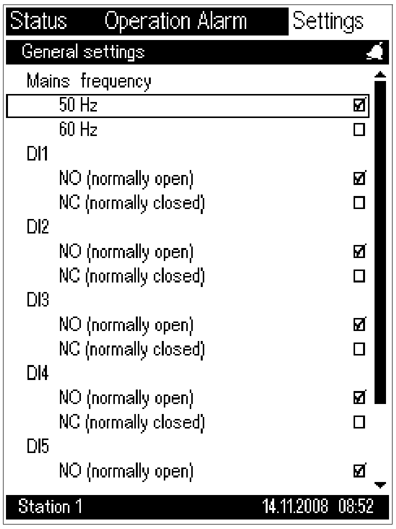
14.7.3 General settings

This display is used for entering the electrical function of the float switches and the mains frequency of the system.

The frequency compensates for mains interference on the analog inputs and must therefore be set. The factory setting is 50 Hz.

Note: The display shown below should be considered as an example.

Path: Settings**>**System**>**I/O settings**>**General settings**>**



Display_520

Description

Select mains frequency and type of float switches.

- Mains frequency:
  - 50 Hz (factory setting)
  - 60 Hz.

Select type of float switch:

- DI1:
  - NO (normally open contact) (factory setting)
  - NC (normally closed contact).
- DI2:
  - NO (normally open contact) (factory setting)
  - NC (normally closed contact).
- DI3:
  - NO (normally open contact) (factory setting)
  - NC (normally closed contact).
- DI4:
  - NO (normally open contact) (factory setting)
  - NC (normally closed contact).
- DI5:
  - NO (normally open contact) (factory setting)
  - NC (normally closed contact).
- DI6:
  - NO (normally open contact) (factory setting)
  - NC (normally closed contact).

14.7.4 CU 401, AI1 analog input

This display is used for setting the AI1 analog input, intended for level measurement in the pit.

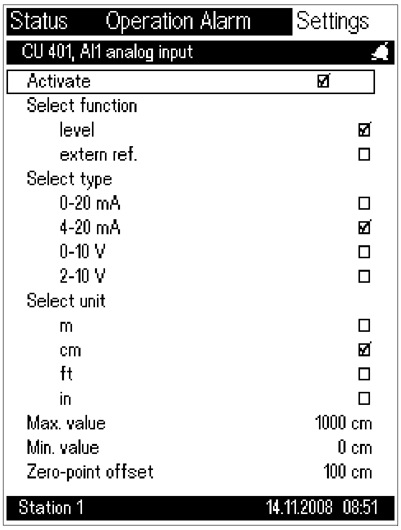
All level measuring units are useful as long as the unit output can be set to one of four options. The level pressure sensor normally shows zero before being immersed.

Example 1: The level pressure sensor is positioned 30 cm above the bottom of the pit. The measuring range is 5 metres and the output 4-20 mA. "Min. value" is set to 30 cm and "Max. value" to 530 cm.

Example 2: An ultrasonic level sensor is positioned 550 cm above the bottom of the pit. The measuring range is 10 metres, and the output is set to 4-20 mA, where 4 mA corresponds to zero metre and 20 mA corresponds to 10 m. "Max. value" is set to 0 and "Min. value" to 1000 cm.
Zero-point offset is set to 1000-550 = 450 cm.

Note: The display shown below should be considered as an example.

Path: Settings**>**System**>**I/O settings**>**CU 401, AI1 analog input**>**



Display_362

Description

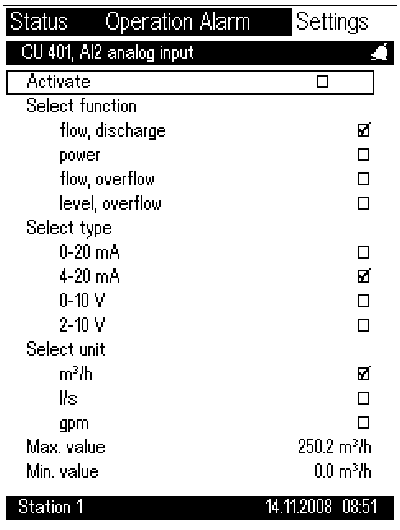
- Select the function of the AI1 analog input:
  - level
  - external ref.
- Select type of standard signal:
  - 0-20 mA
  - 4-20 mA (factory setting)
  - 0-10 V
  - 2-10 V.
- Select unit/indication:
  - m
  - cm (factory setting)
  - ft
  - in.
- Max. value
- Min. value
- Zero-point offset.

14.7.5 CU 401, AI2 analog input

This display is used for setting the AI2 analog input intended for flow measurement, power measurement, overflow level sensor or overflow volume. The setting is adapted to the output signal from the transmitter (meter) connected.

Note: The display shown below should be considered as an example.

Path: Settings>System>I/O settings>CU 401, AI2 analog input>



Display_363

Description

- Select the function of the AI2 analog input:
  - flow, discharge
  - power
  - flow, overflow
  - level, overflow.
- Select type of standard signal:
  - 0-20 mA
  - 4-20 mA (factory setting)
  - 0-10 V
  - 2-10 V.
- Select unit/indication:
  - m^3/h (factory setting)
  - l/s
  - gpm.
- Select:
- Max. value
- Min. value.

14.7.6 CU 401, CNT counter input

This display is used for setting the CNT counter input function. An energy meter or flowmeter with pulse output can be connected. The inputs have internal pull-up resistor; they are designed for connection of relay or open collector outputs.

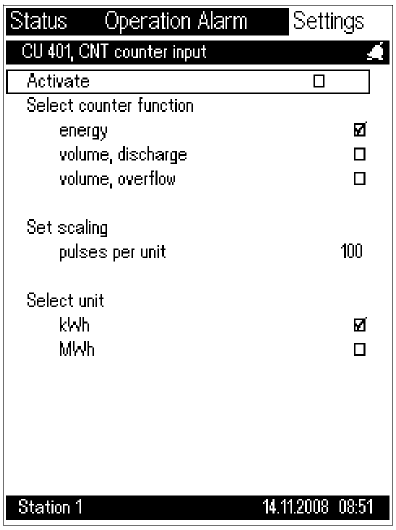
Input voltage: 5 V (activated)

Min. pulse width: 1 mSec.

Measuring range: Max. 999 pulses/second.

Note: The display shown below should be considered as an example.

Path: Settings>System>I/O settings>CU 401, CNT counter input>



Display_361

Description

- Select the function of the CNT counter input:
  - energy
  - volume, discharge (from flowmeter)
  - volume, overflow.
- Set scaling:
  - pulses per unit.
- Select unit:
  – m^3
  – l
  – gal
  – kWh
  – MWh.

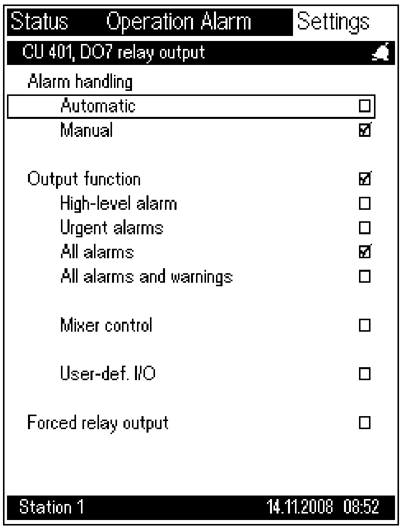
14.7.7 CU 401, DO7 relay output

This display is used for setting the functions of the DO7 relay output of the CU 401.

Note: The physical connection of the relay must match the settings made in this display.

Note: The display shown below should be considered as an example.

Path: Settings**>**System**>**I/O settings**>**CU 401, DO7 relay output**>**



Display_511

Description

- Select type of alarm handling:
  - Automatic
  - Manual.
- Select output function:
  - High-level alarm
  - Urgent alarms
  - All alarms
  - All alarms and warnings
  - Mixer control
  - User-defined I/O.
- Forced relay output.

Description of display texts

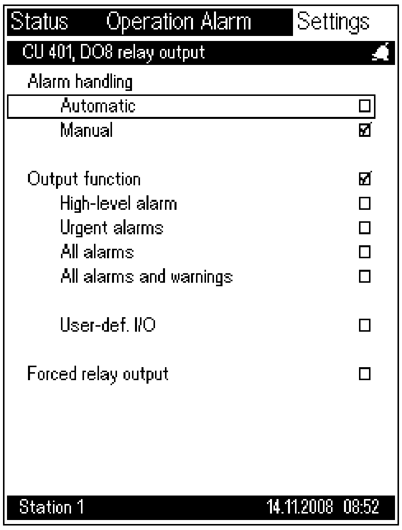
Display text	Function
Automatic	The relay will automatically be deactivated when all alarms are below the alarm limits.
Manual	The relay will be deactivated when the reset button is pressed or the alarms are reset on the operator display. See section *12.1.3 Operation - resetting alarm indicators and alarm relays*. **Note:** All alarms must be below the alarm limits.
High-level alarm	A high level detected by float switches or level sensor will activate the DO7 relay output.
Urgent alarms	The following alarms will activate the DO7 relay output: • high level • alarm level • dry-running level • mains supply, off • missing phase • wrong phase sequence • overflow.
All alarms	All alarms will activate the DO7 relay output.
All alarms and warnings	All alarms and warnings will activate the DO7 relay output.
Mixer control	When enabled, the DO7 relay output will be used as a start/stop function for the mixer.
User-defined I/O	Output controlled by "User IO". See section *5.12 User-defined I/O*.
Forced relay output	When enabled, the DO7 relay output will be activated. This is used for testing equipment connected to the DO7, e.g. mixer or alarm horn.

14.7.8 CU 401, DO8 relay output

This display is used for setting the functions of the DO8 relay output of the CU 401.

Note: The display shown below should be considered as an example.

Path: Settings>System>I/O settings>CU 401, DO8 relay output>



Display_512

Description

- Select type of alarm handling:
  - Automatic
  - Manual.
- Select output function:
  - High-level alarm
  - Urgent alarms
  - All alarms
  - All alarms and warnings
  - User-defined I/O.
- Forced relay output.

Description of display texts

Display text	Function
Automatic	The relay will automatically be deactivated when all alarms are below the alarm limits.
Manual	The relay will be deactivated when the reset button is pressed or the alarms are reset on the operator display. See section *12.1.3 Operation - resetting alarm indicators and alarm relays*. **Note:** All alarms must be below the alarm limits.
High-level alarm	The high-level alarm detected by float switches or level sensor will activate the DO8 relay output.
Urgent alarms	The following alarms will activate the DO8 relay output: • high level • alarm level • dry-running level • mains supply, off • missing phase • wrong phase sequence • overflow.
All alarms	All alarms will activate the DO8 relay output.
All alarms and warnings	All alarms and warnings will activate the DO8 relay output.
User-defined I/O	Output controlled by "User IO". See section *5.12 User-defined I/O*.
Forced relay output	When enabled, the DO8 relay output will be activated. This is used for testing equipment connected to the DO8, e.g. alarm horn.

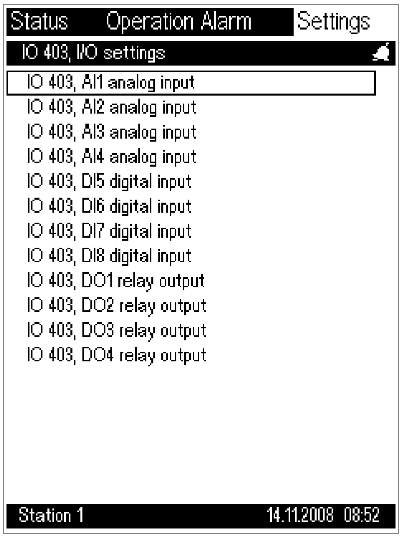
14.7.9 IO 403, I/O settings

This display is used for setting the inputs and outputs of the IO 403.

The user can select a suitable name for the individual input and output. Enter the name using the keypad as described in section *14.7.28 Installation name*.

Note: The display shown below should be considered as an example.

Path: Settings>System>I/O settings>IO 403, I/O settings>



Display_570

Description

Select the input or output to be set:

- IO 403, AI1 analog input
- IO 403, AI2 analog input
- IO 403, AI3 analog input
- IO 403, AI4 analog input
- IO 403, DI5 digital input
- IO 403, DI6 digital input
- IO 403, DI7 digital input
- IO 403, DI8 digital input
- IO 403, DO1 relay output
- IO 403, DO2 relay output
- IO 403, DO3 relay output
- IO 403, DO4 relay output.

14.7.10 Analog inputs

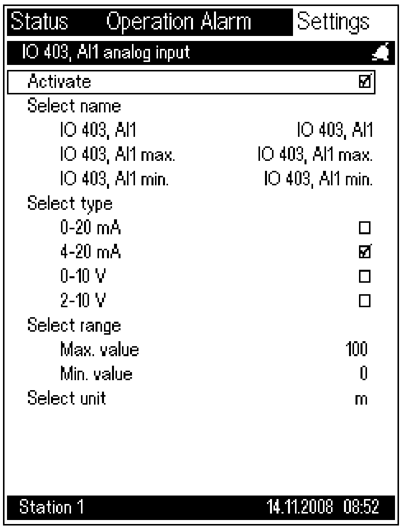
This display is used for setting the function of the individual analog input.

Each of the four analog inputs can be configured according to the user's wishes.

The user can change the name of the input by selecting "Select name". The corresponding alarm values "AI1 max." and "AI1 min." can also be changed. See section *14.2.1 System alarms and warnings - enable/disable*. The example below shows how the AI1, AI2, AI3 and AI4 inputs are configured.

Note: The display shown below should be considered as an example.

Path: Settings**>**System**>**I/O settings**>**IO 403, I/O settings**>**IO 403, AI1 analog input**>**



Display_571

Description

Select the name of the desired analog input (the example shows the AI1 input):

- IO 403, AI1
- IO 403, AI1 max.
- IO 403, AI1 min.

Select type of standard signal:

- 0-20 mA
- 4-20 mA (factory setting)
- 0-10 V
- 2-10 V.

Select range:

- Max. value
- Min. value.

Select unit:

- Enter the desired unit via the unit keypad.

Analog inputs

Terminal designation	AI1	AI2	AI3	AI4
Voltage range	0-10 V / 2-10 V	0-10 V / 2-10 V	0-10 V / 2-10 V	0-10 V / 2-10 V
Current range	0-20 mA / 4-20 mA	0-20 mA / 4-20 mA	0-20 mA / 4-20 mA	0-20 mA / 4-20 mA
Supply sensors	+24 V / 25 mA	+24 V / 25 mA	External supply required	External supply required

For further electrical data, see installation and operating instructions for the IO 403.

Note: Only two 0-20 mA or 4-20 mA transmitters can be powered by the IO 403 module. If more transmitters are to be used, they must be powered via an external 24 V supply.

14.7.11 Digital inputs

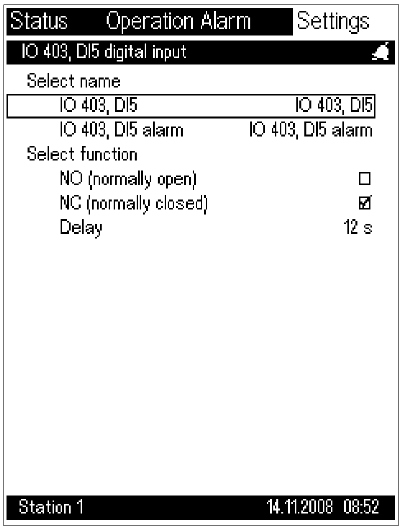
This display is used for setting the function of the individual digital input.

Each of the four digital inputs can be configured according to the user's wishes. The inputs are factory-named IO 403, DI5 to DI8. The DI8 input can be configured so that it can be used as a pulse counter and is described under *14.7.12 Pulse counter input DI8*.

The user can change the name of the input by selecting "Select name". The corresponding alarm name "DI5 alarm" can also be changed. See section *14.2.1 System alarms and warnings - enable/disable*. The example below shows how the DI5, DI6 and DI7 inputs are configured.

Note: The display shown below should be considered as an example.

Path: Settings**>**System**>**I/O settings**>**IO 403, I/O settings**>**IO 403, DI5 digital input**>**



Display_575

Description

Select the name of the desired digital input (the example shows the DI5 input):

- IO 403, DI5
- IO 403, DI5 alarm.

Select the function of the connected equipment:

- NO (normally open)
- NC (normally closed)
- Delay.

	Digital inputs		
Terminal designation	DI5	DI6	DI7
Output voltage	24 V	24 V	24 V
Output current	5 mA	5 mA	5 mA

For further electrical data, see installation and operating instructions for the IO 403.

14.7.12 Pulse counter input DI8

The DI8 input can be configured so that it can be used as a pulse counter. To ensure the correct function of the connected equipment, it is important to select the right pulse counter function.

Two different pulse counter functions are available:

- Units per pulse
- Pulses per unit.

Units per pulse

"Units per pulse" is used in the case of transmitters which emit a pulse when a certain quantity has been measured, e.g. 0.2 mm rainwater. Enter the quantity per pulse in the "Value" line. The decimal figure is entered via a keypad which only contains figures and a comma. Click [OK] in the "Value" line to activate the keypad.

Pulses per unit

"Pulses per unit" is used in the case of transmitters which emit pulses continuously while they measure a quantity, e.g. 1000 pulses per kWh. Enter the number of pulses for each quantity unit in the "Value" line.

Example:

A rainwater gauge is connected to the DI8 input.

Rainwater gauge data: One pulse corresponds to 0.2 mm rainwater.

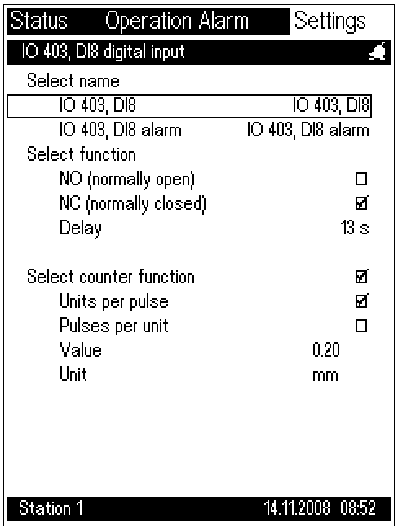
In this example, "Units per pulse" was selected.

To ensure a correct measurement of the rainwater quantity, the DI8 input must be configured. Enter the amount of rainwater per pulse [0.2].

The user can change the name of the input by selecting "Select name". The corresponding alarm name "DI8 alarm" can also be changed. See section *14.2.1 System alarms and warnings - enable/disable*. The example below shows how the DI8 input is configured.

Note: The display shown below should be considered as an example.

Path: Settings>System>I/O settings>IO 403, I/O settings>IO 403, DI8 digital input>



Display_584

Description

Select the name of the DI8 input used as a pulse counter input:

- IO 403, DI8
- IO 403, DI8 alarm.

Select the function of the connected equipment:

- NO (normally open)
- NC (normally closed)
- Delay.

Select counter function:

- Units per pulse
- Pulses per unit
- Value
- Unit.

Note: Select either "Units per pulse" or "Pulses per unit".

	Digital input
Terminal designation	DI8
Output voltage	5 V
Output current	5 mA

For further electrical data, see installation and operating instructions for the IO 403.

14.7.13 Relay outputs

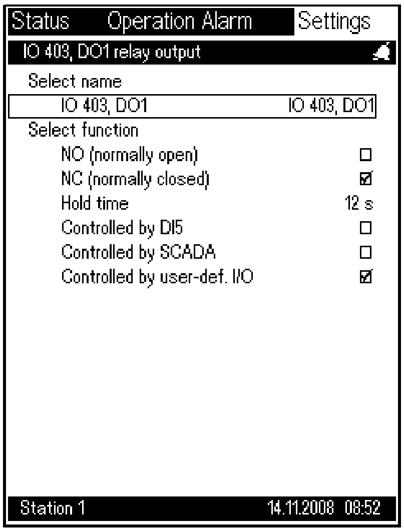
This display is used for setting the function of the individual relay output.

Each of the four outputs can be configured according to the user's wishes.

The user can change the name of the output by selecting "Select name". The example below shows how the DO1, DO2, DO3 and DO4 outputs are configured.

Note: The display shown below should be considered as an example.

Path: Settings**>**System**>**I/O settings**>**IO 403, I/O settings**>**IO 403, DO1 relay output**>**



Display_591

Description

Select the name of the desired relay output (the example shows the DO1 output):

- IO 403, DO1.

Select the function of the desired relay:

- NO (normally open) (factory setting)
- NC (normally closed)
- Hold time
- Controlled by DI5
- Controlled by SCADA
- Controlled by user-defined I/O.

	Relay outputs			
Terminal designation	DO1	DO2	DO3	DO4
Controller	DI5 / SCADA	DI6 / SCADA	DI7 / SCADA	DI8 / SCADA
Maximum load	400 VAC, 2 A	400 VAC, 2 A	400 VAC, 2 A	400 VAC, 2 A
Maximum load power	400 VA	400 VA	400 VA	400 VA

For further electrical data, see installation and operating instructions for the IO 403.

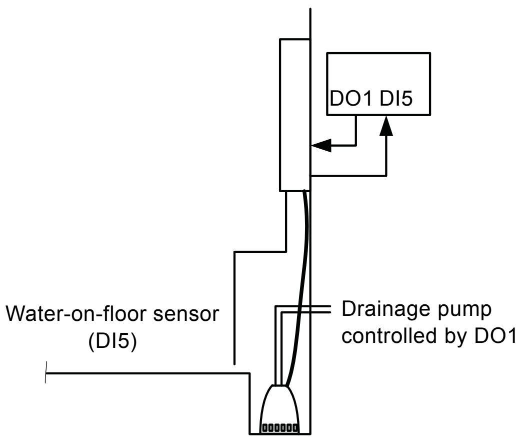


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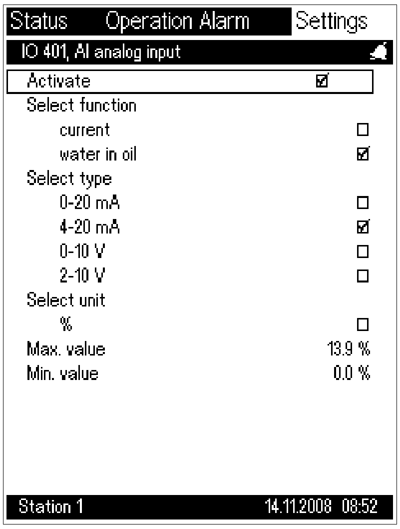
Fig. 28 Example of installation with sensor and drainage pump where DI5 has been set to control DO1

14.7.14 IO 401, AI analog inputs

This display is used for setting the AI inputs of the IO 401.
This setting applies to all IO 401 modules.

Note: The display shown below should be considered as an example.

Path: Settings>System>I/O settings>IO 401, AI analog input>



Display_541

Description

Select the function of the AI analog input:

- current (factory setting)
- water in oil.

Select type of standard signal:

- 0-20 mA
- 4-20 mA (factory setting)
- 0-10 V
- 2-10 V.

Select unit:

- % (water in oil)
- A
- mA.

Select:

- Max. value
- Min. value.

14.7.15 SM 111, analog input

This display is used to set the function of the additional analog input on the SM 111.

The range used for scaling the sensor input is selected with the up and down arrows. The unit is selected via the unit keypad.

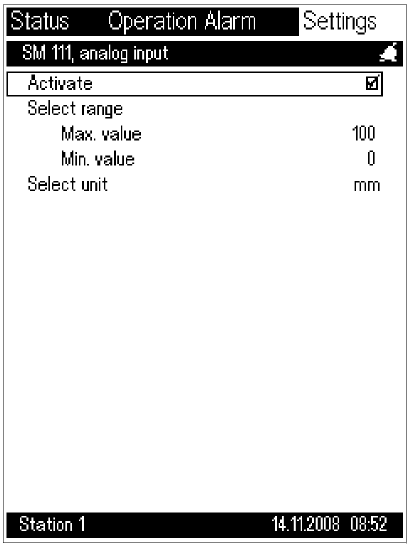
The example below shows how to configure the input.

Note: Alarm and warning monitoring is not available for this input.

Note: For configuration of the SM 111 and IO 111, see section *5.10 SM 111 and IO 111*.

Note: The display shown below should be considered as an example.

Path: Settings**>**System**>**I/O settings**>**SM 111, analog input**>**



Display_542

Description

Select the SM 111 analog input.

- Activate by putting a check mark in the check box.

Select range:

- Max. value
- Min. value.

Select unit:

- Enter the desired unit via the unit keypad.

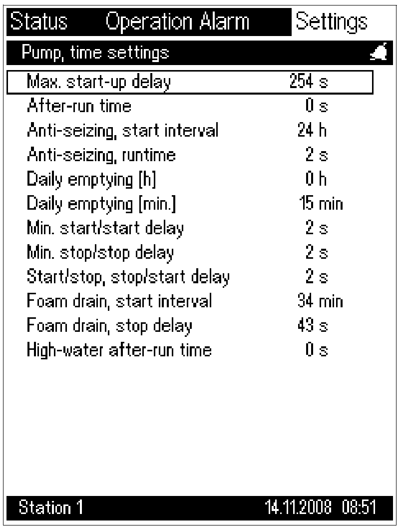
14.7.16 Pump - time settings

This display is used for special pump time settings.

The settings in this display will only rarely need to be altered.

Note: The display shown below should be considered as an example.

Path: Settings>System>Pump, time settings>



Display_371

Description

Make the desired pump settings.

- Max. start-up delay
- After-run time
- Anti-seizing, start interval
- Anti-seizing, runtime
- Daily emptying [h]
- Daily emptying [min.]
- Min. start/start delay
- Min. stop/stop delay
- Start/stop, stop/start delay
- Foam drain, start interval
- Foam drain, stop delay
- High-water after-run time.

Description of display texts

Display text	Function
Max. start-up delay	The time from the system is switched on until the first pump is allowed to start. This random time varies between zero and the maximum time entered.
	Factory setting: 10 seconds.
After-run time	The time from the sensor stop signal is given until the pump actually stops.
	Factory setting: 0 seconds.
Anti-seizing, start interval	The interval between forced pump starts.
	Factory setting: 24 hours.
Anti-seizing, runtime	Operating time when the pump has made a forced start.
	Factory setting: 2 seconds.
Daily emptying [h]	The time of the day when the pump is to start. In major systems, daily emptying can be used for mainly night-time operation when the electricity is often cheaper. Enter number of hours.
	Factory setting: --- h.
Daily emptying [min.]	The time of the day when the pump is to start. In major systems, daily emptying can be used for mainly night-time operation when the electricity is often cheaper. Enter number of minutes.
	Factory setting: --- min.
Min. start/start delay	The minimum time between starts.
	Factory setting: 2 seconds.
Min. stop/stop delay	The minimum time between two pump stops.
	Factory setting: 2 seconds.
Start/stop, stop/start delay	The minimum time between the start of pump X and the stop of pump Y, and vice versa. This delay has higher priority than "min. start/start delay" and "min. stop/stop delay".
	Factory setting: 2 seconds.
Foam drain, start interval *	The interval between the times when the pit has been totally drained. The purpose of the foam drain setting is to pump below the normal stop level to remove the remaining liquid in the pit. The pump runs until it starts to make a slurping sound. This setting should only be made if the pumps are designed to withstand air in the hydraulic chamber. The time is set in minutes. The upper limit is 65.534 minutes.
	Factory setting: --- minutes.
Foam drain, stop delay	The stop delay running from the time when the foam drain level has been reached. Use this function if the foam drain level is below the level sensor in the pit. The time is set in seconds.
	Factory setting: 10 seconds.

Display text	Function
High-water after-run time	To prevent overflow in case of a defective level sensor, a float switch can be installed at the top of the pit. If the float switch is enabled, all pumps will start (the number of pumps can be set in the group configuration display. See section *14.7.19 Group properties*). The subsequent pumping time is called the "high-water after-run time". The length of this time is best found by trial and error in the actual pit. If the pit also has a dry-running float switch, the pump may be allowed to pump to the dry-running float switch level. This emergency operation situation is maintained until the sensor is replaced. The sensor replacement is confirmed in the alarm list display. The time is set in seconds. **Note:** The conflicting-levels alarm must be enabled for a fault to be displayed in the alarm list. The alarm must be reset to return to normal sensor operation.
	Factory setting: 30 seconds.

* **Note:** Do not enable the foam drain setting unless the pumps can withstand dry running. Enabling foam drain causes the dry-running alarm to be disabled; consequently, the pumps will continue to run despite a defective pressure sensor.

Note: The foam drain level must be above the lower pump suction level.

14.7.17 Group configuration - alternation/advanced alternation

This display allows the user to select pump group properties and alternation between the pump groups.

The pumps connected can belong to one of three groups. There is a minimum of one pump group, if no other number has been selected.

Alternation can be enabled or disabled for the individual groups.

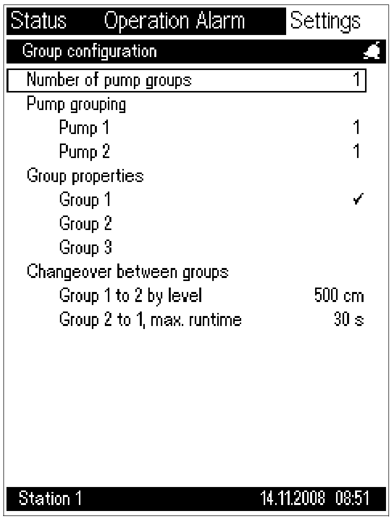
Group 1 is used if the pumps are identical and alternation between the pumps is required.

Group 2 is used if the system consists of small and large pumps. The small pumps are placed in group 1 and the large pumps in group 2.

Group 3 is used for pumps not allowed to start/stop individually. **Example:** Three identical pumps, where pumps 1 and 2 have soft start. Pump 3 uses direct start. The system is provided with a small pressure tank. If pump 3 starts individually, it will cause a start pressure impact. By allocating pumps 1 and 2 to group 1, and pump 3 to group 3, the system will ensure that pumps 1 and 2 run when pump 3 is started/stopped.

Note: The display shown below should be considered as an example.

Path: Settings>System>Pump groups>



Display_390

Description

Select the location of the pumps, number of groups and properties of the individual groups.

Pump grouping:

- Pump 1
- Pump 2.

Group properties:

- Group 1
- Group 2
- Group 3.

Changeover between groups:

- Group 1 to 2 by level
- Group 2 to 1, max. runtime. This is the time T, see *Example of alternation*, page 119.

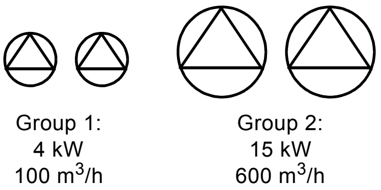
Alternation

If alternation has been selected for a number of pumps in Group 1, for example, the start/stop levels of this group will change. Not always the same pump(s) start at start level 1, the control unit decides the sequence.

If alternation has not been selected, the start/stop level is attached to the pumps.

Example: If start level 1 is 60 cm and stop level 1 is 40 cm, pump 1 always starts at the 60 cm level and stops at the 40 cm level.

Example of alternation



Groups 1 and 2 can be set to alternate individually. This is useful if four pumps are to alternate two by two.

The graph below, fig. 29, shows the functions occurring during a time period, based on the following level settings:

- Start level 1 = 60 cm
- Start level 2 = 90 cm
- Start level 3 = 65 cm
- Start level 4 = 95 cm
- Changeover level = 100 cm
- Common stop level = 35 cm
- Alarm level = 105 cm.

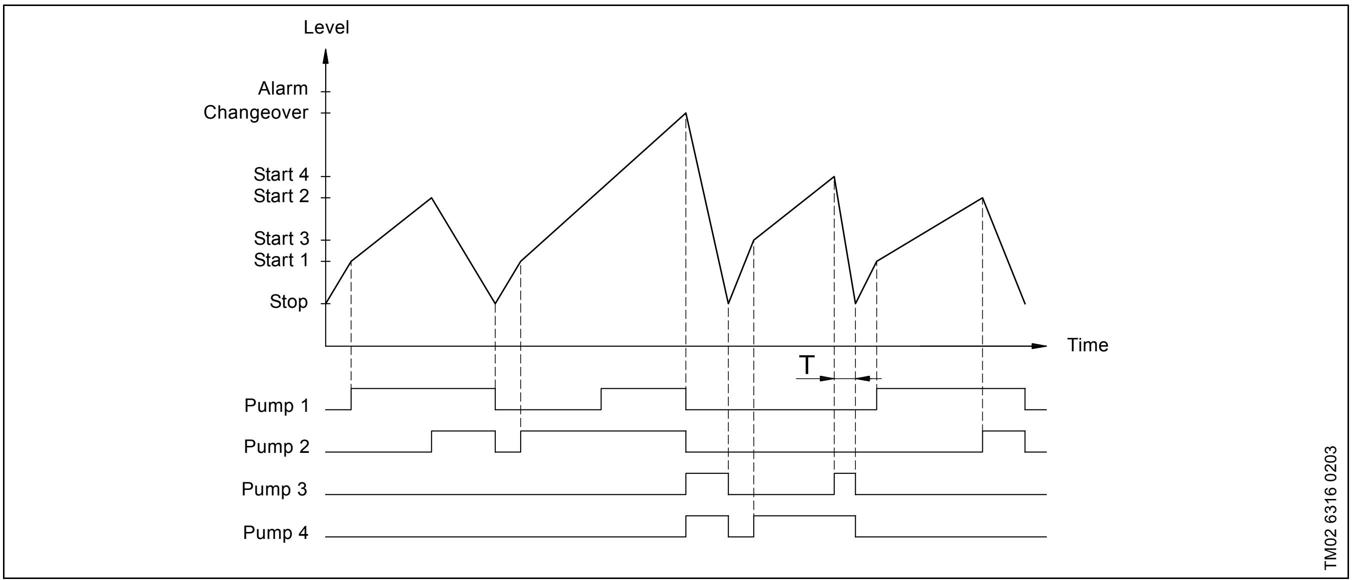


Fig. 29 Example of alternation

The graph shows that pumps 1 and 2 continue to alternate as long as the water level is below the changeover level. When the water level reaches the changeover level, pumps 3 and 4 take over.

Pumps 3 and 4 alternate until one pump has lowered the water level faster than the time T. Subsequently, pumps 1 and 2 alternate.

The time T is fixed by trial and error.

If the alarm level is reached, the user may choose to send an alarm SMS.

If the same value has been set for the two lowest start levels of group 2, there will always be more than one pump running. Consequently, the system cannot change back to group 1.

If the system incorporates float switches only, both pumps must be in group 1.

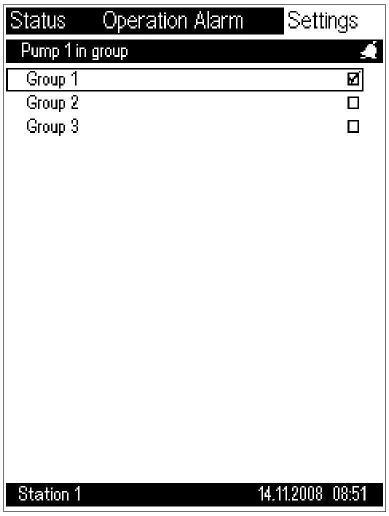
Note: Make the configuration of alternation and groups before the configuration of levels, as alternation and groups affect the use of levels.

14.7.18 Pump grouping

In this display each individual pump is assigned to a group.

Note: The display shown below should be considered as an example.

Path: Settings>System>Pump groups>Pump grouping>



Display_391

Description

In the display shown to the left, pump 1 is assigned to group 1. See section *14.7.17 Group configuration - alternation/advanced alternation*.

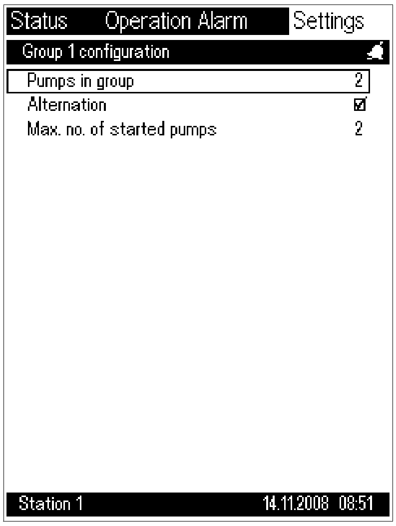
Note: All pumps are assigned to group 1 from factory.

14.7.19 Group properties

This display allows the user to select the functions of pump group 1.

Note: The display shown below should be considered as an example.

Path: Settings>System>Pump groups>Group properties, Group 1>



Display_397

Description

Select from the list:

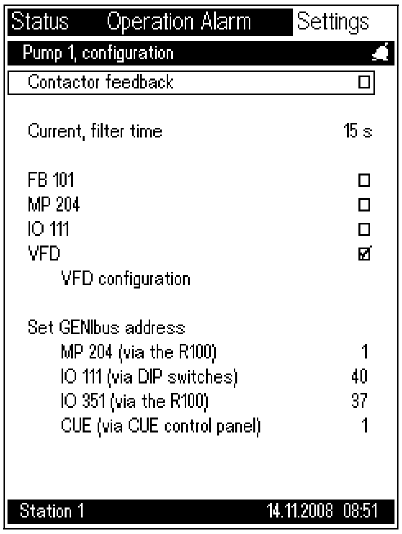
- Pumps in group (only indication)
- Alternation
- Max. number of started pumps (the maximum number of pumps that are allowed to start in the group simultaneously in order not to overload the electricity supply).

14.7.20 Pump configuration (pump 1-6)

This display is used for setting motor/pump configuration parameters.

Note: The display shown below should be considered as an example.

Path: Settings>System>Pump 1>



Display_372

Description

Indicate by check mark whether the motor is fitted with

- Contactor feedback.
  When contactor feedback is selected, the DI5 input is used for contactor feedback and the T8 terminal is used for the moisture switch.
- Current, filter time.
  The period when the alarm state existed before triggering an alarm. The period applies to these pump alarms:
  – Overload
  – Underload
  – Contactor.
- FB 101 phase-sequence module.
- MP 204 motor protector.
  Measurements and settings are set on the MP 204 or via an R100 remote control. All measurements made by the MP 204 are set in the relevant displays.
- IO 111 module.
  Measurements are enabled and limits are set in section *14.2.4 Pump alarms and warnings - enable/disable*.
- VFD.
  A check mark indicates that the frequency converter is active.
  - VFD configuration.
    Setting of frequency converter.
- GENIbus address.
  The information fields show the GENIbus address to be used.

Note: The FB 101 is only set for pump 1 (IO 401). Once the FB 101 is enabled for pump 1, it applies to all the pumps of the system. The FB 101 alarms must also be enabled.

Note: If an MP 204 or IO 111 is used, the GENIbus address below must be used. Set the MP 204 GENIbus address with the Grundfos R100 remote control. Set the IO 111 GENIbus address with the DIP switches on the IO 111 module.

Note: The IO 111 detects the SM 111 module automatically.

GENIbus address

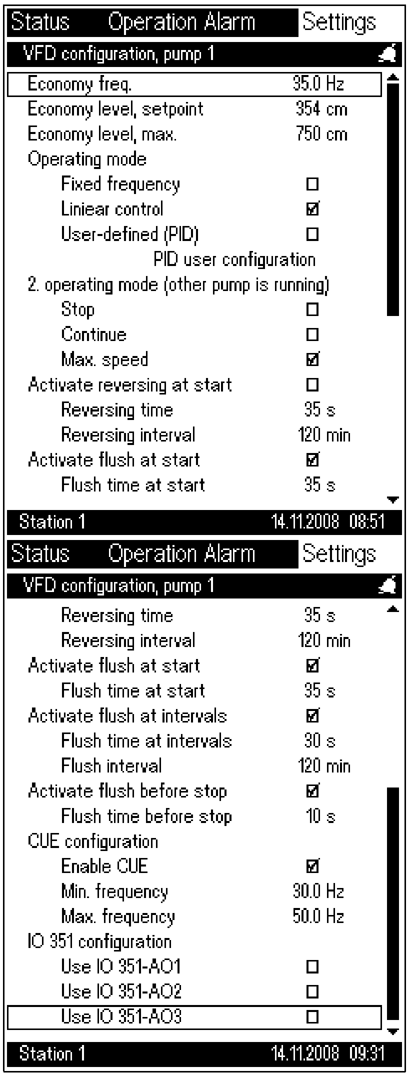
	Pump 1	Pump 2	Pump 3	Pump 4	Pump 5	Pump 6
MP 204	1	2	3	4	5	6
IO 111	1 (40)	2 (42)	3 (43)	4 (44)	5 (45)	6 (46)
CUE	1	2	3	4	5	6
IO 351	37	37	37	(37)	(37)	(37)

14.7.21 VFD configuration

This display allows the user to set the operating parameters of the frequency converter.

Note: The display shown below should be considered as an example.

Path: Settings>System>Pump 1>VFD configuration>



Display_382

Description

Set mixer operating parameters:

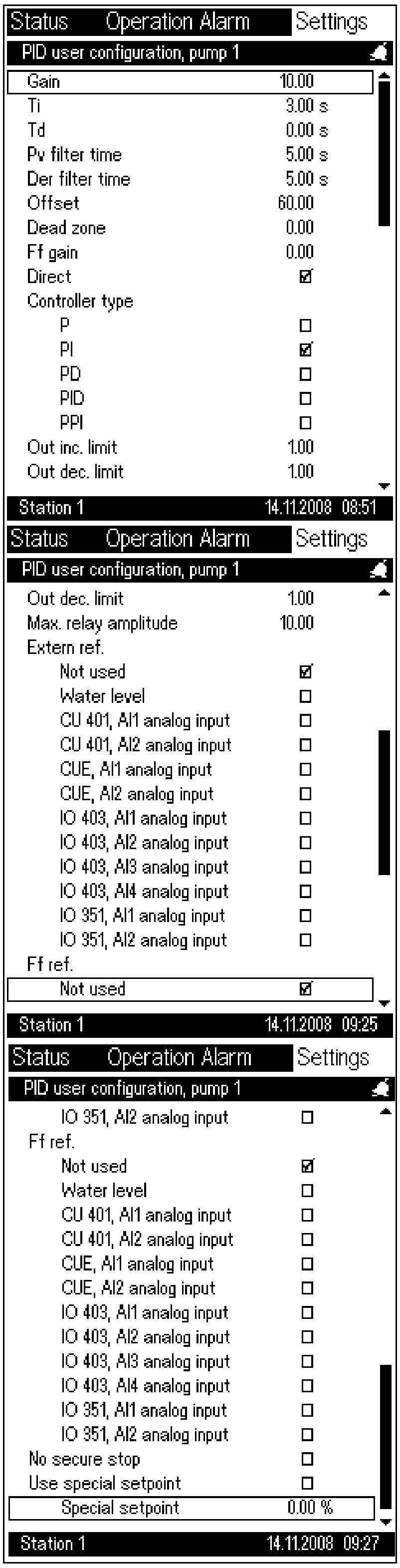
- Economy frequency
  (The frequency with optimum system efficiency).
- Economy level, setpoint
- Economy level, max. (only in connection with linear control)
- Operating mode:
  – Fixed frequency
  – Linear control
  – User-defined (PID)
  (To be set under PID user configuration).
- 2. operating mode (other pump is running)
  – Stop
  Note: If all pumps in the pit are controlled by a frequency converter, all pumps will stop if two or more pumps are started at the same time.
  Example: Two pumps in a pit are both controlled by a frequency converter and sat to "Stop". When the water level reaches start level 2, both pumps will stop.
  – Continue
  (Automatic operation).
  – Max. speed
  (When pump 2 is started, the frequency of pump 1 will be changed to max. frequency. It will return to normal operating frequency when pump 2 or more pumps stop).
- Activate reversing at start.
  (A check mark indicates that the motor is to run with opposite direction of rotation at start).
  – Reversing time
  (The time when the motor is to run with opposite direction of rotation, 50 Hz).
  – Reversing interval
  (Interval between operation with opposite direction of rotation).
- Activate flush at start.
  (The pump runs at 50 Hz to increase the system pressure and flush the system).
  – Flush time at start
  (The time when the motor is to run at 50 Hz).
- Activate flush at intervals.
  (A check mark indicates that the pump is to run at 50 Hz at intervals during operation).
  – Flush time at intervals
  (The time when the motor is to run at 50 Hz at intervals).
  – Flush interval
  (Interval between operation at 50 Hz).
- Activate flush before stop.
  (A check mark indicates that the pump is to run at 50 Hz before stopping).
  Note: The must be enough water in the pit so that the pump does not stop due to dry running.
  – Flush time before stop
  (The time when the motor is to run at 50 Hz before stopping).
- CUE configuration
  – Enable CUE
  – Min. frequency
  – Max. frequency.
  (The frequency is found automatically via the CUE, but can be changed via the CUE).
- IO 351 configuration:
  – Use IO 351-AO1
  – Use IO 351-AO2
  – Use IO 351-AO3.

14.7.22 PID user configuration

In this display, operating parameters for P, PI and PID control are set.

Note: The display shown below should be considered as an example.

Path: Settings>System>Pump 1>VFD configuration>PID user configuration>



Display_364

Description

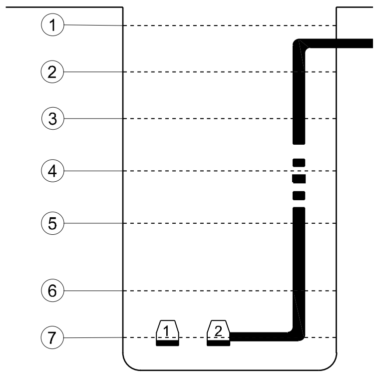
Set mixer operating parameters:

- Gain (relation between input and output signal)
- Ti (integral time)
- Td (differential time)
- Pv filter time (on signal input)
- Der filter time (differential element on signal input)
- Offset (as percentage; 60-100 % = 30-50 Hz, i.e. 60 % = 30 Hz)
- Dead zone (deviation of same unit as setpoint; there is only a positive dead zone)
- Ff gain (feed-forward)
- Direct (if no check mark is set at "Direct", control is inverse)
- Controller type
  – P
  – PI
  – PD
  – PID
  – PPI.
- Out inc. limit (max. permissible change at rising Pv)
- Out dec. limit (max. permissible change at falling Pv)
- Max. relay amplitude (max. output power as percentage point)
- Extern ref. (only one external reference can be selected):
  – Not used
  – Water level
  – CU 401, AI1 analog input
  – CU 401, AI2 analog input
  – CUE, AI1 analog input
  – CUE, AI2 analog input
  – IO 403, AI1 analog input
  – IO 403, AI2 analog input
  – IO 403, AI3 analog input
  – IO 403, AI4 analog input
  – IO 351, AI1 analog input
  – IO 351, AI2 analog input.
- Ff ref. (only one external reference can be selected):
  – Not used
  – Water level
  – CU 401, AI1 analog input
  – CU 401, AI2 analog input
  – CUE, AI1 analog input
  – CUE, AI2 analog input
  – IO 403, AI1 analog input
  – IO 403, AI2 analog input
  – IO 403, AI3 analog input
  – IO 403, AI4 analog input
  – IO 351, AI1 analog input
  – IO 351, AI2 analog input.
- No secure stop (dry running is deactivated; alarm stop is maintained)
- Use special setpoint (in relation to external reference)
  – Special setpoint (as percentage).

14.7.23 Pit levels

This display illustrates the various start and stop levels of the frequency-controlled pump.

Note: The display shown below should be considered as an example.



TM043327 4308

Description

Setpoint

1: High level
2: Start level 2
3: Economy level, max.
4: Start level 1
5: Economy level
6: Stop level 1
7: Dry-running level.

Pump operation

1-2: Pump is running in the operating mode selected.
2-3: The pump is running at max. speed.
3-4: The pump is running with linear control.
4-5: The pump is running with linear control.
5-6: The pump is running at economy frequency (buffer range before stop).
6-7: The pump has been stopped.

14.7.24 CUE configuration

Procedure for configuration of Grundfos CUE frequency converter:

1. Read and carry out the CUE start-up guide.
2. Then make these changes in the CUE menus:

- 1.2 Operating mode = Stop (recommended). Sets how the CUE is to react if there is no signal from the CU 401.
- 3.1 Control mode = Open loop.
- 3.7 Protocol = GENIbus.
- 3.8 Pump number (1-6) = Pumps to be controlled by the frequency converter must be allocated a number.

3. Jumper between CUE terminals 18 and 20 and screened 3-conductor cable between the CU 401 and the CUE (GENIbus).

Connections

Note: Read the installation and operating instructions of the individual units before making any connections.

Grundfos IO 351 terminals 76-81 and 82-89 may be connected to dangerous contact voltage. External control voltage from other groups may occur.

IO 351 and frequency converter

Grundfos IO 351 can be connected to up to three frequency converters. Below is an example of connection.

Terminal, Grundfos IO 351			Danfoss VLT	
1. VLT	2. VLT	3. VLT	FC 102	Signal type
10	12	14	29	Alarm, normally high
15	15	15	20	D-GND
17	25	21	55	Analog GND
18	22	26	53	0-10 V
76	76, 83	85, 87	12	VLT 24 V
77	81, 82	84	18	Start
79	82	86	27	Revers

CU 401 and CUE

CU 401	CUE	
Output	**Terminal**	**Signal type**
Y	61	Bus
A	68	Bus
B	69	Bus

CU 401 and IO 351

CU 401	IO 351	
Output	**Terminal**	**Signal type**
Y	Y	Bus
A	A	Bus
B	B	Bus

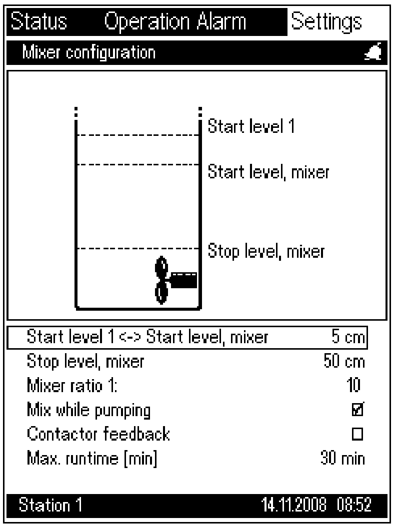
14.7.25 Mixer configuration

This display is used for setting the mixer operating parameters.

The mixer configuration display will only appear when the mixer is enabled. See section *14.7.7 CU 401, DO7 relay output*.

Note: The display shown below should be considered as an example.

Path: Settings > System > Mixer configuration >



Display_530

Description

Set the mixer operating parameters:

- Start level 1 <-> Start level, mixer (difference between the two levels)
  - The mixer will start at "Start level, mixer". When the level reaches "Start level 1", the pump will start, thus ensuring mixing before pump start.
- Stop level, mixer
  - Make sure the mixer is completely submerged during operation.
- Mixer ratio 1:
  - The mixer starts every 10 pump starts.
- Mix while pumping
  - When enabled, the mixer will continue to run until the level reaches the "Stop level, mixer" or "Max. runtime".
- Contactor feedback
  - When enabled, the operational status is determined by a feedback signal from the contactor.
- Max. runtime [min]
  - When the mixer has been running for the set number of minutes, it will stop.

14.7.26 Userlog - selection of data to log

This display is used for enabling logging of data for the parameters selected. As a principal rule, logging of data should be confined to measurable parameters; if, for example, the MP 204 motor protector is not used for temperature measurement on any of the pumps, it should be disabled.

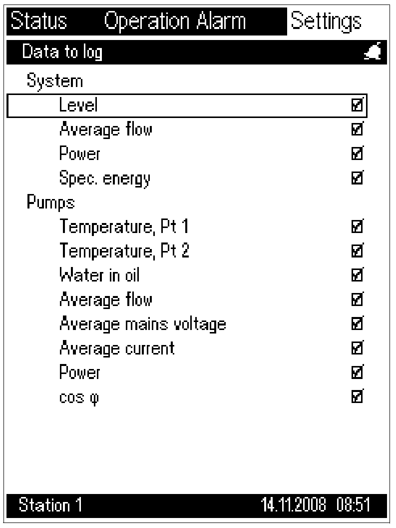
To prevent the waste of internal storage space, the logging of data should be limited to the parameters logged in the system.

The purpose of logging is to monitor

- the operation
- the pump condition.

Note: The display shown below should be considered as an example.

Path: Settings**>**System**>**Userlog data selection**>**



Display_348

Description

Select the type of data to be logged from system and pumps:

- System:
  - Level
  - Average flow
  - Power
  - Specific energy.
- Pumps:
  - Temperature, Pt 1 (IO 401)
  - Temperature, Pt 2 (from MP 204 or IO 111. If both units are connected, only IO 111 data will be logged.)
  - Water in oil
  - Average flow
  - Average mains voltage
  - Average current
  - Power
  - cos φ.

It is the responsibility of the SCADA system to collect the data relevant to the overall operation of the system.

The SCADA system is usually capable of storing all logged data.

The logged data can be used for trend curves. Trend curves can be viewed on the basis of the majority of the logged data. See section *11.11 Trend curves*.

14.7.27 Userlog intervals

This display is used for setting the log interval of a specific parameter. This means that the logging of, for example, all temperature parameters are made at the interval indicated under temperature.

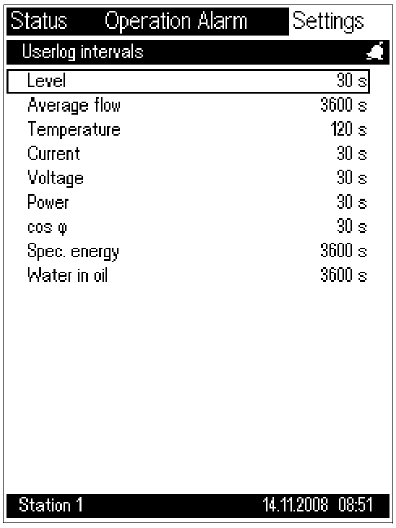
Generally, the log interval should depend on the period of time (log period) covered by the graph to be displayed. The time depends on the parameter in question.

The log period can be extended by choosing a longer log interval.

For example, the level parameter could be followed over a period of 24 hours, whereas changes in water-in-oil sensor parameters occur slowly and could be followed weekly or even monthly.

Note: The display shown below should be considered as an example.

Path: Settings>System>Userlog intervals>



Display_349

Description

Select the desired log intervals for the parameters listed:

- Level
- Average flow
- Temperature
- Current
- Voltage
- Power
- cos φ
- Specific energy
- Water in oil.

The storage space is limited as described below.

Each individual parameter can be logged in 2880 data points, resulting in the following optimum intervals:

Log item	Maximum number of samples	Can be changed by user
Level	3000 (25 hours at 30-second intervals)	Yes
Average flow	720 (30 days at 1-hour intervals)	No
Pumps 1-6: Average flow		
Pumps 1-6: Temperature Pt100	750 (25 hours at 120-second intervals)	Yes
Pumps 1-6: MP 204, Pt temperature (If an IO 111 is connected, the temperature from the IO 111 will be used.)		
Pumps 1-6: MP 204, current	3000 (25 hours at 30-second intervals)	Yes
Pumps 1-6: MP 204, voltage	3000 (25 hours at 30-second intervals)	Yes
Power	3000 (25 hours at 30-second intervals)	Yes
Pumps 1-6: MP 204, power		
Pumps 1-6: MP 204, power factor (cos φ)	3000 (25 hours at 30-second intervals)	Yes
Specific energy	720 (30 days at 1-hour intervals)	No
Pumps 1-6: Water in oil	720 (30 days at 1-hour intervals)	No
AI1 input, IO 403	3000 (2-minute interval)	Yes
AI2 input, IO 403	3000 (2-minute interval)	Yes
AI3 input, IO 403	3000 (2-minute interval)	Yes
AI4 input, IO 403	3000 (2-minute interval)	Yes
DI8 input, IO 403	3000 (2-minute interval)	Yes

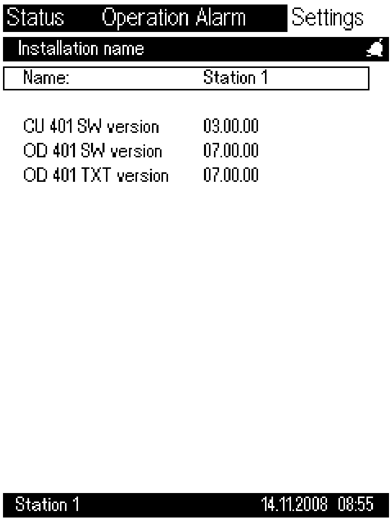
14.7.28 Installation name

This display is used for entering the name (identification) of the installation.

Displayed at the bottom of the display, the name can be used for communicating with the SCADA system.

Note: The display shown below should be considered as an example.

Path: Settings>System>Installation name>



Display_480

Description

Press **ok** to display a qwerty keypad for entering the installation name.

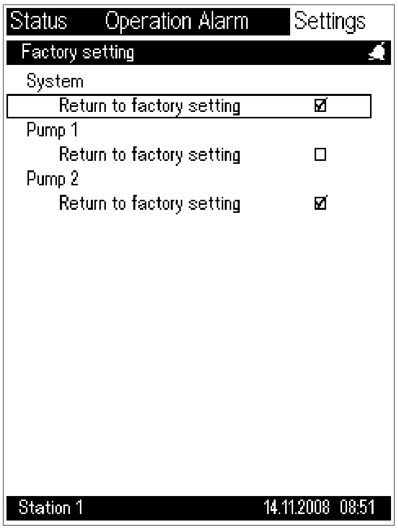
14.8 Factory settings

This display is used for restoring the original factory settings. See the factory settings for system and pumps in section *9. Factory settings*.

Note: When returning to the system factory settings, the system will be configured to a two-pump system. See section *10. Quick start* for additional information.

Note: The display shown below should be considered as an example.

Path: Settings>Factory setting>



Display_380

Description

Procedure:

1. Select "System" or the pump to return to factory settings.
2. Press **ok** (the focus frame is flashing).
3. Press the **plus** button (a check mark **V** appears).
4. Press **ok** to approve.

With the exception of the parameters listed below, all factory settings will be reestablished.

- Counter values
- Telephone book for SCADA and interlock
- Profile access code (only used by SCADA system)
- PUK code.

14.8.1 Units

This display is used for the selection of measuring units.
The selection of units is also possible in connection with the configuration of the individual sensors.

Changing standard will cause the system to change the default units for each of the given parameters.

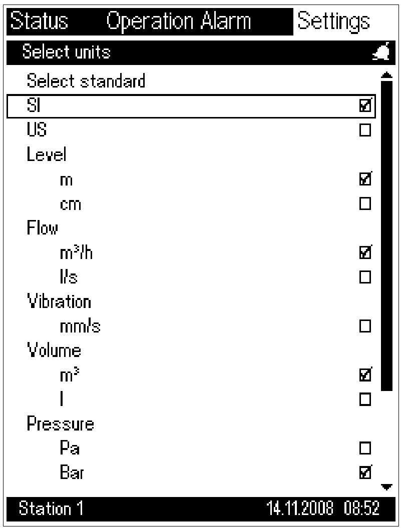
The unit selected will be displayed on the OD 401 displays.

The choice of units does not affect the data displayed on a SCADA system.

Note: The IO 403 treats units independently of these settings.

Note: The display shown below should be considered as an example.

Path: Settings>Units>



Display_485

Description

Select SI or US standard. Then select the unit to be displayed on the OD 401 displays.

- Level:
  - – m
  - – cm
  - – ft
  - – in.
- Flow:
  - – m^3/h
  - – l/s
  - – gpm.
  - – Vibration:
  - – mm/s
  - – in/s.
- Volume:
  - – m^3
  - – l
  - – gal.
- Pressure:
  - – Pa
  - – Bar
  - – mBar
  - – kPa
  - – psi.
- Temperature:
  - – °C
  - – °F.

14.8.2 Access code

This display allows the user to define and enter an access code, if required.

The code consists of a combination of five buttons on the operator display.

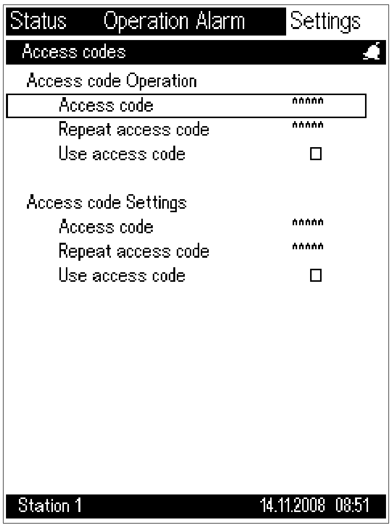
The purpose of the access code is to prevent unintended changes in a given configuration by an unauthorised person.

There are two different access codes:

- access code for settings
- access code for operation.

Note: The display shown below should be considered as an example.

Path: Settings>Access codes>



Display_330

Description

Recommended procedure for the entry of an access code:

1. Write down the desired access code on a piece of paper: Five characters consisting of "arrow up", "arrow down", plus and minus.
2. Enter the code in the line against "Access code" and press **ok**.
3. Enter the code once more in the line against "Repeat access code" and press **ok**.
   If the two access codes entered are identical, a check mark appears in the "Use access code" check box. The access code is now enabled.

After the access code has been enabled, the OD 401 will prompt for an access code each time settings are to be made. When the access code has been entered, it will be active for 8 minutes after the last button press.

To disable the access code, remove the check mark in the "Use access code" check box.

To abort the entry of access code, press **home** or wait for the OD 401 to abandon the display automatically after approx. 3 minutes.

Note: If you forget your access code, you will have to contact Grundfos for assistance.

14.8.3 GSM modem configuration

14.8.4 GSM modem

The display can be used to configure a G 401 module as a communication module.

The G 401 must be physically present.

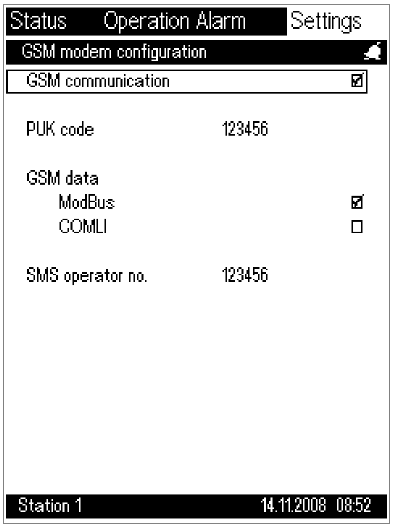
The CU 401 must be connected to an Ethernet module and to the G 401 via a crossed network cable.

If the G 401 is installed in an area of weak coverage as regards GSM signal, an external antenna must be installed.

A COMLI or ModBus protocol can be used.
For further information about the protocols, contact Grundfos.

Note: The display shown below should be considered as an example.

Path: Settings>GSM modem>



Display_450

Description

- A check mark in the check box enables GSM communication. GSM communication can only be enabled if GPRS communication is disabled.
- PUK code: Enter the PUK code to unlock the SIM card. The G 401 generates a PIN code so that the card cannot be moved to another GSM unit. The PUK code is handed out by the teleoperator from whom the SIM card was bought.
- GSM data: A check mark in the check box indicates which protocol the G 401 should use for outgoing calls.

  - ModBus

  - COMLI.

Note: If the protocol is changed, the system, the CU 401 and the G 401 modem need a restart (off/on).

- SMS operator number: The operator number is handed out by the teleoperator from whom the SIM card was bought. In most cases, the number is stored on the SIM card.
  Note: The teleoperator must be informed that the SIM card is to be used for a Siemens modem.

14.8.5 GPRS configuration

The display can be used to configure a G 401 module as a communication module via GPRS.

The G 401 must be physically present.

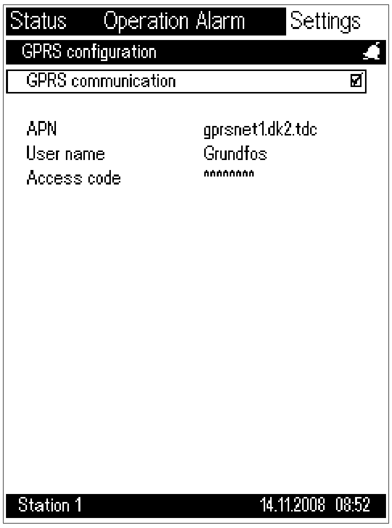
The CU 401 must be connected to an Ethernet module and to the G 401 via a crossed network cable.

If the G 401 is installed in an area of weak coverage as regards GPRS signal, an external antenna must be installed.

OPC is used as communication protocol.
For further information about the protocols, contact Grundfos.

Note: The display shown below should be considered as an example.

Path: Settings>GPRS configuration>



Display_540

Description

- A check mark in the check box enables GPRS communication. GPRS communication can only be enabled if GSM communication is disabled.
- APN (Access Point Node): Enter the APN address.
- User name: Enter the user name.
- Access code: Enter the access code.

Note: The APN address, user name and access code are handed out by the teleoperator from whom the SIM card was bought.

The IP address on the SM card can be viewed. See section *11.7 GSM/GPRS modem*.

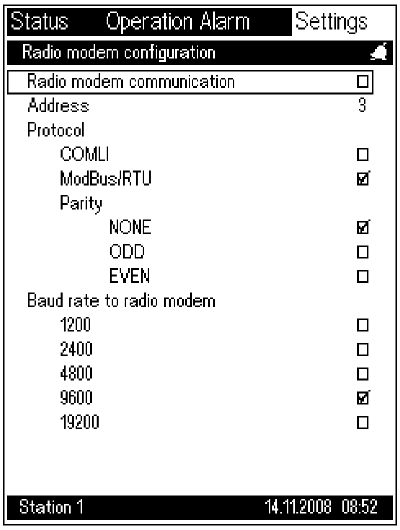
14.8.6 Radio modem configuration

The radio modem configuration menu is accessed via the OD 401 operator display.

The G 403 module is configured with communication settings matching the actual radio modem.

Note: The display shown below should be considered as an example.

Path: Settings>Radio modem configuration>



Display_545

Description

Select radio modem communication by putting a check mark in the check box.

- Address 3 (slave address of G 403 ModBus/COMLI).

Select protocol:

- COMLI (1 start bit, 8 data bits and ODD parity)
- ModBus/RTU (parity can be set by the user)
- Parity:
  - NONE
  - ODD
  - EVEN.

Select baud rate to modem:

- 1200
- 2400
- 4800
- 9600
- 19200.

Note: The baud rate is the communication speed between the G 403 and the radio modem. The baud rate for the radio modem must be set manually according to the specifications for the actual radio modem.

Factory settings

Function	Setting
Protocol	ModBus/RTU
Address	3
Baud rate	9600
Parity	NONE
Data bits	8
Stop bits	2
Range	
Address	1-247 (ModBus requirement)
Baud rate	1200-19200 (between G 403 and radio modem)
Parity	NONE-ODD-EVEN for ModBus and ODD for COMLI
Data bits	8 (ModBus requirements, COMLI recommendation)

14.9 Telephone book

This display is the opening display for setting parameters concerning communication:

- SCADA (setting of SCADA alarm dialup and feedback to SCADA)
- Interlock (setting of stop between controllers in the event of alarm level)
- SMS (enables the system to send SMS messages to, for example, a mobile or cellular phone)
- SMS (setting of telephone number, schedule, heartbeat, etc.).

Note: The display shown below should be considered as an example.

Path: Settings>Telephone book>

Display_460

Description

A check mark indicates that the channel of communication has been selected.

Note: In order to see specific settings, for instance if advanced or simple SMS has been selected, open the individual submenus.

14.9.1 SCADA network

This display is used for setting the system in a SCADA network and various dialup functions.

In the "Handling system alarms" display, sections *14.2.3 System alarms and warnings - alarm handling* and *14.2.6 Pump alarms and warnings - alarm handling*, it is possible to set the system to send an alarm telegram to a SCADA system in the event of an alarm.

In addition to setting the alarms, the parameters for dialup to the SCADA system must be set and a CompactFlash modem installed.

It is possible to enter several SCADA systems. Normally, dialup is made to only one SCADA system. However, if dialup to the priority 1 SCADA system fails, dialup to the priority 2 SCADA system will be attempted, etc.

Possible settings:

- SCADA dialup active
- number of dialup attempts to each SCADA system
- delay between each alarm dialup attempt.

When a dialup is successful, no more dialups will be made until a new alarm dialup is active.

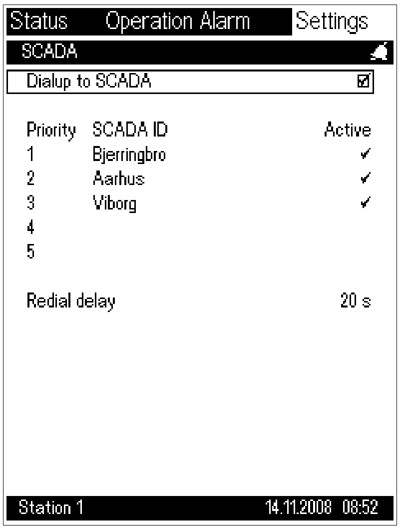
An analog modem communicates with the SCADA system via ModBus protocol via the modem connection.

A GSM modem can communicate with the SCADA system via ModBus or COMLI protocol.

Note: Calls should only be set for the GSM and analog modems. The GPRS modem has always connection and does not have to be set separately.

Note: The display shown below should be considered as an example.

Path: Settings**>**Telephone book**>**SCADA**>**



Display_461

Description

Select the SCADA systems connected to the controller.

Possible settings:

- Dialup to SCADA (determines whether alarm dialups are active)
- Priority 1 (the first SCADA system to be dialled in the event of alarm)
- Priority 2 (the next SCADA system to be dialled if dialup to priority 1 fails)
- Priority 3 (the next SCADA system to be dialled if dialup to priority 2 fails)
- Priority 4 (the next SCADA system to be dialled if dialup to priority 3 fails)
- Priority 5 (the next SCADA system to be dialled if dialup to priority 4 fails)
- Redial delay (delay between two alarm dialups).

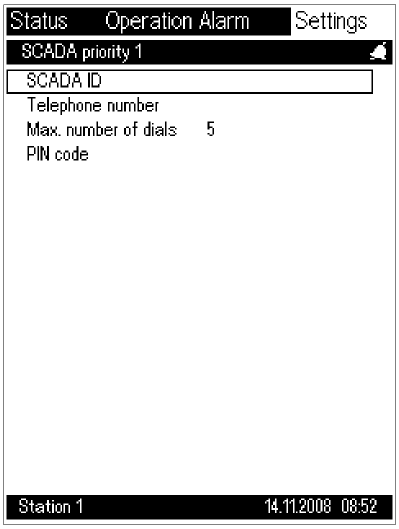
14.9.2 SCADA

This display is used for setting the user in a SCADA network.

The telephone number is entered by means of the keypad with numbers.

Note: The display shown below should be considered as an example.

Path: Settings>Telephone book>SCADA>Priority 1>



Display_462

Description

Enter data on the actual SCADA system:

- SCADA ID (identification of the system)
- Telephone number
- Max. number of dials (set the number to zero to disable the actual SCADA system)
- PIN code (code to be used by the SCADA system for dialups for the purpose of changing parameters).

14.9.3 Interlock

The interlock is enabled when the alarm level is active and disabled when the water level has fallen below the highest stop level for the selected pumps.

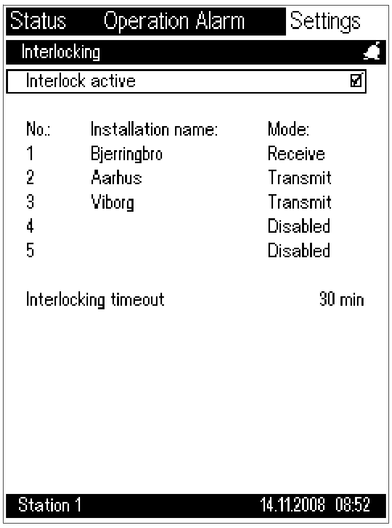
Interlock means that the pumping station can dial one or more pumping stations upstream to stop the flow of liquid from these stations. The function is used if a pumping station is almost full.

The interlock dial is protected by an access code, allowing only the station with the right access code to stop other stations.

If interlock is enabled, the alarm level alarm must also be enabled as this alarm controls the interlock function.

Note: The display shown below should be considered as an example.

Path: Settings>Telephone book>Interlock>



Display_471

Description

Select the installations/stations to dial or from which to receive dialups.

Possible settings:

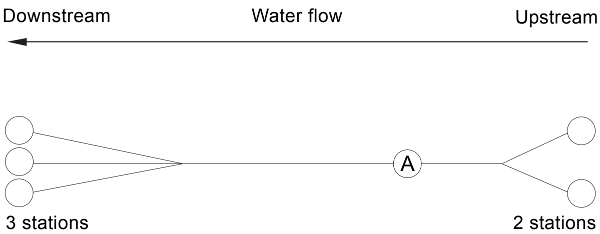
- Interlock active: A check mark in the check box enables the interlock function.
- No. 1 (the first station to be dialled or from which to receive a dialup).
- No. 2 (the second station to be dialled or from which to receive a dialup).
- No. 3 (the third station to be dialled or from which to receive a dialup).
- No. 4 (the fourth station to be dialled or from which to receive a dialup).
- No. 5 (the fifth station to be dialled or from which to receive a dialup).

Interlocking timeout:

If no new interlock dialup is received within this time, interlock will be disabled.

Note: The interlocking timeout is determined by the sending station.

Example:



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In a system, station A can, for example, receive a dialup from up to three downstream stations to stop the flow of liquid (a stop command). Station A can then dial up to two upstream stations. See above figure.

A station can handle five connections.

14.9.4 Interlock - user settings

This display is used for setting the interlock function. The function is factory-set to "not active".

The telephone number is only required if the pumping station is to dial another pumping station.

The PIN code is used for dialups received or dialups transmitted to another station (identical PIN codes are allowed).

If interlock is enabled, the alarm level alarm must also be enabled as this alarm controls the interlock function.

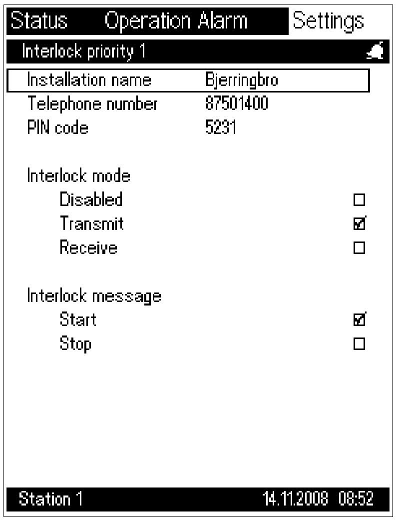
Note: The PIN code must be valid in order for the station to activate other stations and for the station to be controlled by other stations.

Procedure:

1. Enter the name of the station to be controlled.
2. Enter the telephone number and PIN code of the station (the PIN code must be entered under "Receive" in the station to be contacted).
3. The dialup function is enabled.

Note: The display shown below should be considered as an example.

Path: Settings>Telephone book>Interlock>No. 1 Bjerringbro>



Display_472

Description

Enter the data on the actual installation/controller:

- Installation name (identification of the controller)
- Telephone number (GSM and analog)
  - Enter the number to be dialled by the CU 401 (not required for receiving dialups).
- Telephone number (GPRS)
  - Enter the IP address and the network number to which the CU 401 is to be connected.
    The IP address can be read from the status display of the station to be controlled. See section *11.7*.
    The network number can be read from the status display of the station to be controlled. See section *11.5*.
    Example:
    IP number of the station to be controlled: 172.16.1.3
    Network number of the station to be controlled: 2
    Enter "172.16.1.3:2".
- PIN code
  - If check mark in "Transmit": PIN code to be transmitted to another controller.
  - If check mark in "Receive": PIN code to be transmitted from another controller to stop the actual controller.
  - The PIN code has no function during GPRS communication.
    Use the standard PIN code "0".
- Interlock mode
  - Disabled: A check mark indicates that the installation/controller is disabled.
  - Transmit: A check mark indicates that the installation/controller is to be dialled.
  - Receive: A check mark indicates that dialups are to be received from the installation/controller.
- Interlock message
  - Start
  - Stop.
    Command to be sent to the controller in the event of interlock; only required if the controller is to be dialled.

The entries are made by means of a keypad. See below.

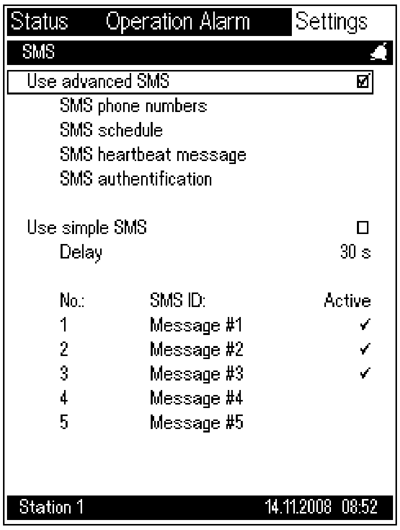
Note: Interlock only works if a check mark has been put in "Receive" for the "controlling" station and in "Transmit" for the "controlled" station.

14.10 SMS

This display is used to enable/disable SMS forwarding and to configure up to five outgoing SMS messages.

Note: The display shown below should be considered as an example.

Path: Settings>Telephone book>SMS>



Display_490

Description

- Use advanced SMS
  – SMS telephone numbers (SMS can be sent to these telephone numbers)
  – SMS schedule (states recipients and time for sending of SMS)
  – SMS heartbeat message (information saying that the pumping station is able to communicate)
  – SMS authentication (setting of access control)
- Use simple SMS
- Delay
- Message #1-5.

14.10.1 SMS phone numbers

In this display, up to three mobile phone numbers can be entered.

SMS numbers

Together with the schedule, the mobile phone numbers are used to send SMS messages in case of a warning or an alarm set to send an SMS.

Under SMS schedule, the user selects which of the phone numbers entered are to be the primary number and the secondary number.

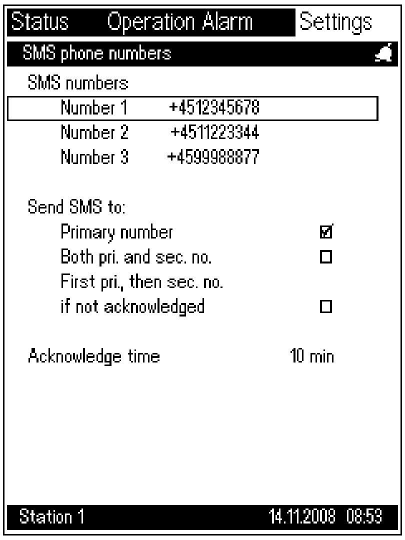
The SMS numbers are also the number used in connection with SMS and communication via SMS. See section *14.10.4 SMS authentication*.

An SMS can be sent in three ways:

- to the primary number
- to the primary and the secondary number
- to the primary number and then to the secondary number if the primary number does not acknowledge receipt of the SMS message.

Note: The display shown below should be considered as an example.

Path: Settings>Telephone book>SMS>SMS phone numbers>



Display_701

Description

- SMS numbers:
  – Click the **ok** button next to number 1, 2 and 3. A new menu appears where the mobile phone numbers can be entered. End by pressing the **ok** button.
- Send SMS to:
  – Primary number
  If there is an alarm or a warning, an SMS will be sent to the primary number.
  – Both pri. and sec. no.
  If there is an alarm or a warning, an SMS will be sent to the primary and the secondary number.
  – First pri., then sec. no. if not acknowledged
  If the primary number does not acknowledge the SMS message within the time set, an SMS will be sent to the secondary number. The primary and secondary number a new time limit for acknowledging the SMS message.
- Acknowledge time
  Is the time within the user with the primary number must acknowledge receipt of the SMS message before it is sent to the secondary number. The acknowledge time is factory-set at 10 seconds and can be changed by the user.
  The user acknowledges receipt of the SMS message by returning the content of the SMS message received.

14.10.2 SMS schedule

In this display and the following submenus, the SMS schedule is set. The user sets to which numbers SMSes are to be sent depending on day of the week and period.

Selection day of the week

When a day of the week has been selected, it is possible to set these periods:

- Work
- Off
- Sleep
- Change 1
- Change 2
- Change 3.

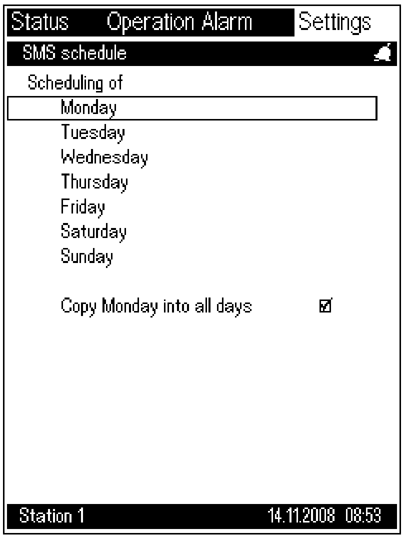
The settings under Monday can be copied to all days of the week by putting a check mark in the box "Copy Monday into all days". Every time settings under Monday are changed, the check mark must be put to update settings made under Monday.

If almost similar settings are required for the week, it is advisable to make the desired settings for Monday and then copy them to the rest of the week. Finally, minor modifications can be made for individual day, e.g. Saturday and Sunday.

Note: Every time Monday is copied to all days of the week, the settings made for the other days of the week will be changed.

The display shown below should be considered as an example.

Path: Settings**>**Telephone book**>**SMS**>**SMS schedule**>**



Display_702

Description

Select the day of the week to be set.

- Scheduling of
  - Monday
  - Tuesday
  - Wednesday
  - Thursday
  - Friday
  - Saturday
  - Sunday.

Setting of day of the week

First set the start time of the three periods, then the times for the three changes.

Period

To enable the sending of an SMS in the three periods, set a check mark at the individual periods.

State also at what time of the day the period is to start (hours and minutes).

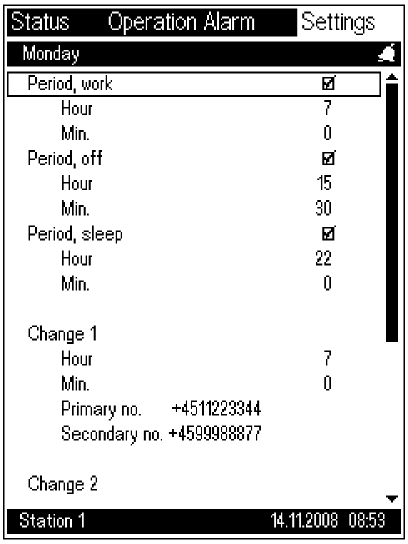
Change

It is possible to set three times for change during the day (24 hours). Under every change, both the primary and the secondary phone number are to be selected.

When the **ok** button is pressed at the primary and the secondary number, a new menu will appear where it is possible to select between the three numbers in the phone book.

Note: The display shown below should be considered as an example.

Path: Settings>Telephone book>SMS>SMS schedule>Monday>



Display_710

Description

Example: The work day begins on Monday at 7:00, the off period at 15:30 and the sleep period at 22:00. The display shows that SMS messages are allowed in all three periods.

A change can cover several periods.

Note: If the same times are set for the three changes, the phone numbers under change 3 will be used.

14.10.3 SMS heartbeat message

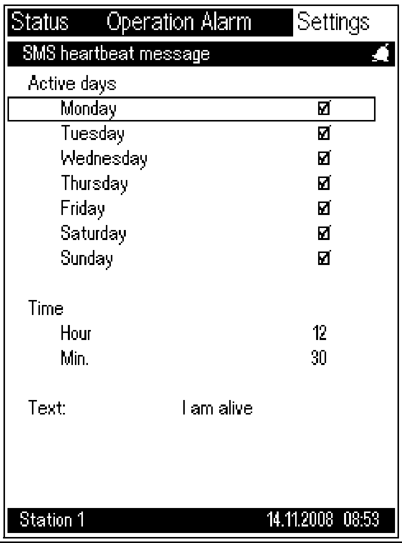
This display allows the user to set when an SMS heartbeat message is to be sent.

A heartbeat message tells the user that the system and the mobile phone is connected.

A heartbeat message is only sent to the primary phone number.

Note: The display shown below should be considered as an example.

Path: Settings**>**Telephone book>SMS**>**SMS heartbeat message>



Display_703

Description

- Active days
  - Here the days are selected on which a heartbeat message is to be sent. It is possible to select all days.
- Time
  - Any time can be selected. In the example below, the time for heartbeat message is set to 12:30.
- Text
  - The text "I am alive" can be changed into another text. In addition to the text, the name of the station is shown. In this case, "Station 1".

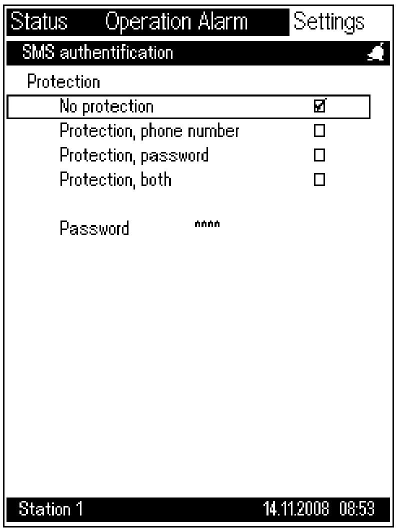
14.10.4 SMS authentication

This display allows the user to set the access control for communication between the mobile phone and the system.

The access control ensures that only the three mobile numbers in the phone book or users knowing the access code, are able to communicate with the system.

Note: The display shown below should be considered as an example.

Path: Settings>Telephone book>SMS>SMS authentication>



Display_704

Description

- No protection
  If no protection is selected, everybody who can send an SMS to the system, can change the settings.
- Protection, phone number
  Only the three mobile numbers in the phone book can change systems settings.
- Protection, password
  All SMS messages sent to the system must start with the password.
- Protection, both
  The user can carry out changes in the system by sending an SMS from a mobile number in the phone book or by sending an SMS from a "foreign" mobile phone and start the text with the password.
- Password
  The password must consist of four digits between 0 and 9. It is always shown as ****.

14.10.5 Simple SMS message #1

This display allows the user to set outgoing SMS messages.

It is possible to distinguish between alarms and warnings from different pumps and the system by setting:

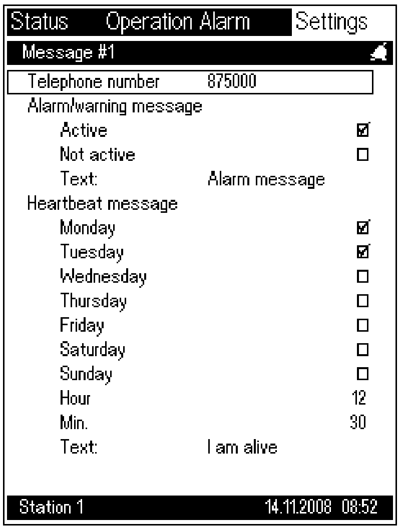
Message #1 = alarm from pump 1

Message #2 = alarm from pump 2

Message #3 = alarm from system.

Note: The display shown below should be considered as an example.

Path: Settings>Telephone book>SMS>Message #1>



Display_492

Description

Set the outgoing SMS messages:

- Telephone number
  - The number must be of international format with '+' followed by the country code and the local number.
- Alarm/warning message
  - Active: If the function has been activated, an SMS will be sent if an alarm or warning occurs.
  - Not active: If the function has been activated, an SMS will be sent when the alarm or warning disappears.
    Note: An "OK" will be added automatically at the end of the "Alarm message".
- Heartbeat message
  - Here the days are selected on which a heartbeat message is to be sent. It is possible to select all days.
  - Any time can be selected. In the example below, the time for heartbeat message is set to 12:30.
  - Text
    The text "I am alive" can be changed into another text. In addition to the text, the name of the station is shown. In this case, "Station 1".

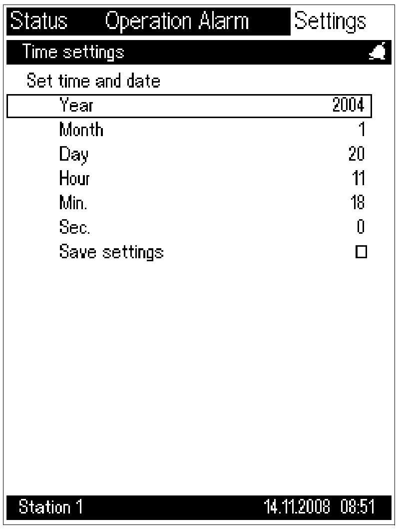
14.11 Time and date settings

This display is used for making the internal year, date and time settings indicated at the bottom of all displays. The time settings are also used for the time registration in the alarm log.

Note: To ensure data logging in the SCADA system, it is important that the clock in the CU 401 and the clock in the SCADA system are synchronous.

Note: The display shown below should be considered as an example.

Path: Settings**>**Time, date and year**>**



Display_340

Description

Procedure:

1. Set the date and time:
   - Year
   - Month
   - Day
   - Hour
   - Min.
   - Sec.
2. Put a check mark in the "Save settings" check box.
3. Press **ok** to enable (synchronise) the date and time.

If the date reverts to 1979 after a power cut, the internal battery of the CU 401 is probably not charged and needs replacement. The battery life is approx. 3 years.

Synchronising

1. Insert the button cell battery into the CU 401 and make sure that the PU 102 is disconnected.
2. Switch on the CU 401.
3. Set the CU 401 clock.
4. Change the userlog intervals, if required. See section *14.7.27 Userlog intervals*.
5. Switch off the CU 401 and connect the PU 102.
6. Switch on the CU 401.
7. Check that the clock is correct after start-up.

14.11.1 Replacement of internal battery of the CU 401

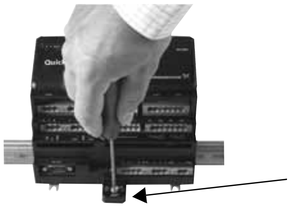
(expected service life 3 years)

TM02 7146 2703

- Switch off the power supply to the CU 401.
- Turn the locking screw to "slide position".
  Use a 6 mm screwdriver for slotted screws.

Slide position

- Push the modules, which are placed next to the CU 401 and have contact via module bus, away from the CU 401 (approx. 15 mm to each side).

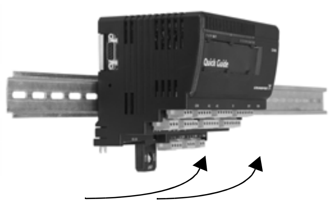


TM02 7135 2703

- Remove the terminals and turn the locking screw to position "unlocked".

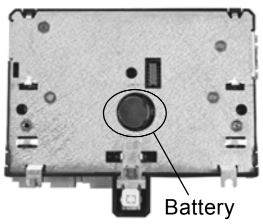
Locked

Unlocked



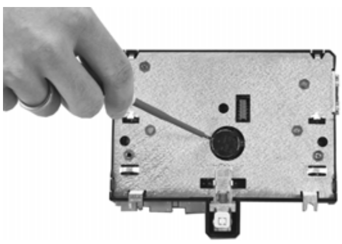
TM02 6667 1303

- Remove the CU 401 unit from the DIN rail by lifting it out at the bottom.



TM02 6674 1303

- Turn the CU 401 unit upside down.

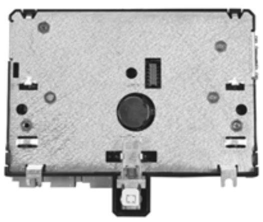


TM02 6664 1303

The battery is kept into place by four "locking elements".

- Use a small screwdriver or a similar tool to lift the battery out of the CU 401.

Note: The tool used should not be made of a conducting material and it must be suitable for a straight slot.



TM02 6674 1303

- Fit a new battery according to the specifications.

Note: The text on the battery should be visible after installation.

TM02 7146 2703

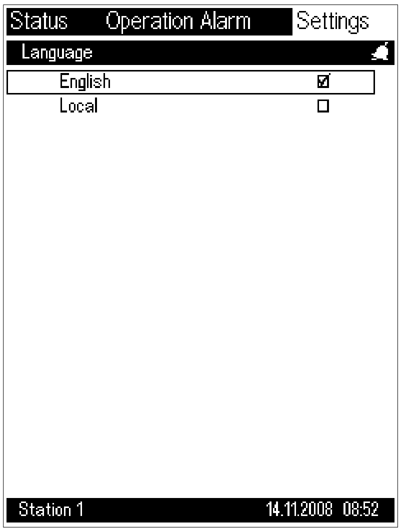
- Mount the CU 401 on the DIN rail, push the modules together and fit the terminals.
- Turn the locking screw to position "locked".
- Switch on the power supply (PU 101).
- Set the clock using the operator display (OD 401).

14.12 Language

This display allows the user to select the language to be used by the OD 401 operator display.

Note: The display shown below should be considered as an example.

Path: Settings>Language>



Display_609

Description

Local language is default.
English can be selected instead of local language.
Put a check mark in the check box of the desired language to enable the language.

15. Configuration of SMS communication via mobile phone

15.1 Entering of phone numbers and access code

SMS communication between a mobile phone and the system always requires that phone numbers are entered into the phone book or that an access code is used.

The first time the system is used, it has to be initialised with this command:

INIT <access code>

The number of the mobile phone used for initialising the system will be added to the internal phone book of the module. Access via SMS commands can now take place from this number or by using the access code, which must consist of four digits between 0 and 9.

Before carrying on the configuration, wait until the system has acknowledged the change with this positive SMS message:

INIT: <phone number> added to Phone No. list

Access code: <access code>

Note: If you forget the access code, you can create a new one via the OD 401 display or Grundfos PC Tool.

If the system has already been initialised from another phone number (the phone book is not empty), it will send a negative SMS message:

INIT: Already initialised from another phone No.

15.1.1 Adding of phone numbers

It is possible to add a number to the system phone book from the mobile phone used for initialisation. The number must be of international format with '+' followed by the country code and the local number.

ADDnumber <phone number>

It is also possible to add a number via another mobile phone with this command:

<access code> ADDnumber <phone number>

When a new number has been added, the system will acknowledge the change with this positive SMS message:

ADDNUMBER: <phone number> added to Phone No. list

Note: The phone book can contain up to three phone numbers.

15.1.2 Deletion of phone numbers

It is possible to delete a number in the phone book with this command:

DELNUMBER: <phone number>

If the number exists, it will be deleted, and the system will acknowledge the change with this SMS message:

DELNUMBER: <phone number> deleted from the Phone No. list

Note: It is possible to delete the number from which the SMS message is sent.

15.2 Change of acknowledge time

It is possible to change the acknowledge time with this command:

DELAY <time in minutes>

The system will acknowledge the change with this positive SMS message:

DELAY changed to <time in minutes>

15.2.1 Change of access option

It is possible to change the access option with this command:

ACCESSCTR <access option>

- **LIST**
  Only numbers in the phone book can send configuration and control messages.
- **CODE**
  The 4-digit access code must be used in front of all configuration and control commands.
- **BOTH**
  Both LIST and CODE can be used for access control.
  Only one is required to get access.
- **NONE**
  There is no access control. All SMS commands can be sent from a mobile phone without access control.

The system will acknowledge the change with this positive SMS message:

Accessctr changed to <access option>

15.2.2 Change of access code

It is possible to change the access code with this command:

CODE <access code>

The system will acknowledge the change with this positive SMS message:

SMS CODE changed to <access code>

15.3 Status information

15.3.1 Status1

Send an SMS with this command to the system in order to get the first part of status information:

STATUS1 <pump number>

The system sends an SMS with this text:

<installation name> : Pump Running: True : Setpoint: xxx.x cm : Level yyy.y cm>

If no pump number is entered, the system will send status1 for pump 1.

Installation name	Pit name
Pump Running: True	The pump is running
Pump Running: False	The pump has been stopped
Starting	The pump has received a start signal
Stopping	The pump has received a stop signal
Setpoint	Start level in centimetres
Level:	Actual water level in centimetres

15.3.2 Status2

Send an SMS with this command to the system in order to get the second part of status information:

STATUS2 <pump number>

The system sends an SMS with this text:

<installation name> : Voltage: xxx.xx V : Current: xx.xx A : Power: xxx.xx kW :
Energy: **xxx.xx kWh : Temp: xxx.xx C : Hours xx h xx min : NoOfStarts: xxxx"**

If no pump number is entered, the system will send status2 for pump 1.

Installation name	Pit name
Voltage	Actual mains voltage
Current	Actual current consumption
Power	Input power
Energy	Total consumption in kWh
Temp	Motor winding temperature
Hours	Number of pump operating hours
NoOfStarts	Number of pump starts

15.4 Start of pump

Send an SMS with this command to the system in order to start a pump:

START <pump number>

The system acknowledges receipt by sending an SMS with this text: "Starting".

If no pump number is entered, pump 1 will be started.

Note: To check if the pump has started, send a status1 command. See section *15.3 Status information*.

15.5 Stop of pump

Send an SMS with this command to the system in order to stop a pump:

STOP <pump number>

The system acknowledges receipt by sending an SMS with this text: "Stopping".

If no pump number is entered, pump 1 will be stopped.

Note: To check if the pump has stopped, send a status1 command. See section *15.3 Status information*.

15.6 AUTO

Send an SMS with this command to the system in order to set a pump to AUTO:

AUTO <pump number>

If no pump number is entered, pump 1 will be set to AUTO.

The system acknowledges receipt by sending an SMS with status1 information for the pump selected.

15.7 Fault messages

- **<phone number>: Not added, Phone No.list full**
  The phone book is full. The phone book can contain up to three phone numbers.
- **<phone number>: Not in Phone No. list**
  The phone number which the user wants to delete is not in the phone book.
- **ADDNUMBER: Illegal phone No. format**
  The phone number which the user wants to add to the phone book has a format which the system cannot read. For instance too many or too few digits.
- **INIT: Already initialised from another Phone No.**
  The SMS system has already been initialised by another phone number.
- **INIT: Illegal Access code format. Must be 4 figures.**
  The access code sent does not consist of four digits.
- **Unknown command**
  General fault which is often by misspelling of commands or missing spaces.

16. Number of inputs and outputs

The table below shows all the system inputs and outputs.

For connection of the individual inputs and outputs. See the wiring diagrams, pages 5 to 8.

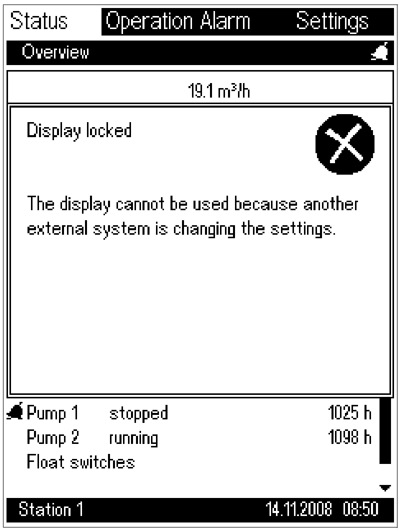
For technical data on module inputs and outputs. See the installation and operating instructions on the modules.

Inputs and outputs	Number of inputs and outputs		
	2 pumps (3 modules)	**4 pumps (4 modules)**	**6 pumps (5 modules)**
Digital inputs (24 V)	19	29	39
Digital inputs (5 V) (pulse counter or digital input)	1	1	1
Analog inputs for connection to sensors with current (4-20 mA) or voltage (0-10 V)	8	10	12
24 V output for sensor electricity supply	6	8	10
10 V output for simple potentiometer connection	1	1	1
Relay outputs for alarms for control of external devices (400 VAC/2 A), e.g. rotor flash, horn, etc.	6	6	6
Relay outputs for pump control via contactors	2	4	6
Pulse counter input (max. 10 kHz)	1	1	1
Pt100/Pt1000 sensor inputs	2	4	6
PTC thermistor inputs, double-insulated (5 V)	2	4	6
Phase sequence detector*	1	2	3
GENIbus communication connections	1	1	1
Connection of OD 401 operator display	1	1	1
Connection of UPS PU 102 battery back-up*		1	
Anybus CU 401 connection*		1	
Ethernet CU 401 connection*		1	
Modem connection		1	
Modem for CU 401 control unit*		1	
Anybus module for CU 401 control unit*		1	

* accessory (requires FB 101 module)

17. Fault finding

17.1 Locked display



Display_40802

Description

If this display appears, an external system is changing the settings, e.g. a SCADA system.

The display can be operated again when the external system is finished.

17.2 CU 401 indicator lights

TM02 6670 1303

Fault (F):

The indicator light is red if the CU 401 unit does not function correctly.

TM02 6670 1303

Run (R):

The indicator light is green when the application program is running correctly.

TM02 6670 1303

Power (P):

The indicator light is green when the power supply to the CU 401 unit is OK.

TM02 6670 1303

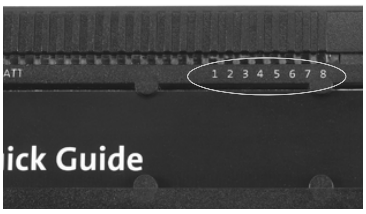
UPS battery back-up (BATT):

The indicator light is green when

- the battery for the UPS battery back-up (PU 102) is charged and capable of powering the CU 401 in case of supply failure.

The indicator light is off when

- the UPS battery back-up (PU 102) has not been fitted
- the battery for the UPS battery back-up (PU 102) has not been charged.



TM02 6670 1303

Digital inputs 1 to 6:

The indicator light is yellow when the input is active.

Relay outputs 7 and 8:

The indicator light is yellow when the output is active.

17.3 IO 403 indicator lights

TM02 6690 1303

Status (S):

The indicator light is green when the IO 403 module is active and OK.

The indicator light is red if the IO 403 module does not function correctly.

Note: The indicator light is red until the program has been loaded into the CU 401.

TM02 6690 1303

Relay outputs 1 to 4:

The indicator light is yellow when the output is active.

Digital inputs 5 to 8:

The indicator light is yellow when the input is active.

17.4 IO 401 indicator lights

TM02 6684 1303

Status (S):

The indicator light is green when the IO 401 module is active and OK.

The indicator light is red if the IO 401 module does not function correctly.

Note: The indicator light is red until the program has been loaded into the CU 401.

TM02 6684 1303

Digital inputs 1 to 5:

The indicator light is yellow when the input is active.

Relay output 8:

The indicator light is yellow when the output is active.

TM02 6684 1303

Inputs for PTC sensor 6 and 7:

The indicator light is green when

- the input is used with PTC sensor (< 500 Ω).
- the input is used with thermal switch (closed).

The indicator light is red when

- the input is used with PTC sensor (MΩ).
- the input is used with thermal switch (open).

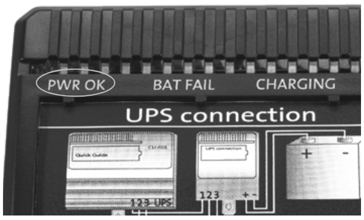
17.5 PU 101 indicator lights

TM02 7166 2703

Status:

When the PU 101 module is connected correctly, the green indicator light on the front will indicate the "OK" status of the module as long as the module is powered.

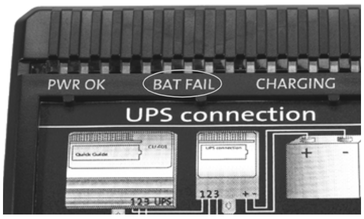
17.6 PU 102 indicator lights



TM02 6102 1303

PWR:

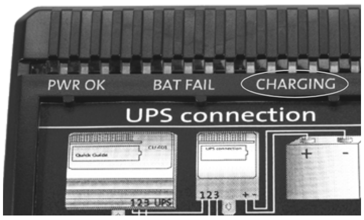
The indicator light is green when the PU 102 is ready to supply power to the CU 401.
(The battery should be half charged, at the minimum.)



TM02 6102 1303

BAT FAIL:

The indicator light is red if the battery has not been fitted or the battery is not charged.



TM02 6102 1303

CHARGING:

The indicator light is yellow when the battery is being charged and the indicator light goes out when the battery is fully charged.

17.7 CUE indicator lights

TM043328 4308

Indicator light	Function
On (green)	The pump is running or has been stopped by a stop function.
	If flashing, the pump has been stopped by the user (CUE menu), external start/stop or bus.
Off (orange)	The pump has been stopped with the on/off button.
Alarm (red)	Indicates an alarm or a warning.

18. Dimensional sketches, panel and board

18.1 Installation example 1 (one row)

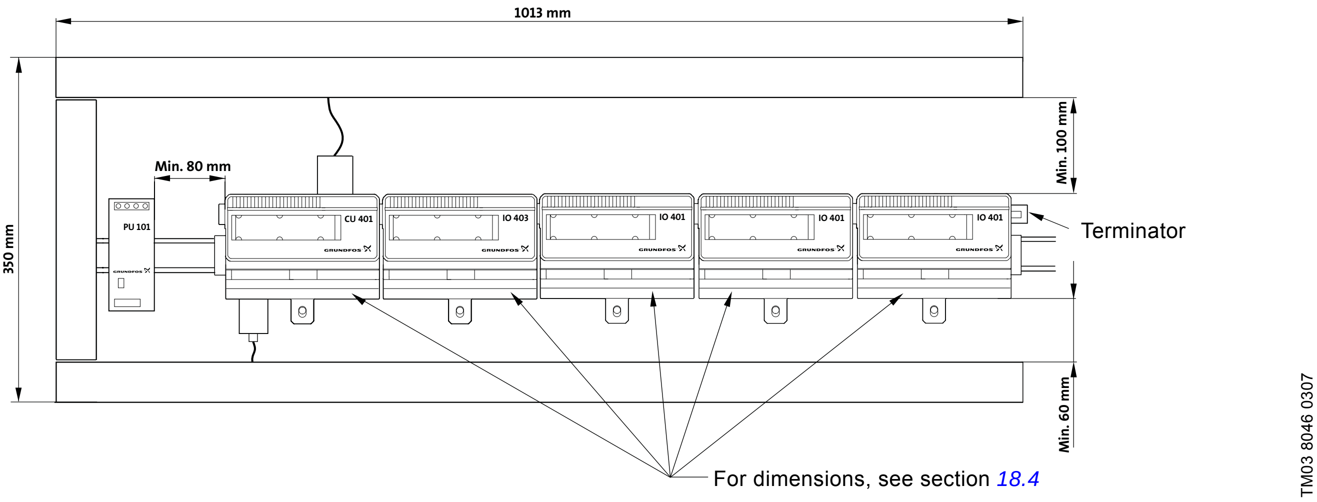


Fig. 30 Dimensional sketch, installation example 1

18.2 Installation example 2 (two rows, with extension cable)

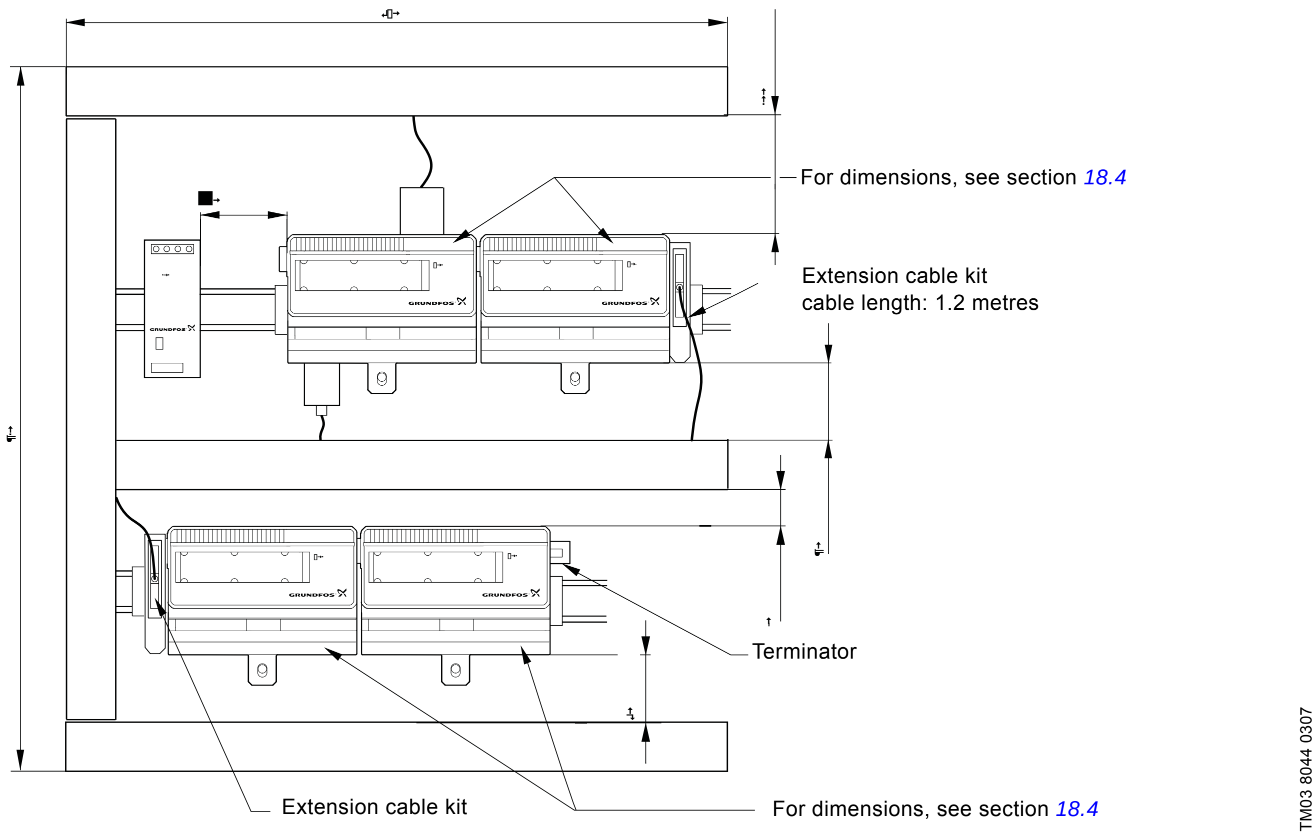


Fig. 31 Dimensional sketch, installation example 2

Note: Example 2 shows a pump controller with the possibility of connecting four pumps.

18.3 Installation example 3 (two rows, with extension cable and UPS module)

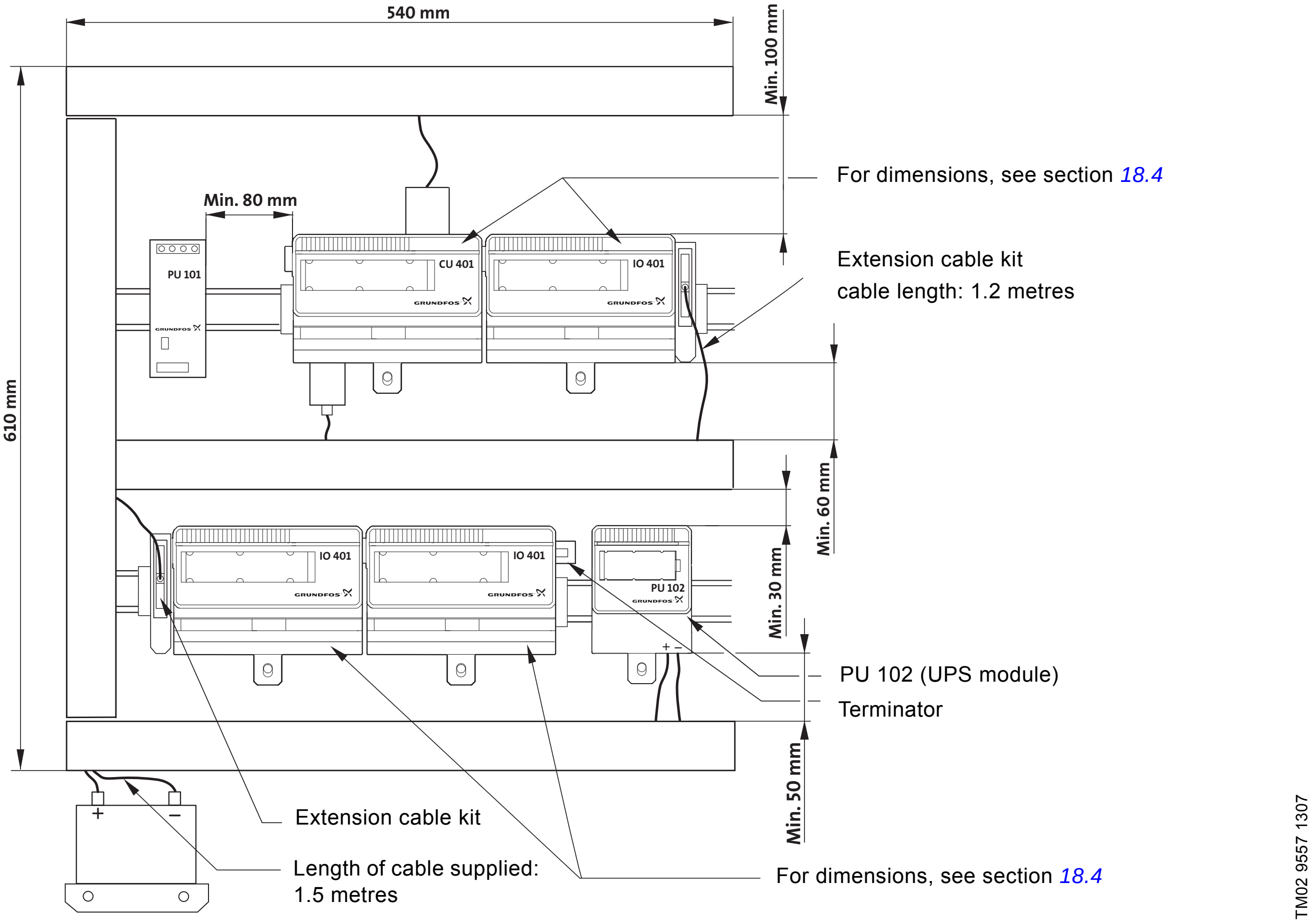
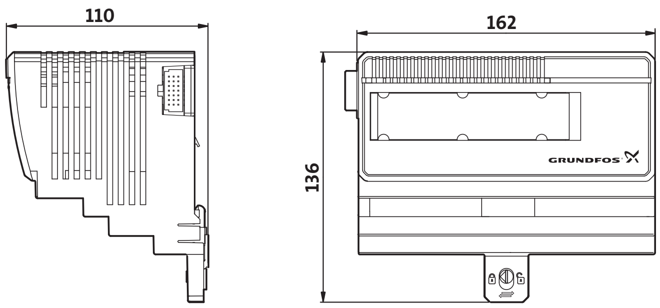


Fig. 32 Dimensional sketches, installation example 3

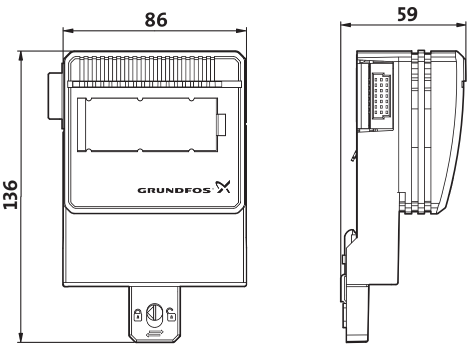
Note: Example 3 shows a pump controller with the possibility of connecting four pumps.

18.4 Dimensional sketches, control unit and modules

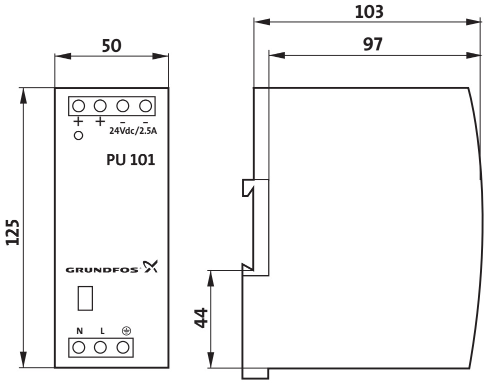
CU 401, IO 403 and IO 401



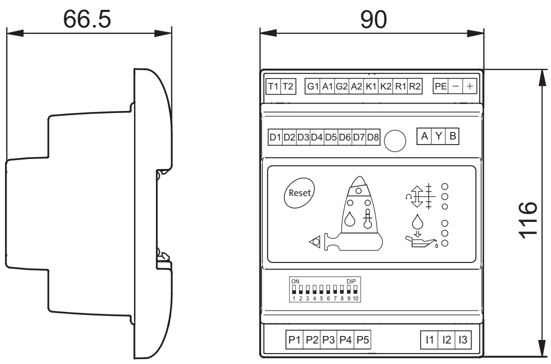
PU 102



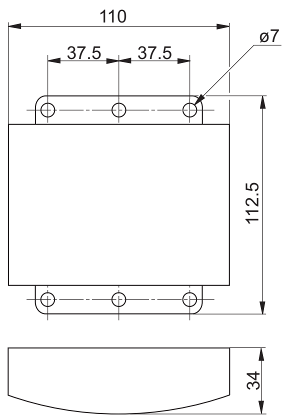
PU 101



IO 111 and IO 111 with communication module



SM 111



TM02 6715 1403 / TM02 6714 1403 / TM02 7106 2306 / TM03 0879 0705 / TM03 5837 4006

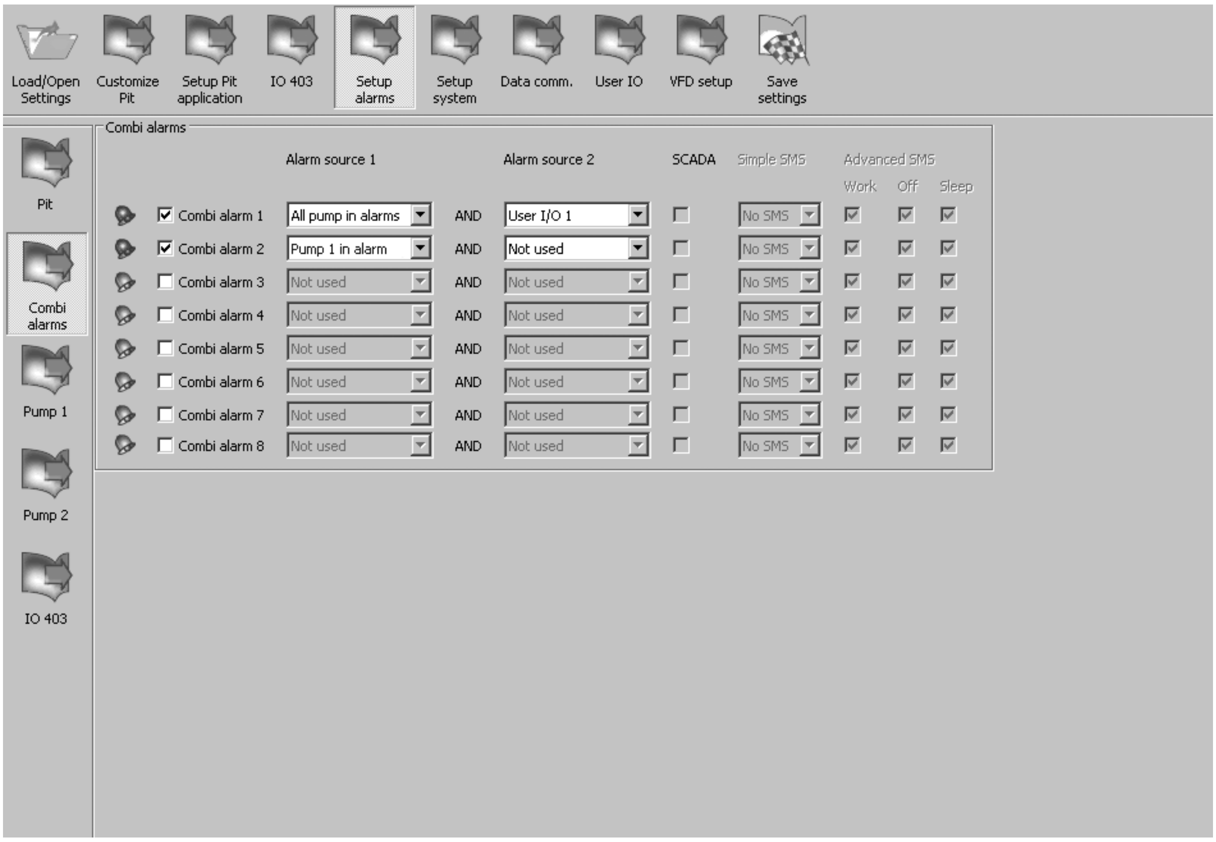
Fig. 33 Dimensional sketches, control unit and modules

Note: The dimensional sketch is only to be considered as a guideline and not a requirement for the construction of a control panel. The distances marked with arrows are the minimum values recommended.

18.5 Appendix

18.5.1 Setting of combi alarms

See section *5.11 Combi alarms*.



TM04 3500 4508

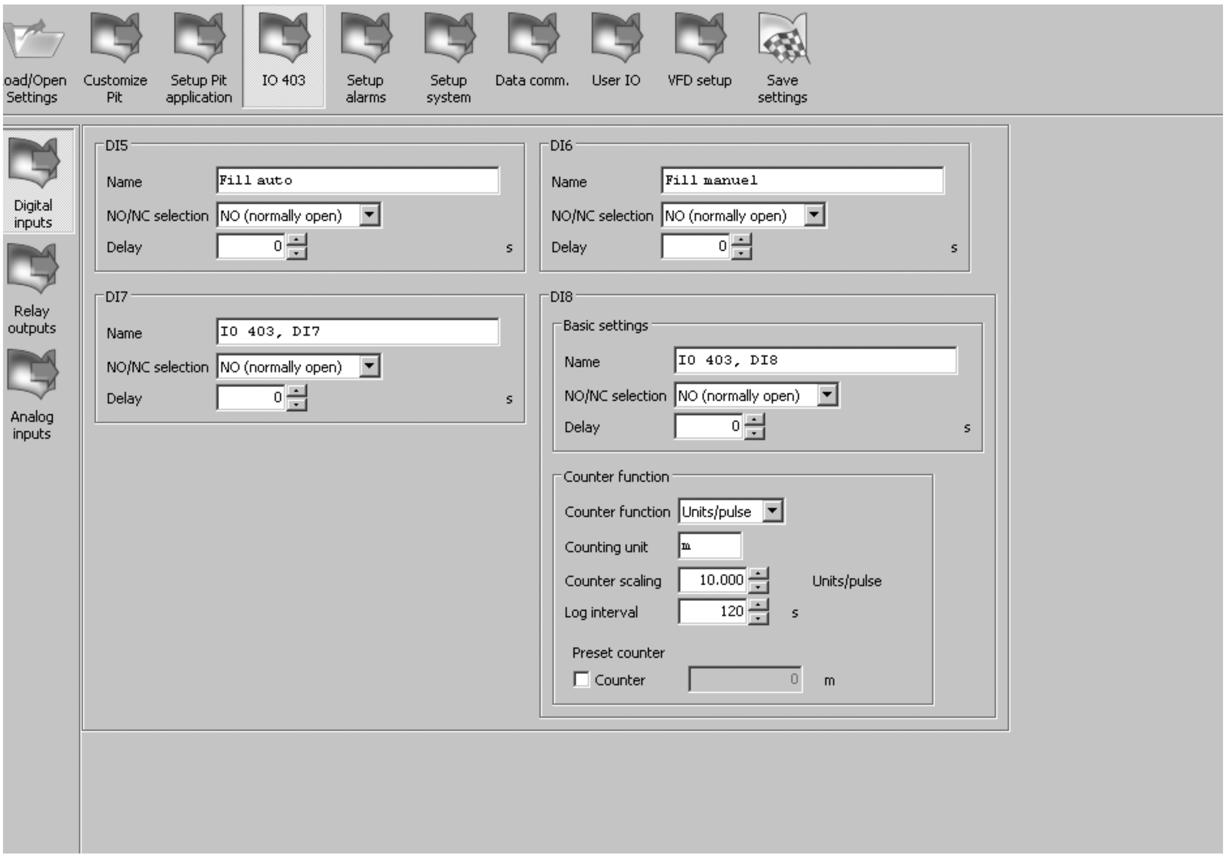
Fig. 34 Setting of combi alarms using Grundfos PC Tool for Modular Controls

18.5.2 Example of user-defined I/O

Settings to be made in the example in section *5.12 User-defined I/O* are described below.

IO 403 input/output	Function
AI1	Pressure transmitter in stormwater retention basin
DO5	Start/stop of drain pump
DO6	Start/stop of mixer 1
DO7	Start/stop of mixer 2
DI1	Auto, drain pump
DI2	Manual, drain pump

In this display, name and contactor settings are entered.

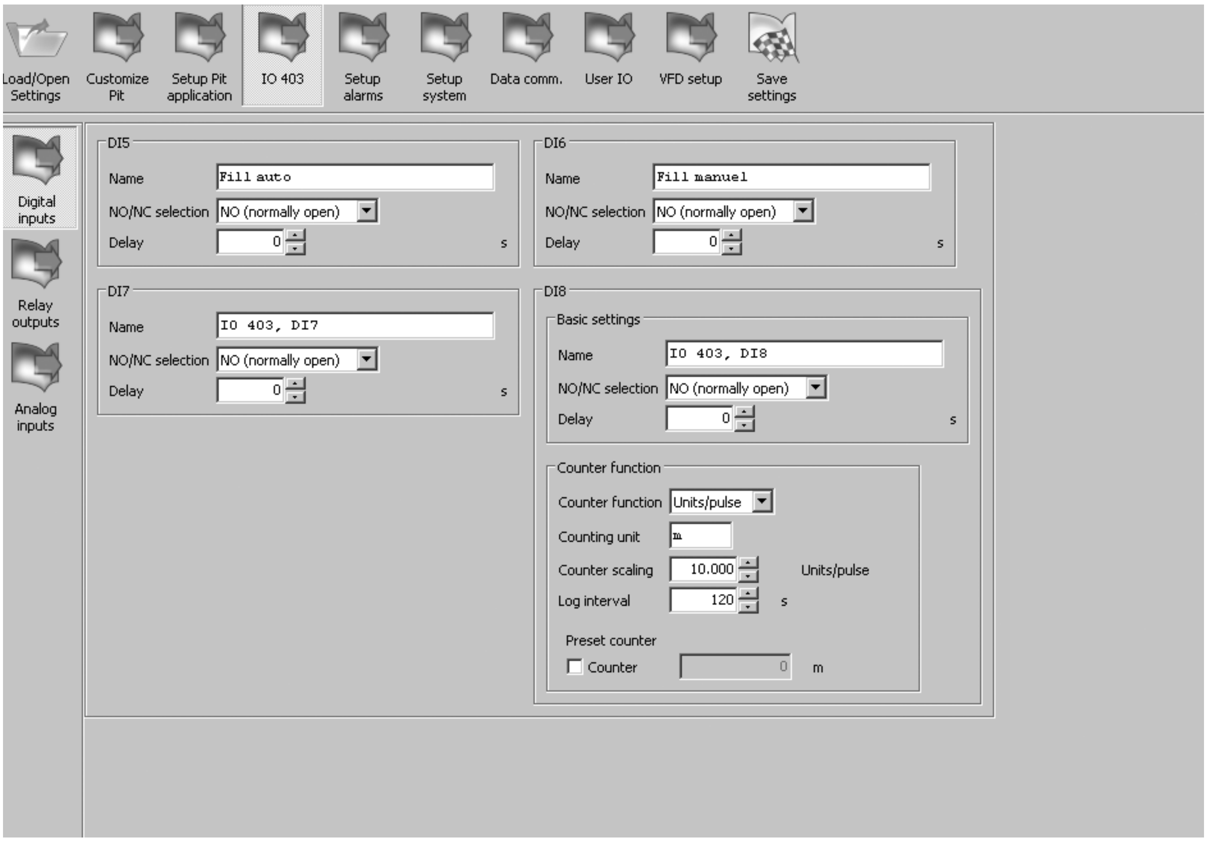


TM04 3715 7908

Fig. 35 Setting of digital inputs of IO 403

Setting of relay outputs of IO 403

In this display, pressure transmitter and auto-function is combined so that a water level over 0.5 m for more than 30 seconds will make the internal flag "User IO 1" active.

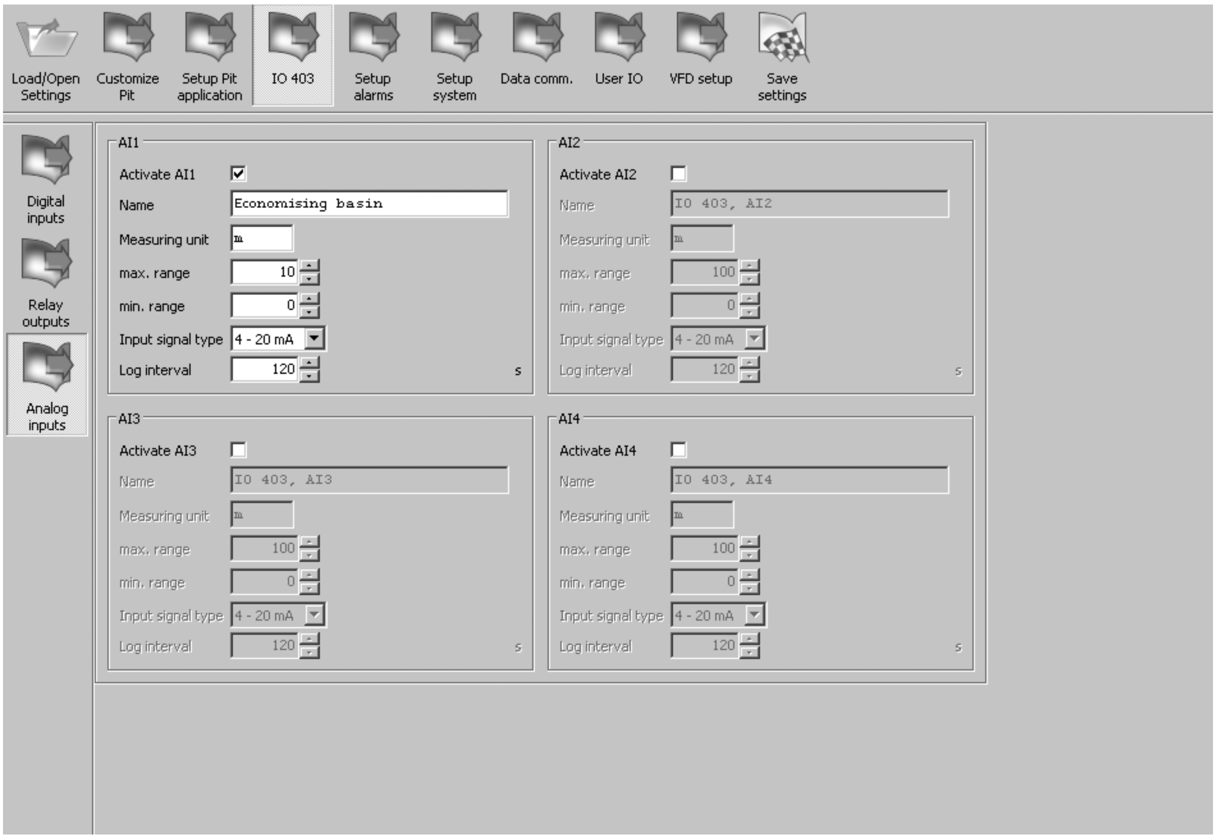


TM04 3714 4908

Fig. 36 Setting of digital inputs of IO 403

Setting of analog inputs of IO 403

Name for function is entered together with control signal and duty range.

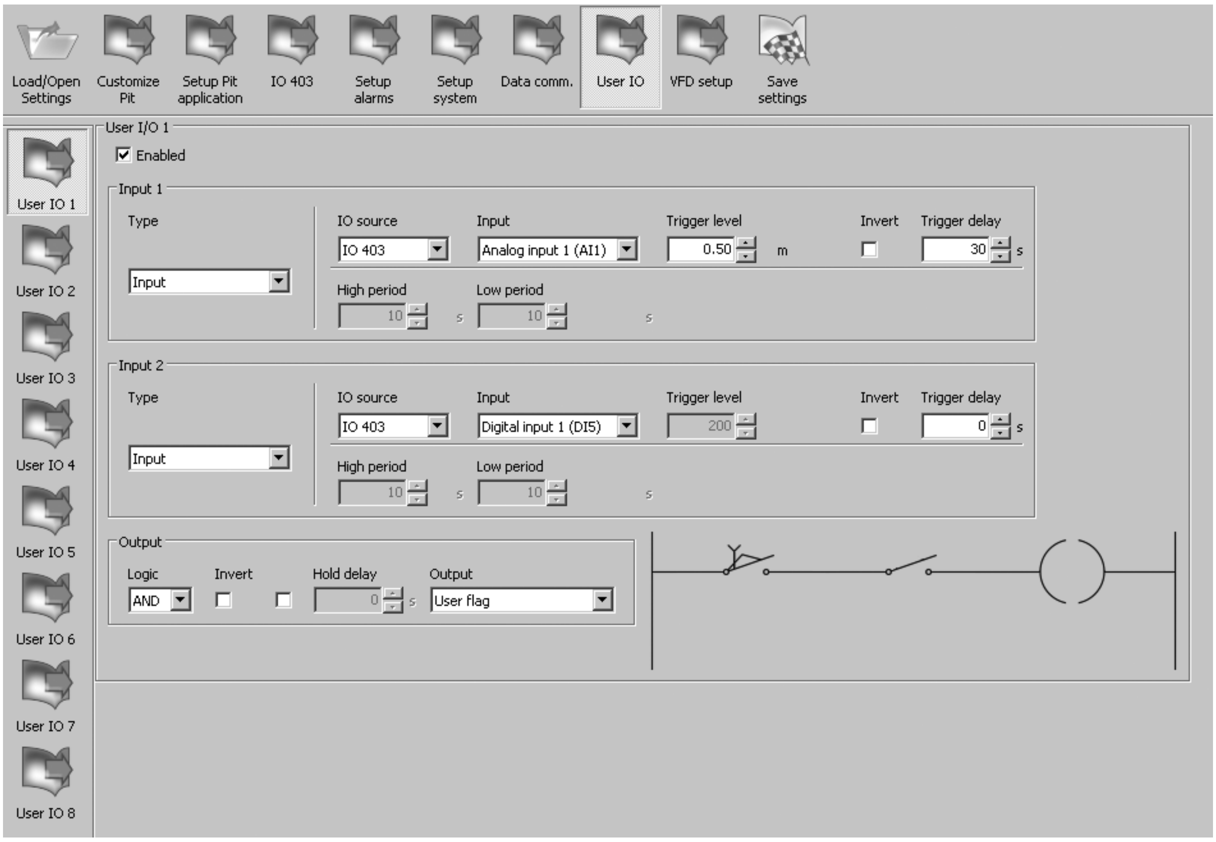


TM04 3713 4908

Fig. 37 Setting of analog inputs of IO 403

Setting of user-defined I/O 1

In this display, pressure transmitter and auto-function is combined so that a water level over 0.5 m for more than 30 seconds will make the internal flag "User IO 1" active.

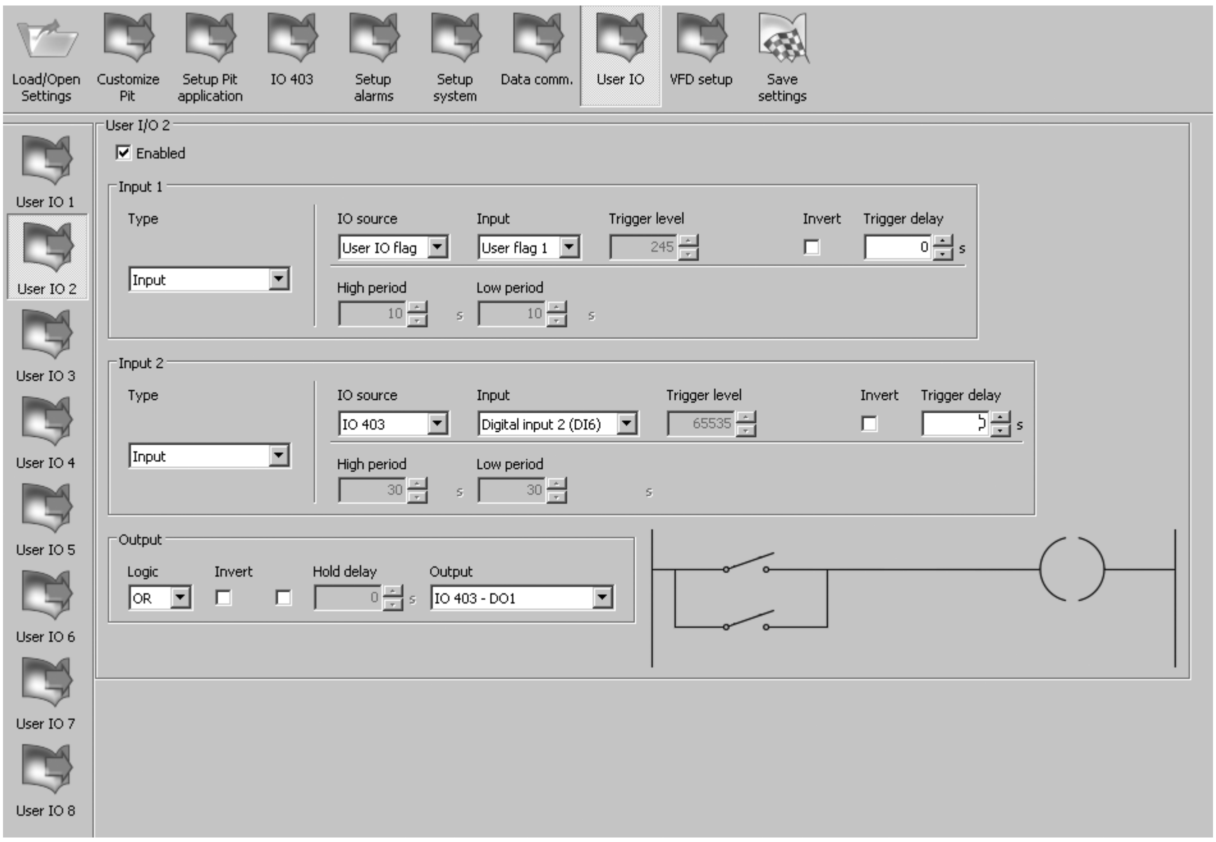


TM04 3495 4508

Fig. 38 Setting of user-defined I/O 1

Setting of user-defined I/O 2

In this display, the drain pump is activated, as manual is set to active or "User IO 1" is active.

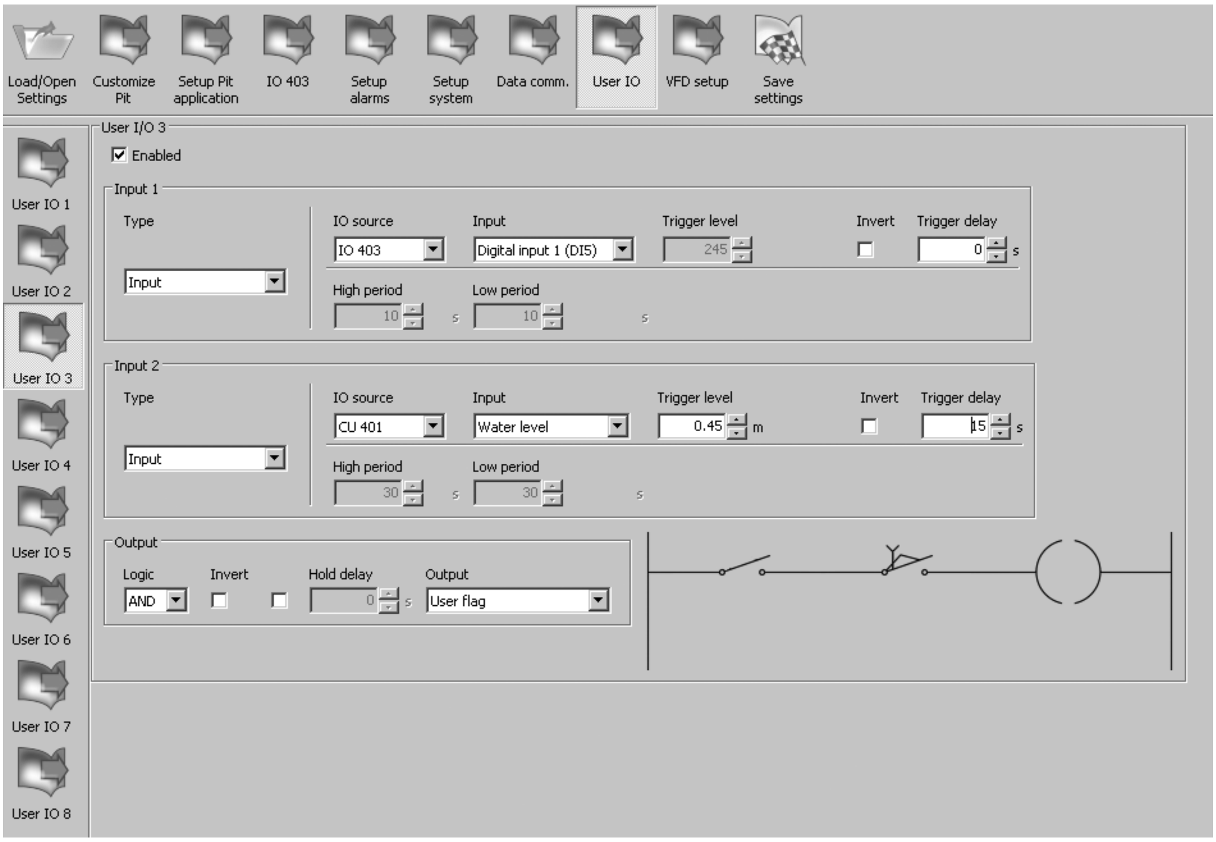


TM04 3496 4508

Fig. 39 Setting of user-defined I/O 2

Setting of user-defined I/O 3

In this display, the internal flag "User IO 3" is activated, as the system is in AUTO and the water level is over 0.45 m for more than 15 seconds. This flag is used for mixer control.

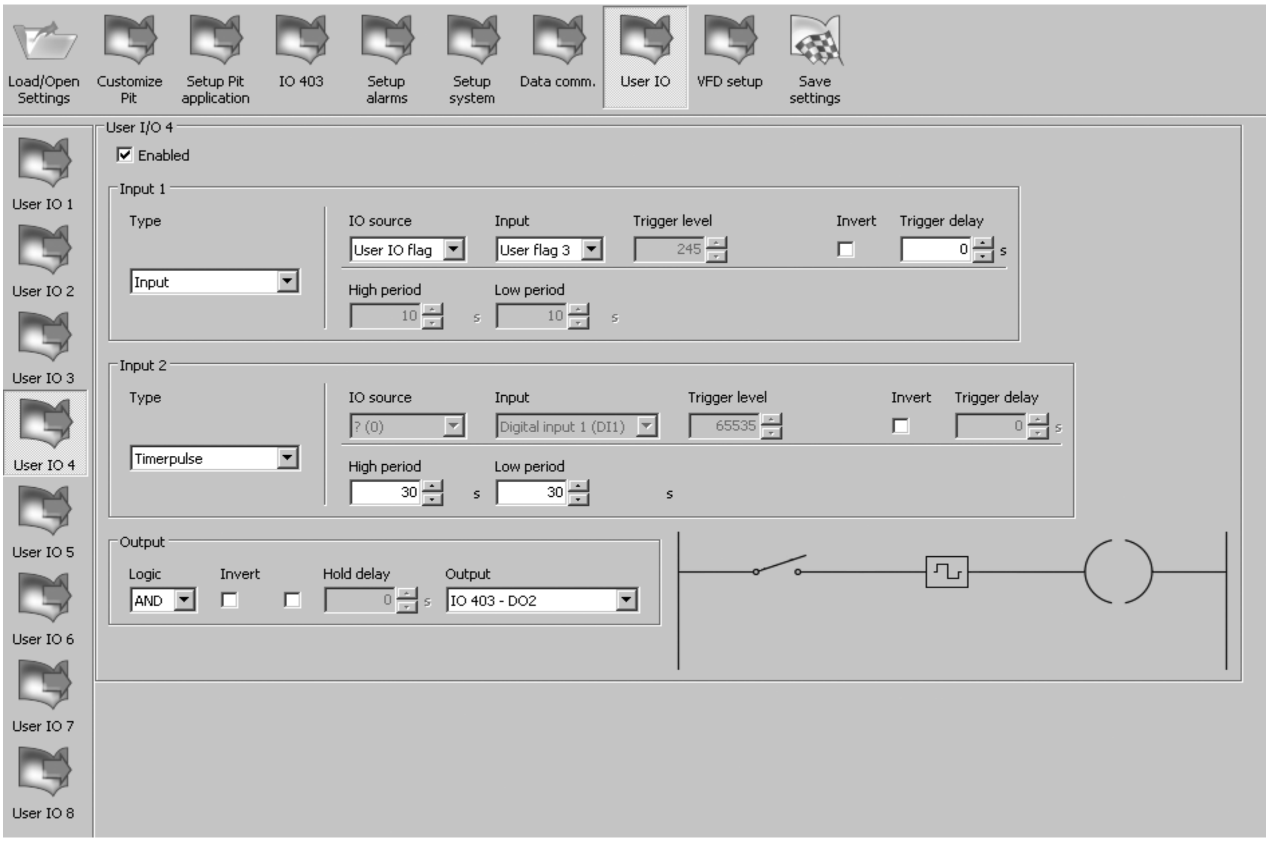


TM04 3497 4508

Fig. 40 Setting of user-defined I/O 3

Setting of user-defined I/O 4

In this display, mixer 1 is activated if "User IO 3" is active and the mixer runs with a cycle of 30 seconds of operation and 30 seconds of stop.

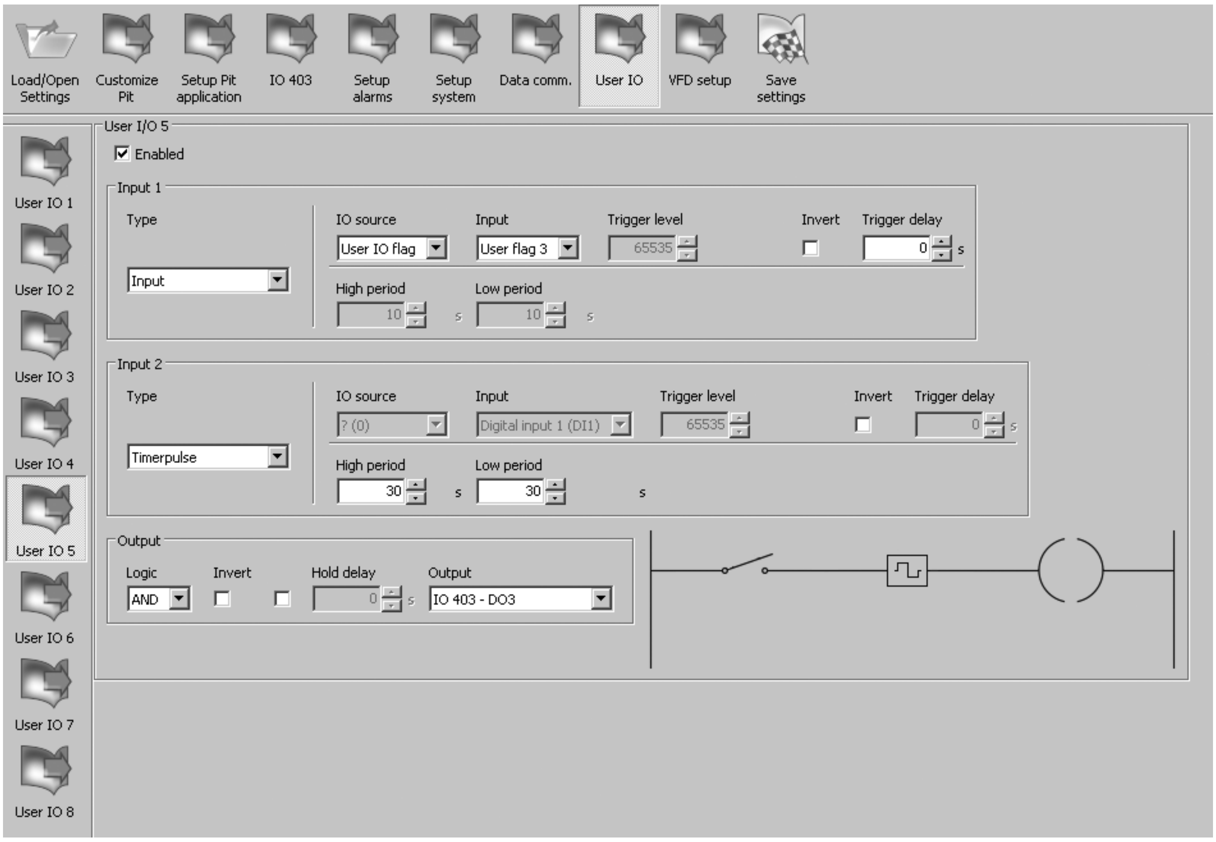


TM04 3498 4508

Fig. 41 Setting of user-defined I/O 4

Setting of user-defined I/O 5

In this display, mixer 2 is activated if "User IO 3" is active. The mixer runs with a cycle of 30 seconds of operation and 30 seconds of stop.



TM04 3492 4508

Fig. 42 Setting of user-defined I/O 5

18.5.3 Combi alarms for user-defined I/O

PC Tool - Input name	PC Tool - Input module	OD 401 - Selected index
Digital input 1 (DI1)	CU 401	256
Digital input 2 (DI2)	CU 401	257
Digital input 3 (DI3)	CU 401	258
Digital input 4 (DI4)	CU 401	259
Digital input 5 (DI5)	CU 401	260
Analog input 1 (AI1)	CU 401	261
Analog input 2 (AI2)	CU 401	262
Counter input (CNT)	-	-
Interlocked	CU 401	264
Water level	CU 401	265
Digital input 1 (DI5)	IO 403	512
Digital input 2 (DI6)	IO 403	513
Digital input 3 (DI7)	IO 403	514
Digital input 4 (DI8)	IO 403	515
Counter input (DI8)	IO 403	516
Analog input 1 (AI1)	IO 403	517
Analog input 2 (AI2)	IO 403	518
Analog input 3 (AI3)	IO 403	519
Analog input 4 (AI4)	IO 403	520
Digital input 1 (DI1)	Pump 1 / 2 / 3 / 4 / 5 / 6	768 / 1024 / 1280 / 1536 / 1792 / 2048
Digital input 2 (DI2)	Pump 1 / 2 / 3 / 4 / 5 / 6	769 / 1025 / 1281 / 1537 / 1793 / 2049
Digital input 3 (DI3)	Pump 1 / 2 / 3 / 4 / 5 / 6	770 / 1026 / 1282 / 1538 / 1794 / 2050
Digital input 4 (DI4)	Pump 1 / 2 / 3 / 4 / 5 / 6	771 / 1027 / 1283 / 1539 / 1795 / 2051
Digital input 5 (DI5)	Pump 1 / 2 / 3 / 4 / 5 / 6	772 / 1028 / 1284 / 1540 / 1796 / 2052
Therm. switch T6	Pump 1 / 2 / 3 / 4 / 5 / 6	773 / 1029 / 1285 / 1541 / 1797 / 2053
Therm. switch T7	Pump 1 / 2 / 3 / 4 / 5 / 6	774 / 1030 / 1286 / 1542 / 1798 / 2054
PT 100/PT 1000	Pump 1 / 2 / 3 / 4 / 5 / 6	775 / 1031 / 1287 / 1543 / 1799 / 2055
SM111 Extra AI	-	-
WIO	Pump 1 / 2 / 3 / 4 / 5 / 6	777 / 1033 / 1289 / 1545 / 1801 / 2057
Vibration	-	-
Pump is running	Pump 1 / 2 / 3 / 4 / 5 / 6	779 / 1035 / 1291 / 1547 / 1803 / 2059
Pump is stopped	Pump 1 / 2 / 3 / 4 / 5 / 6	781 / 1037 / 1293 / 1549 / 1805 / 2061
User flag 1	User IO flag	2304
User flag 2	User IO flag	2305
User flag 3	User IO flag	2306
User flag 4	User IO flag	2307
User flag 5	User IO flag	2308
User flag 6	User IO flag	2309
User flag 7	User IO flag	2310
User flag 8	User IO flag	2311
Combi Alarm 1	Combi alarm	2560
Combi Alarm 2	Combi alarm	2561
Combi Alarm 3	Combi alarm	2562
Combi Alarm 4	Combi alarm	2563
Combi Alarm 5	Combi alarm	2564
Combi Alarm 6	Combi alarm	2565
Combi Alarm 7	Combi alarm	2566
Combi Alarm 8	Combi alarm	2567
PC Tool - Output name	OD 401-Output channel	
User flag	0	
Relay output 7 (DO7)	1	
Relay output 8 (DO8)	2	
IO 403 - DO1	3	
IO 403 - DO2	4	
IO 403 - DO3	5	
IO 403 - DO4	6	

P1 relay output 8 (DO8)	-
P2 relay output 8 (DO8)	8
P3 relay output 8 (DO8)	9
P4 relay output 8 (DO8)	10
P5 relay output 8 (DO8)	11
P6 relay output 8 (DO8)	12
IO 351_D00	
IO 351_D01	
IO 351_D02	
IO 351_D03	
IO 351_D04	
IO 351_D05	
IO 351_D06	
IO 351_D07	
IO 351_D08	

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